Chapter 19 — Case Studies: Commerce & Marketplaces (Part 2 of Platforms)

> **Where this fits.** Chapter 18 designed platforms where people *talk* (Q&A sites, social networks, sports scores). This chapter designs platforms where people *buy, sell and trade*. Money moves, stock runs out, prices change every second, and two people often want the same last thing at the same moment. Those three pressures — **money must be exact, state must follow rules, and shared things must not be double-sold** — shape every design here.

What you will learn

#	Problem	The one idea that makes it work	Main patterns
19.1	Amazon — online shopping	An order is an immutable copy of the cart, moved through a **state machine**	State machine (enum), Strategy (payment), Observer (notifications), Facade
19.2	Online stock brokerage + stock-exchange matching engine	Four order types differ in only **two questions**; a book matches by **price, then time**	Template Method, Observer, Strategy (price policy)
19.3	Online auction	"Read highest bid, compare, replace" must be **one atomic step**	Monitor lock, Observer, Specification
19.4	Music streaming (Spotify)	Play order is a **pluggable strategy**; plan rules live **in the plan**	Strategy, Observer, enum-with-behaviour
19.5	Food delivery (Swiggy) + FoodKart	Capacity and riders are **claimed atomically**; selection is a **strategy**	Strategy, Observer, State machine, CAS
19.6	Grocery store (POS & checkout)	A **pure pricing engine**: best offer per product, then a coupon spread over items so tax and refunds stay exact	Strategy (offers, tax, payment), Facade, Observer, enum state machine

How to run every example in this chapter

```
cd chapters/19-case-platforms-2/code
javac --release 21 -d /tmp/lld_19 $(find . -name '*.java')
java -cp /tmp/lld_19 lld.ch19.amazon.Main          # or brokerage, stockexchange, auction, music, fooddelivery, foodkart, grocery
```

`javac` is the Java compiler (it turns `.java` files into runnable `.class` files); `-d` names the output folder, and `java -cp` runs a class, where the **classpath** (`-cp`) is the list of folders Java searches for compiled classes.

All eight `Main` classes compile with Java 21 (standard library only — only the classes that ship with Java) and run. The output under each implementation is the real output of that program. The stock-exchange `Main` also runs a small test suite (a set of automatic checks that confirm the code behaves as expected; no JUnit — the usual Java testing library — needed).

> **Code-style note.** To keep each case readable, one file sometimes holds several small package-private types (for example `Payments.java` holds `Payment`, its two subclasses and `Shipment`). That is legal Java. In production you would usually give each top-level type its own file.

> **Key term — Java vocabulary used in this chapter:** A **package-private** type has no access keyword and is visible only inside its own package (folder of classes). A **top-level type** is a class, interface, enum or record not nested in another. An **interface** lists methods a class promises to provide. An **enum** is a type with a fixed list of named values. A **record** is a compact class that just carries data that never changes. A **sealed** type lists exactly which subclasses are allowed (the `permits` clause), so the compiler knows every possible kind. **`synchronized`** means only one thread at a time may run that code on the same object.

> **Key term — the design patterns named in this chapter: Strategy** puts each interchangeable way of doing a job (each payment method, each play order) behind one interface so it can be swapped or added without editing the caller. **Observer** lets objects subscribe to events from another object and be called back automatically, so the publisher does not know who listens. **Facade** is one simple class that hides many classes behind a few easy methods. **Template Method** is a base-class method that fixes the outline of an algorithm and lets subclasses fill in one or two steps. **Specification** turns each business rule or search condition into a small yes/no object that can be combined with others.

Four rules that apply to every design in this chapter

1. **Money is `BigDecimal`, never `double`.** `double` is Java's floating-point number type: it stores numbers in binary, so most decimal fractions are only approximated. `BigDecimal` is Java's exact decimal-number class. A `double` cannot store 0.1 exactly, so `0.1 + 0.2` prints `0.30000000000000004`. With `BigDecimal` you choose the scale (number of decimal places) and the rounding mode yourself. Two traps: build it from a `String` (`new BigDecimal("0.10")`), not from a `double`; and compare with `compareTo`, not `equals` — `new BigDecimal("2.0").equals(new BigDecimal("2.00"))` is `false` because the scales differ.
2. **Status changes follow a state machine.** A **state machine** is a fixed list of states plus the allowed moves between them. We put the allowed moves *inside the enum*, so an illegal move (for example "cancel a delivered order") throws immediately instead of silently corrupting data.
3. **Check-then-act on shared stock must be atomic.** "Is there one left? Then take it" is two steps. If two threads (independent paths of execution running at the same time inside one program) run step 1 together, both see "yes" and both take it — a **race condition**, where the result depends on which thread happens to get there first. The fix is to do both steps under one lock (a guard that only one thread can hold at a time; the others wait), or with one atomic compare-and-set (CAS: "set to B only if it is still A").
4. **Copy prices into orders.** An order remembers the price that was paid, and is **immutable** (it cannot be changed after it is created). If the menu or catalog price changes later, old orders must not change.

Design Amazon — Online Shopping System

Problem & clarifying questions

Amazon started as an online bookshop and now sells almost every kind of consumer product and digital media. We design the **online retail part**: people list products, other people find them, put them in a cart, pay, and track the parcel.

Questions worth asking before you draw anything (with the answers we assume):

Question	Assumed answer
Can someone buy without an account?	No. Browsing and filling a cart are open to everyone; placing an order needs a **registered member** account.
Who sells?	Members can list products for sale (a marketplace, not only first-party — goods sold by the platform itself).
How do people pay?	Credit card or electronic bank transfer. Cards clear at once, transfers clear later.
Can an order be cancelled?	Yes, until it has shipped.
Do we notify people?	Yes, on every order or shipping status change.
Returns, wish lists, seller payouts, recommendations, stock spread over several warehouses?	Out of scope — a deliberate choice, and you should say so out loud.

Requirements

Functional

1. Members can add new products to sell.
2. Anyone can search products by **name** or **category**.
3. Anyone can search and view products; only registered members can buy.
4. Users can add, remove and change the quantity of items in their shopping cart.
5. Users can check out and buy everything in the cart.
6. Members can rate and review a product.
7. Members give a shipping address for delivery.
8. A member can cancel an order that has not shipped.
9. Members are notified each time the status of their order or of its shipment changes.
10. Members pay with a credit card or with an electronic bank transfer.
11. Members can track the shipment and see the current order state.

Non-functional

- **Correctness of stock:** two buyers can never both get the last unit.
- **Exact money:** totals are computed with `BigDecimal`.
- **Auditability** (being able to prove later exactly what happened and when)**:** every order and shipment keeps a history of its states, not just the latest one.
- **Extensibility** (how easily new features can be added without rewriting old code)**:** adding a payment method or a notification channel must not change the checkout code.

Use cases

Four actors:

- **Admin** — manages accounts (block / unblock) and adds or changes product categories.
- **Guest** — searches the catalog, adds and removes cart items, and can register to become a member.
- **Member** — everything a guest can do, plus placing orders and listing products to sell.
- **System** — sends order and shipment notifications.

Top use cases: add or update products (each change also updates the catalog), search products (by name, or by category), put items into the cart or take them out, check out, pay, add a product category, and send shipment notifications.

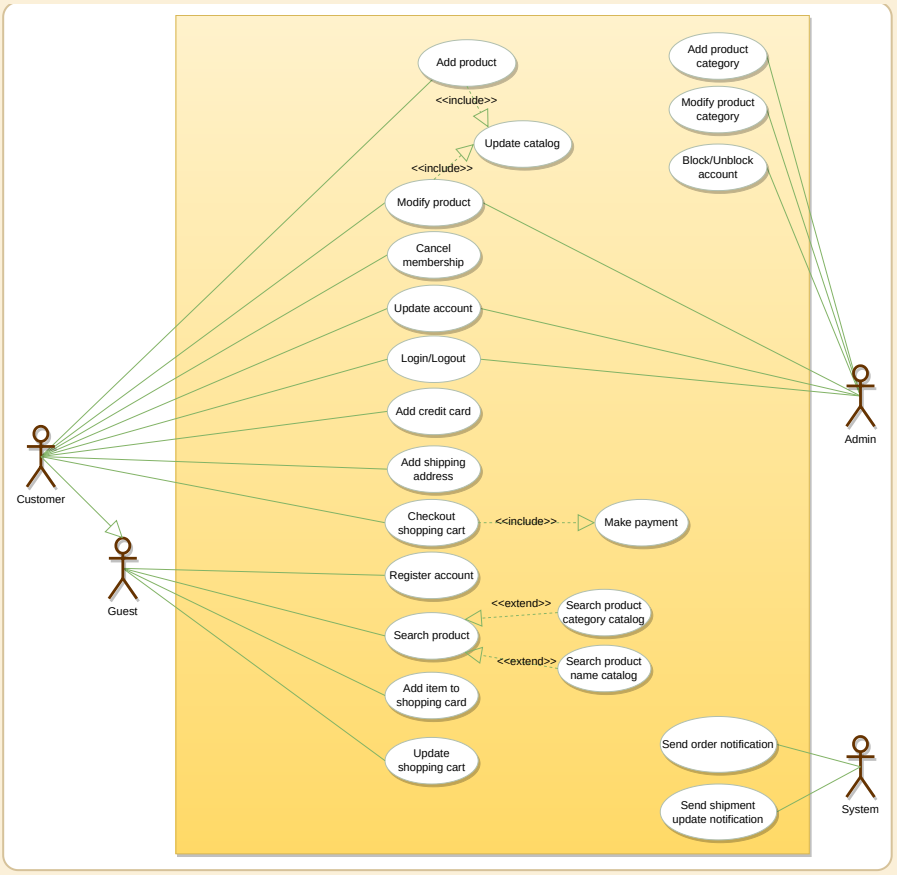


Figure — Use-case diagram for the online shopping system: Admin, Guest, Member and System and what each may do (course diagram)

In the figure above, notice that "Checkout" *includes* "Make payment" and "Add shipping address": you cannot check out without them. Search by name and search by category *extend* the general "Search product" use case.

Core entities

Class	Responsibility
`ProductCategory`	A named group such as Books or Electronics. Exists whether or not any product uses it.
`Product`	The thing sold: price, seller, stock count. **Owns** its reviews and guards its own stock (`reserve` / `release`).
`ProductReview`	Rating 1–5 plus text. Records the reviewer **by id**, so `Product` never depends on `Member` .
`Search` / `Catalog`	Interface with two search methods. The catalog keeps one index (a lookup map built in advance) per search (name words → products, category → products), so a lookup reads a map instead of scanning every product, and `addProduct` updates both indexes together so they cannot drift apart. The catalog only *points to* products; it never owns them.
`ShoppingCart`	Mutable (changeable) working area. **Owns** its items; clearing the cart destroys them.
`Item`	One line: product id, name, quantity, unit price (an immutable `record`). A pen is a *product*; ten pens in stock are ten product *items*.
`Order`	Immutable **copy** of the cart lines + address + a status that moves only along legal edges.
`OrderLog`	One entry per status change (unshipped, pending, complete, cancelled...).
`Payment` (`CreditCardPayment` , `BankTransferPayment`)	Takes the money. A card clears immediately; a bank transfer comes back `PENDING` from `process()` and clears later.
`Shipment` / `ShipmentLog`	A parcel and its tracking history (pending, shipped, delivered, on hold). The last log entry is "where is it now".
`Notification` channel (`EmailChannel` , `SmsChannel`)	One way of delivering a message. (The course draws an abstract `Notification` class with one subclass per channel; we use an interface — same idea.)
`Account`	Login data, addresses, status (active, blocked, banned...).
`Customer` → `Guest` , `Member`	`Customer` has only what *anyone* may do; ordering lives on `Member` .
`Admin`	Adds categories, blocks and unblocks members.
`OrderService`	Application service (a class that coordinates domain objects to carry out one user action) that turns a cart into an order and drives its state machine.

Class diagram

The course builds the model in five stages (products → catalog → payment/shipment/notification → cart/order → accounts). Here is the finished picture:

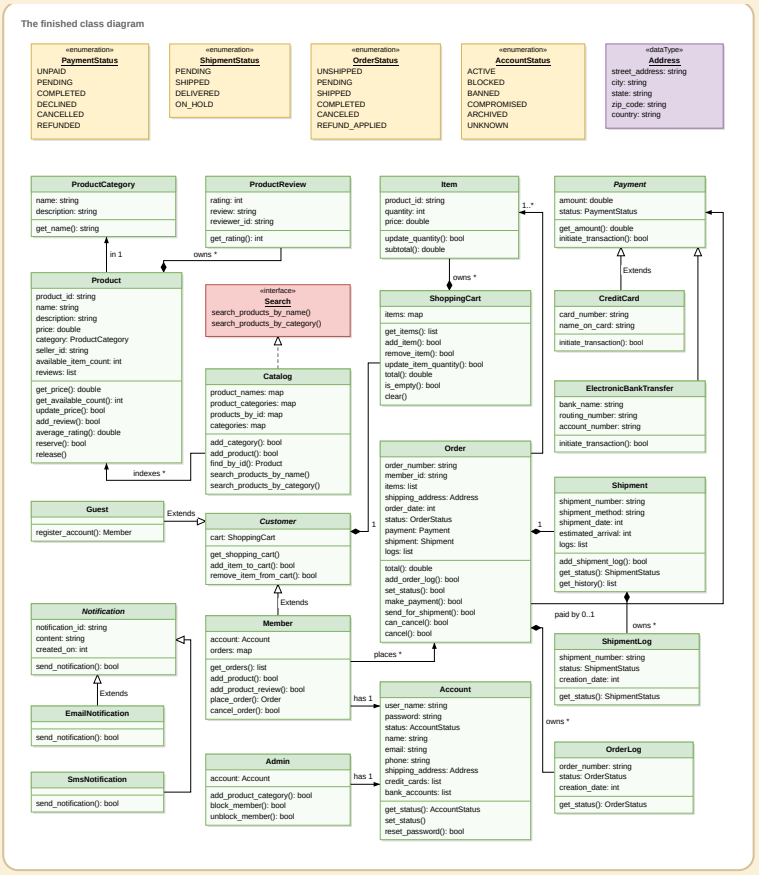


Figure — The finished class diagram of the online shopping system (course diagram)

In the figure above, read the arrows this way: a **filled diamond** means *composition* (the part belongs to the whole and is destroyed along with it — `Product` owns its reviews, `ShoppingCart` owns its items, `Order` owns its log and shipment). A **plain arrow** is *association* (a link without ownership — a product points at its category; an order only *refers to* the payment it was handed). The hollow triangle is inheritance (`Guest` and `Member` extend `Customer`).

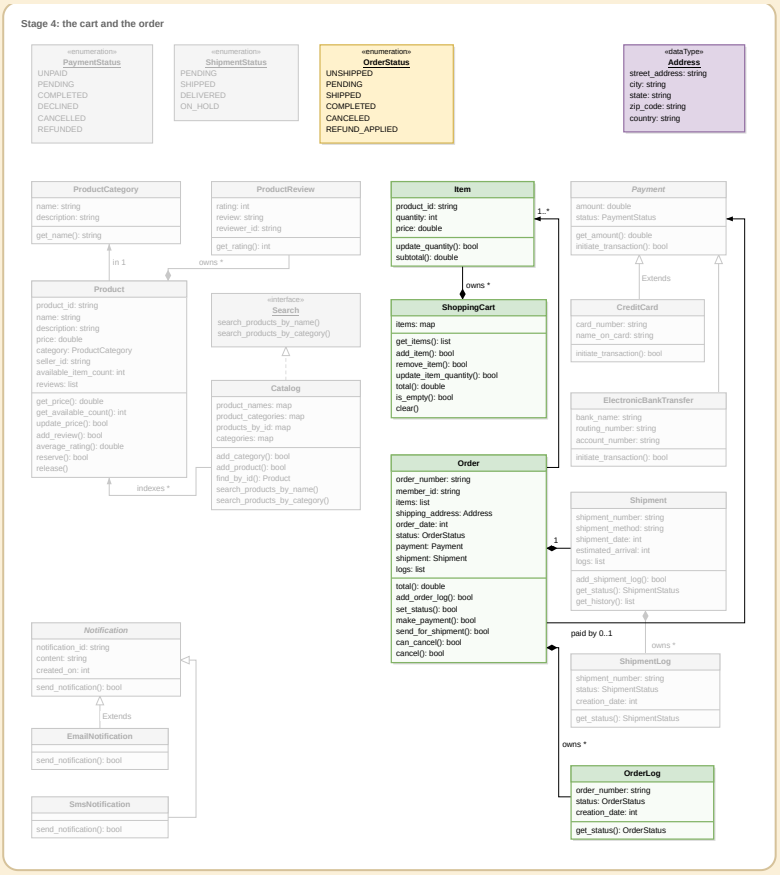
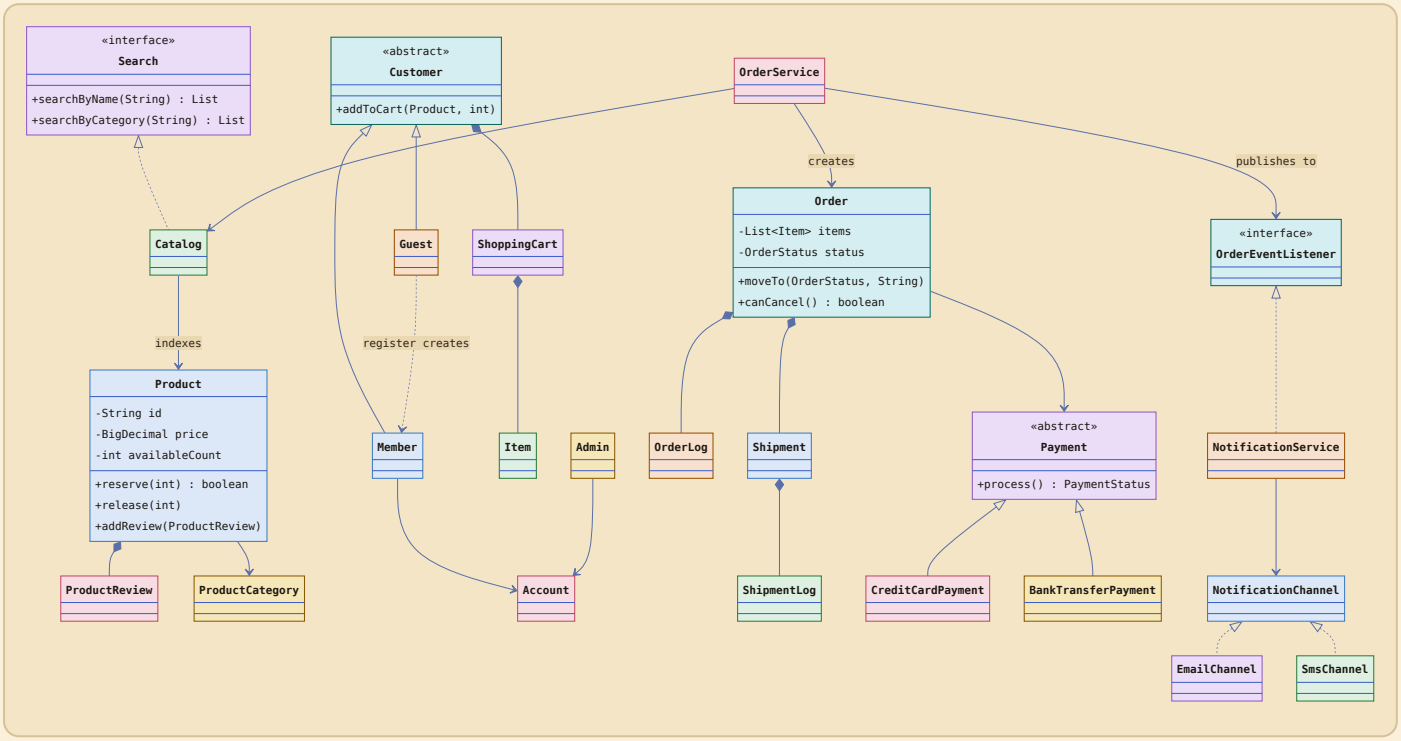


Figure — Stage 4: the cart owns its items, while the order holds its own copy of what was bought (course diagram)

Our Java design follows the same shape:



Key flows

Shopping activity. The course's activity diagram shows the loop a shopper goes round: search, view, add to cart, update the cart, "more shopping?", and finally checkout.

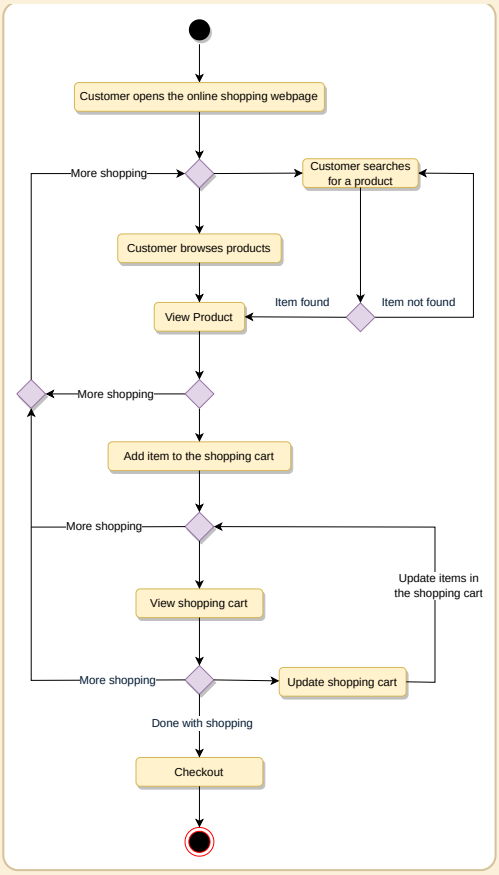


Figure — Activity diagram of a customer shopping online, from opening the site to checkout (course diagram)

Searching the catalog, and adding to the cart. The first two course sequence diagrams show the simple read path (customer → catalog → matching products) and the cart update path (customer → cart → item).

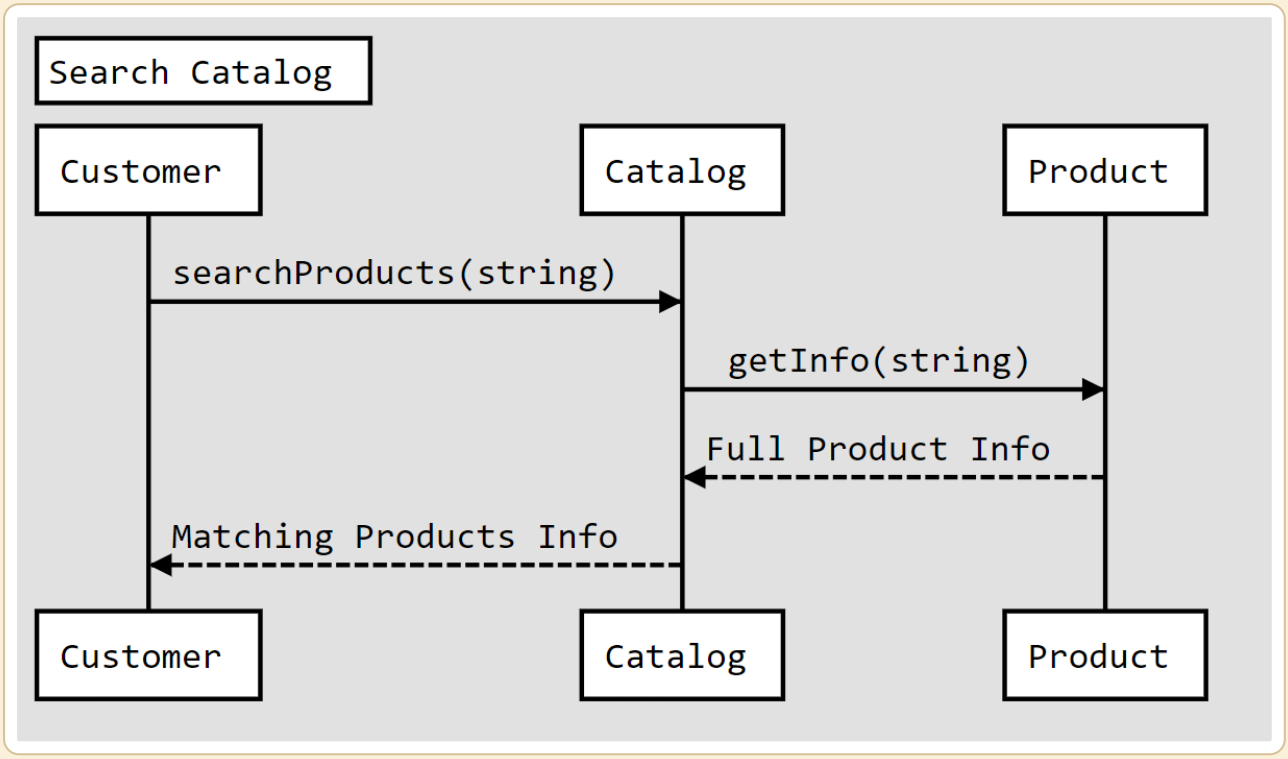


Figure — Sequence diagram for searching the catalog (course diagram)

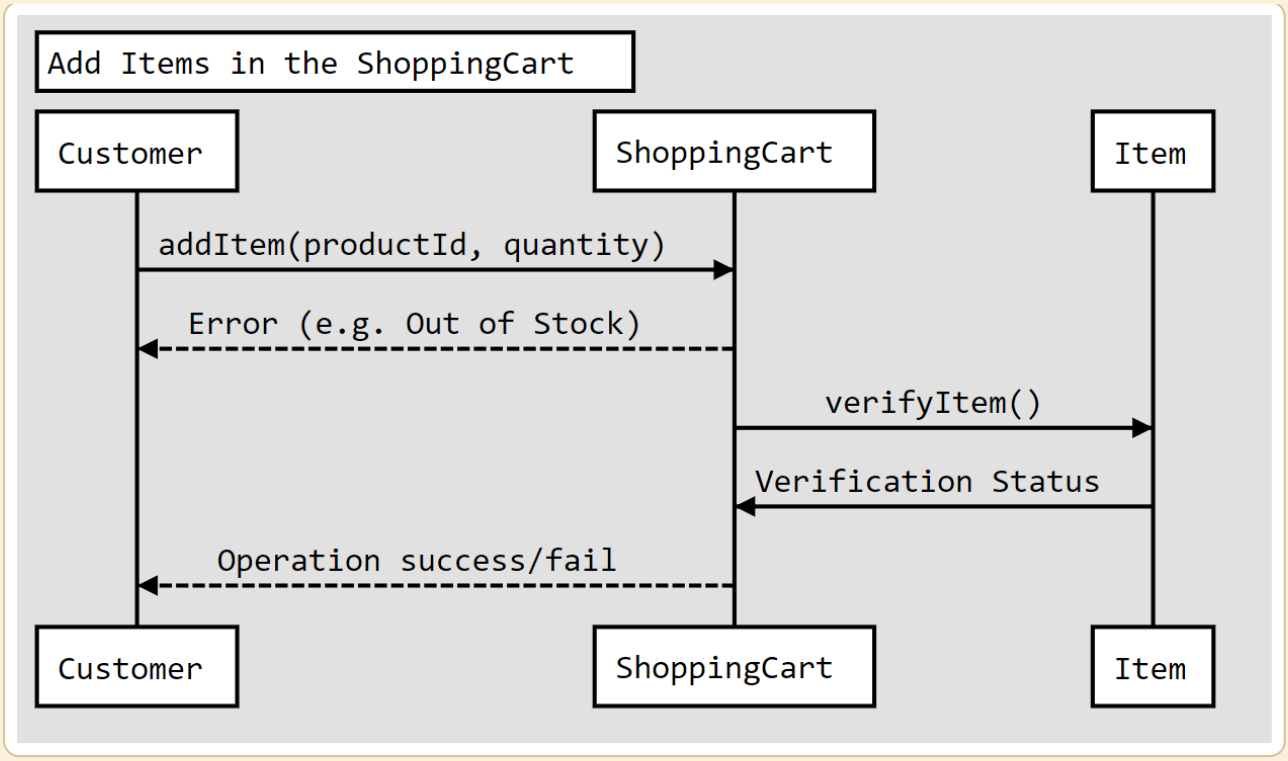


Figure — Sequence diagram for adding an item to the shopping cart (course diagram)

Checkout. The third course sequence diagram shows the important one: get the cart items, get billing and shipping info, create the order, verify every item, process the payment, create the shipment, and return the order status — with an error path after item verification and after payment.

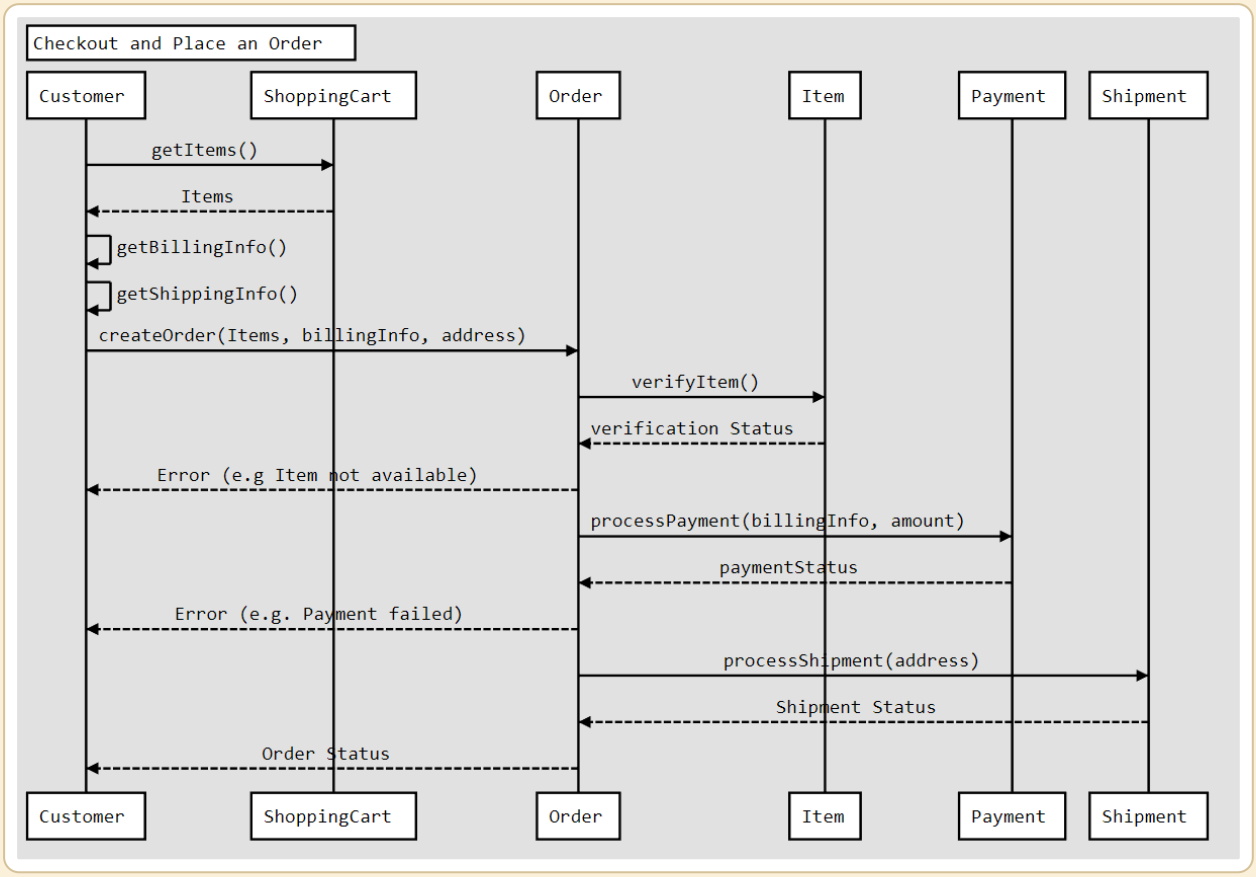
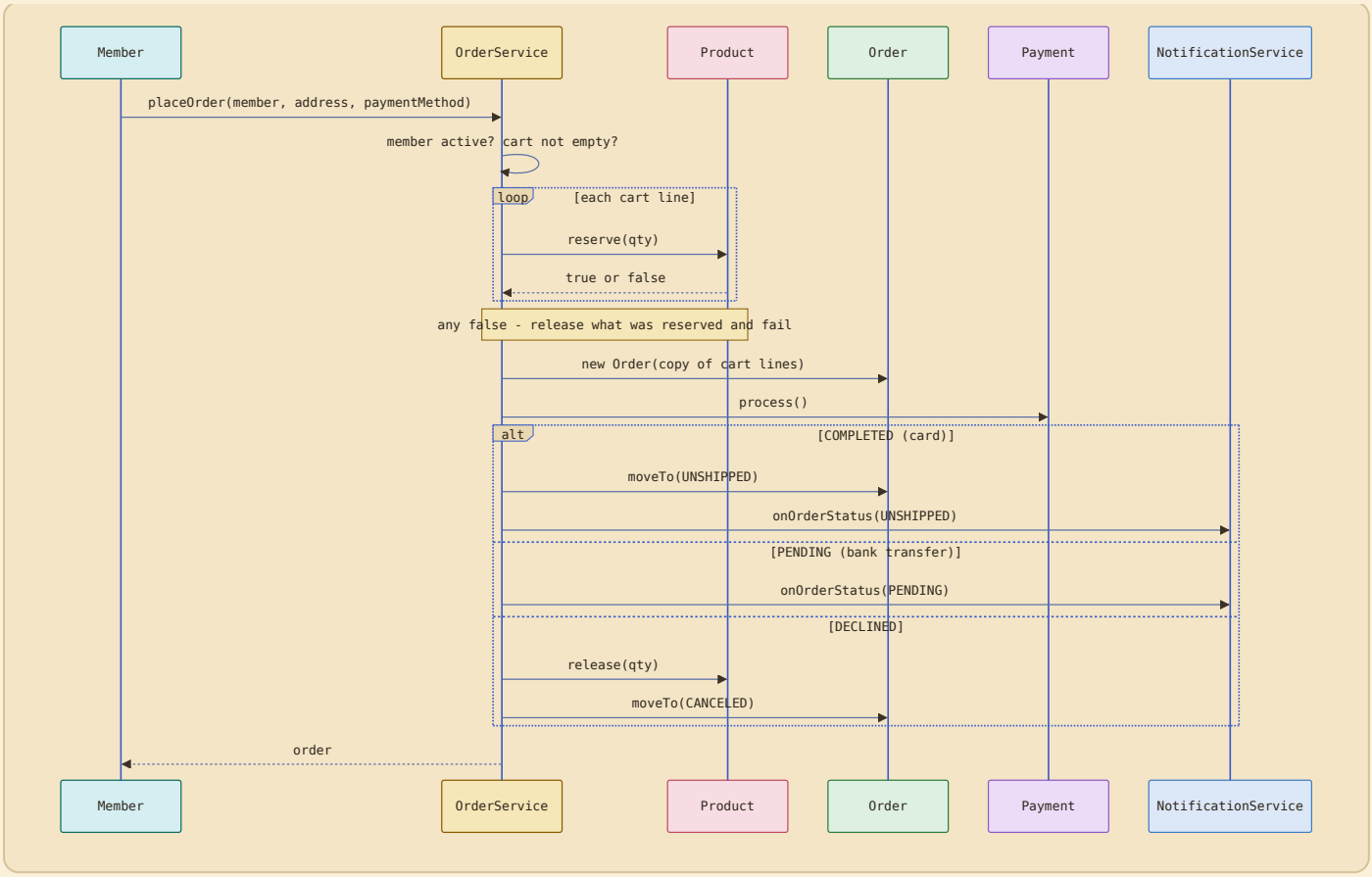
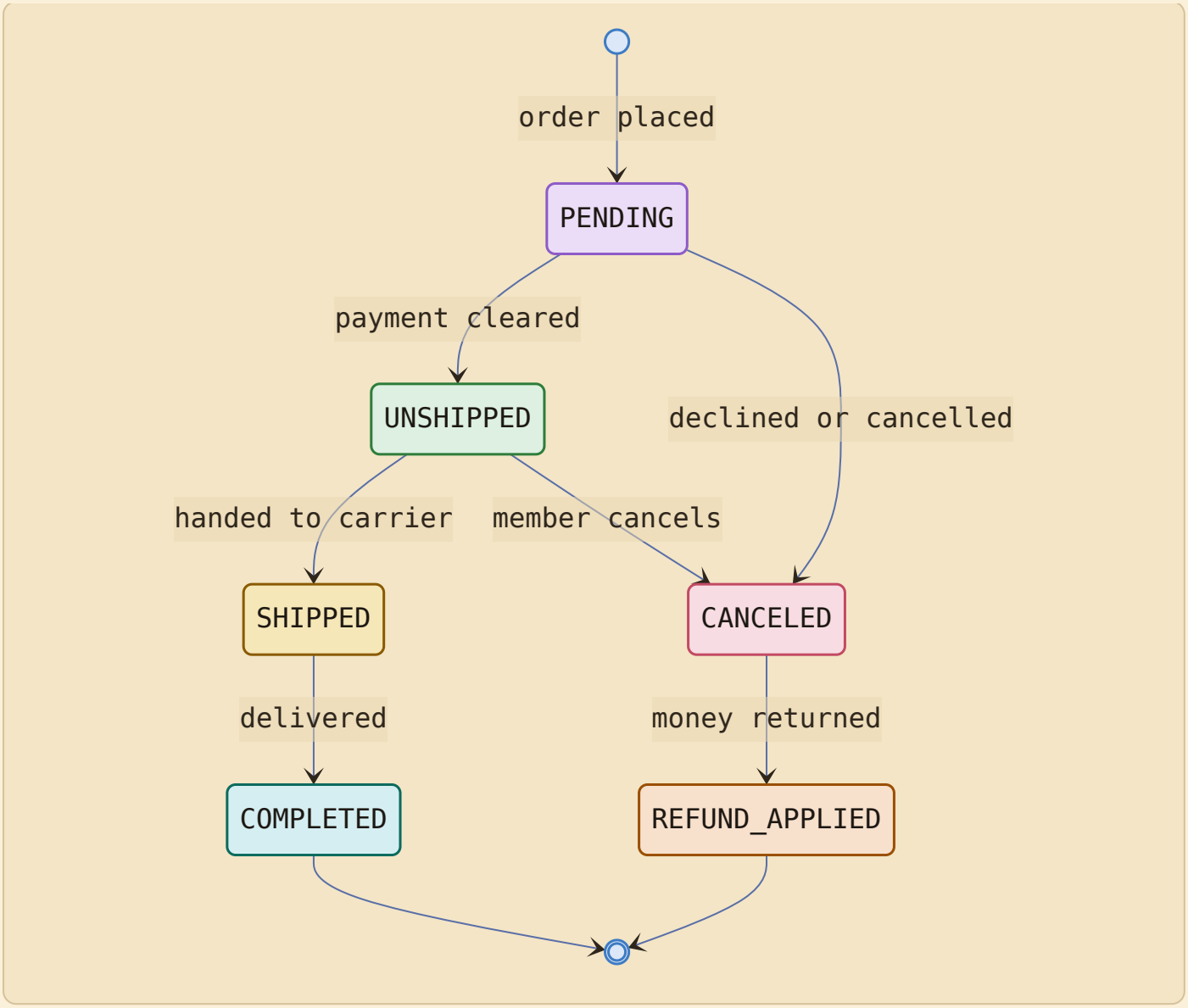


Figure — Sequence diagram for checkout: verify items, take payment, create shipment (course diagram)

Our `OrderService.placeOrder` follows that order, with one addition: stock is **reserved** during verification (not merely checked), so the check cannot go stale before payment.



The order state machine. Every legal move is listed in `OrderStatus.next()`. Anything else throws.



Requirement 8 ("cancel only if not shipped") is now one line: `canCancel()` returns `status.canMoveTo(CANCELED)`. There is no `SHIPPED → CANCELED` edge, so it is impossible by construction.

Design decisions & patterns used

Decision	Why	Alternative rejected
Ordering lives on `Member`, not `Customer`	A `Guest` *is a* `Customer`. If `Customer` had `placeOrder`, every guest would inherit an operation it has to reject — breaking the **Liskov Substitution Principle** (a subclass must be usable wherever its parent is, without surprises).	`Customer.placeOrder()` that throws for guests (the classic LSP smell).
`Guest.register()` returns a new `Member` and carries the cart across	Registration is a real-world moment where the cart must survive.	Mutating a guest into a member (Java objects cannot change class).
`Order` copies the cart lines	An order documents a *finished purchase*; it must stay fixed even if the cart is edited later.	Sharing `Item` objects — cheaper, but editing the cart would silently rewrite a finished order.
Stock reserved at order time, not at payment time	Prevents two shoppers from both getting the final unit. Cost: an abandoned order holds stock until cancelled; real systems release it on a timer.	Check stock, then pay, then decrement (a race window between check and decrement).
State machine inside `OrderStatus`	One place owns the rules. `moveTo` rejects illegal edges.	`if` checks scattered over `cancel`, `ship`, `deliver`.
`Payment` is an abstract (sealed) class, one subclass per method — a **Strategy**	The two methods *behave* differently (card clears now, transfer is pending). An **abstract class** cannot be instantiated itself — only its subclasses can — so every payment is a real card or transfer. Each subclass passes the substitution test (it can genuinely stand in for `Payment`: both really take money), and the same holds for the notification channels. A new method = a new subclass; checkout does not change (Open/Closed Principle: code should be open to extension but closed to modification). The course calls this out: it is why `Payment` is not an enum.	`enum PaymentType` + `switch` in checkout.
Payment passed in as `Function<BigDecimal, Payment>` (a function object: give it an amount, get a `Payment` back)	Checkout knows the total only after building the order, so it asks the caller's factory (a piece of code whose job is to create objects) for a payment of that amount.	Creating the payment before the total is known.
Observer for notifications	`OrderService` publishes events; `NotificationService` decides who to tell and how. Adding push notifications = a new `NotificationChannel`.	Calling email and SMS code directly inside `OrderService`.
`Shipment` keeps a log	Requirement 11 = read the last entry; the whole history stays for support and disputes.	A single `status` field (history lost).
Reviews store `reviewerId`	Avoids a two-way dependency `Product ↔ Member`; each class can be understood alone.	`ProductReview` holding a `Member` object.
`OrderService` as a Facade	One simple entry point hides stock, payment, shipment and notification steps.	Callers orchestrating five objects themselves.

Java implementation

Files: `Money.java` (BigDecimal helpers), `Catalog.java` (category, product, review, search, catalog), `Order.java` (address, item, cart, order status machine, order), `Payments.java` (payments, shipment), `Accounts.java` (account, customer, guest, member, admin), `OrderService.java` (listeners, notifications, service), `Main.java` (demo).

```java
// File: lld/ch19/amazon/Accounts.java
package lld.ch19.amazon;

import java.util.ArrayList;
import java.util.List;

enum AccountStatus { ACTIVE, BLOCKED, BANNED, COMPROMISED, ARCHIVED, UNKNOWN }

/** Login data shared by members and admins. */
final class Account {
    private final String userName;
    private final String email;
    private final String phone;
    private final List<Address> addresses = new ArrayList<>();
    private volatile AccountStatus status = AccountStatus.ACTIVE;

    Account(String userName, String email, String phone) {
        this.userName = userName;
        this.email = email;
        this.phone = phone;
    }

    String userName() { return userName; }
    String email() { return email; }
    String phone() { return phone; }
    AccountStatus status() { return status; }
    void setStatus(AccountStatus s) { status = s; }
    synchronized void addAddress(Address a) { addresses.add(a); }
    synchronized List<Address> addresses() { return List.copyOf(addresses); }
}

/**
 * Everything any shopper may do - browse and fill a cart. Placing an order is
 * deliberately NOT here, because a Guest is a Customer and must not inherit it (LSP).
 */
abstract sealed class Customer permits Guest, Member {
    private final ShoppingCart cart = new ShoppingCart();

    ShoppingCart cart() { return cart; }
    void addToCart(Product p, int qty) { cart.addItem(p, qty); }
    void removeFromCart(String productId) { cart.removeItem(productId); }
}

final class Guest extends Customer {
    /** Registering turns a guest into a member and carries the cart across. */
    Member register(String memberId, Account account) {
        Member m = new Member(memberId, account);
        m.cart().mergeFrom(cart());
        cart().clear();
        return m;
    }
}

final class Member extends Customer {
    private final String id;
    private final Account account;

    Member(String id, Account account) {
        this.id = id;
        this.account = account;
    }
```

```
    String id() { return id; }
    Account account() { return account; }
    boolean canBuy() { return account.status() == AccountStatus.ACTIVE; }
}

final class Admin {
    private final Account account;

    Admin(Account account) { this.account = account; }

    ProductCategory addCategory(String name, String description) { return new ProductCategory(name,
description); }

    void block(Member m) { m.account().setStatus(AccountStatus.BLOCKED); }

    void unblock(Member m) { m.account().setStatus(AccountStatus.ACTIVE); }

    String name() { return account.userName(); }
}
```

```java
// File: lld/ch19/amazon/Catalog.java
package lld.ch19.amazon;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.Collections;
import java.util.HashSet;
import java.util.List;
import java.util.Locale;
import java.util.Map;
import java.util.Optional;
import java.util.Set;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.CopyOnWriteArrayList;

/** A category such as "Books". Products point at it; it does not own them. */
record ProductCategory(String name, String description) { }

/** A review names its reviewer by id, so Product never depends on Member. */
record ProductReview(String reviewerId, int rating, String text) {
    ProductReview {
        if (rating < 1 || rating > 5) throw new IllegalArgumentException("rating must be 1..5");
    }
}

/** The thing being sold. Owns its reviews and guards its own stock count. */
final class Product {
    private final String id;
    private final String name;
    private final ProductCategory category;
    private final String sellerId;
    private volatile BigDecimal price;
    private int availableCount;                       // guarded by this
    private final List<ProductReview> reviews = new CopyOnWriteArrayList<>();

    Product(String id, String name, ProductCategory category, BigDecimal price,
            String sellerId, int availableCount) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = Money.of(price);
        this.sellerId = sellerId;
        this.availableCount = availableCount;
    }

    String id() { return id; }
    String name() { return name; }
    ProductCategory category() { return category; }
    String sellerId() { return sellerId; }
    BigDecimal price() { return price; }
    synchronized int availableCount() { return availableCount; }

    void updatePrice(BigDecimal newPrice) {
        if (newPrice.signum() < 0) throw new IllegalArgumentException("negative price");
        price = Money.of(newPrice);
    }

    /** Take stock out of the pool. False means "not enough" - nothing changes. */
    synchronized boolean reserve(int quantity) {
        if (quantity <= 0 || quantity > availableCount) return false;
        availableCount -= quantity;
```

```
        return true;
    }

    synchronized void release(int quantity) { availableCount += quantity; }

    void addReview(ProductReview review) { reviews.add(review); }

    double averageRating() {
        return reviews.stream().mapToInt(ProductReview::rating).average().orElse(0.0);
    }

    @Override public String toString() { return name + " (" + Money.fmt(price) + ")"; }
}

/** Requirement: search by name or by category. */
interface Search {
    List<Product> searchByName(String text);
    List<Product> searchByCategory(String categoryName);
}

/**
 * The catalog points to products but never owns them. It keeps one index per search
 * (words of the name, category name), so a lookup is a map read, not a scan of every
 * product. Every index is updated in one method (addProduct) so they can never drift apart.
 */
final class Catalog implements Search {
    private final Map<String, Product> byId = new ConcurrentHashMap<>();
    private final Map<String, Set<Product>> byNameWord = new ConcurrentHashMap<>();
    private final Map<String, List<Product>> byCategory = new ConcurrentHashMap<>();

    void addProduct(Product p) {
        byId.put(p.id(), p);
        for (String word : words(p.name()))
            byNameWord.computeIfAbsent(word, k -> ConcurrentHashMap.newKeySet()).add(p);
        byCategory.computeIfAbsent(key(p.category().name()), k -> new CopyOnWriteArrayList<>
()).add(p);
    }

    Optional<Product> find(String productId) { return Optional.ofNullable(byId.get(productId)); }

    /** Products whose name contains every word of the query (e.g. "clean code"). */
    @Override public List<Product> searchByName(String text) {
        List<String> query = words(text);
        if (query.isEmpty()) return List.of();
        Set<Product> hits = new HashSet<>(byNameWord.getOrDefault(query.get(0), Set.of()));
        for (String w : query.subList(1, query.size())) hits.retainAll(byNameWord.getOrDefault(w,
Set.of()));
        List<Product> out = new ArrayList<>(hits);
        out.sort((a, b) -> a.name().compareTo(b.name()));
        return out;
    }

    @Override public List<Product> searchByCategory(String categoryName) {
        return Collections.unmodifiableList(byCategory.getOrDefault(key(categoryName), List.of()));
    }

    private static List<String> words(String s) {
        return Arrays.stream(key(s).split("[^\\p{Alnum}]+")).filter(w -> !w.isEmpty()).toList();
    }

    private static String key(String s) { return s.toLowerCase(Locale.ROOT).trim(); }
}
```

```java
// File: lld/ch19/amazon/Main.java
package lld.ch19.amazon;

import java.math.BigDecimal;
import java.util.List;
import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.CountDownLatch;
import java.util.concurrent.atomic.AtomicInteger;

public class Main {
    public static void main(String[] args) throws Exception {
        Admin admin = new Admin(new Account("root", "admin@shop.test", "000"));
        ProductCategory books = admin.addCategory("Books", "Printed and e-books");
        ProductCategory electronics = admin.addCategory("Electronics", "Gadgets");

        Catalog catalog = new Catalog();
        catalog.addProduct(new Product("P1", "Clean Code", books, Money.of("32.50"), "seller-9", 5));
        catalog.addProduct(new Product("P2", "Refactoring", books, Money.of("41.00"), "seller-9", 2));
        catalog.addProduct(new Product("P3", "USB-C Cable", electronics, Money.of("9.99"), "seller-4",
100));
        catalog.addProduct(new Product("P4", "Design Patterns (last copy)", books, Money.of("55.00"),
"seller-2", 1));

        System.out.println("== 1. Search ==");
        System.out.println("name 'code'      -> " + catalog.searchByName("code"));
        System.out.println("name 'usb cable' -> " + catalog.searchByName("usb cable"));
        System.out.println("category 'books' -> " + catalog.searchByCategory("books"));

        Map<String, Account> accounts = new ConcurrentHashMap<>();
        OrderService orders = new OrderService(catalog);
        orders.subscribe(new NotificationService(accounts::get, List.of(new EmailChannel(), new
SmsChannel())));

        System.out.println("\n== 2. Guest fills a cart, then registers (cart carries over) ==");
        Guest guest = new Guest();
        guest.addToCart(catalog.find("P1").orElseThrow(), 1);
        guest.addToCart(catalog.find("P3").orElseThrow(), 3);
        Account aliceAcc = new Account("alice", "alice@mail.test", "+1-555-0101");
        Member alice = guest.register("M1", aliceAcc);
        accounts.put(alice.id(), aliceAcc);
        alice.cart().updateQuantity("P3", 2);
        alice.cart().snapshot().forEach(i -> System.out.println("  cart: " + i.quantity() + " x " +
i.name()
                + " = " + Money.fmt(i.lineTotal())));

        System.out.println("\n== 3. Checkout with a credit card ==");
        Address home = new Address("1 Main St", "Seattle", "98101", "US");
        Order o1 = orders.placeOrder(alice, home,
                amt -> new CreditCardPayment(amt, "4242", Money.of("1000.00")));
        System.out.println("  " + o1.orderNumber() + " total " + Money.fmt(o1.total()) + ", status " +
o1.status()
                + ", stock of Clean Code now " + catalog.find("P1").orElseThrow().availableCount());

        System.out.println("\n== 4. Ship, track, deliver ==");
        orders.ship(o1.orderNumber(), "UPS Ground");
        orders.updateShipment(o1.orderNumber(), ShipmentStatus.SHIPPED, "Portland hub");
        orders.updateShipment(o1.orderNumber(), ShipmentStatus.DELIVERED, "front door");
        System.out.println("  tracking now: " + o1.shipment().current() + ", order " + o1.status());
        System.out.println("  cancel after delivery? " + orders.cancel(o1.orderNumber()));

        System.out.println("\n== 5. Bank transfer (pending) then cancel before shipping -> refund
```

```
==");
        alice.addToCart(catalog.find("P2").orElseThrow(), 2);
        Order o2 = orders.placeOrder(alice, home, amt -> new BankTransferPayment(amt, "Chase"));
        System.out.println("  " + o2.orderNumber() + " status " + o2.status() + ", payment " +
o2.payment().status());
        orders.onPaymentSettled(o2.orderNumber());
        System.out.println("  cancel before shipping? " + orders.cancel(o2.orderNumber())
                + " -> " + o2.status() + ", payment " + o2.payment().status()
                + ", Refactoring stock back to " + catalog.find("P2").orElseThrow().availableCount());
        System.out.println("  order log: " + o2.log().stream().map(OrderLog::status).toList());

        System.out.println("\n== 6. Illegal transition is rejected by the state machine ==");
        try {
            orders.ship(o2.orderNumber(), "FedEx");
        } catch (IllegalStateException e) {
            System.out.println("  " + e.getMessage());
        }

        System.out.println("\n== 7. Review ==");
        Product cleanCode = catalog.find("P1").orElseThrow();
        cleanCode.addReview(new ProductReview(alice.id(), 5, "Changed how I name things"));
        cleanCode.addReview(new ProductReview("M7", 4, "Good, a bit long"));
        System.out.println("  Clean Code average rating = " + cleanCode.averageRating());

        System.out.println("\n== 8. Blocked member cannot buy ==");
        admin.block(alice);
        alice.addToCart(catalog.find("P3").orElseThrow(), 1);
        try {
            orders.placeOrder(alice, home, amt -> new CreditCardPayment(amt, "4242",
Money.of("1000")));
        } catch (IllegalStateException e) {
            System.out.println("  rejected: " + e.getMessage());
        }
        admin.unblock(alice);
        alice.cart().clear();

        System.out.println("\n== 9. Ten members race for the last copy ==");
        OrderService quiet = new OrderService(catalog);          // no listeners: keep output short
        Product last = catalog.find("P4").orElseThrow();
        CountDownLatch start = new CountDownLatch(1);
        AtomicInteger winners = new AtomicInteger();
        AtomicInteger losers = new AtomicInteger();
        Thread[] threads = new Thread[10];
        for (int t = 0; t < threads.length; t++) {
            Member m = new Member("R" + t, new Account("racer" + t, "r" + t + "@mail.test", "-"));
            m.addToCart(last, 1);
            threads[t] = new Thread(() -> {
                try {
                    start.await();
                    quiet.placeOrder(m, home, amt -> new CreditCardPayment(amt, "1111", new
BigDecimal("500")));
                    winners.incrementAndGet();
                } catch (IllegalStateException e) {
                    losers.incrementAndGet();
                } catch (InterruptedException e) {
                    Thread.currentThread().interrupt();
                }
            });
            threads[t].start();
        }
        start.countDown();
        for (Thread t : threads) t.join();
```

```
        System.out.println("  winners=" + winners + " losers=" + losers + " stock left=" +
last.availableCount());
    }
}
```

```
// File: lld/ch19/amazon/Money.java
package lld.ch19.amazon;

import java.math.BigDecimal;
import java.math.RoundingMode;

/** Money helpers: always BigDecimal, always 2 decimal places, never double. */
final class Money {
    private Money() { }

    static final BigDecimal ZERO = of("0");

    static BigDecimal of(String amount) { return new BigDecimal(amount).setScale(2,
RoundingMode.HALF_EVEN); }

    static BigDecimal of(BigDecimal amount) { return amount.setScale(2, RoundingMode.HALF_EVEN); }

    static String fmt(BigDecimal amount) { return "$" + amount.setScale(2,
RoundingMode.HALF_EVEN).toPlainString(); }
}
```

```java
// File: lld/ch19/amazon/Order.java
package lld.ch19.amazon;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.Collections;
import java.util.EnumSet;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;
import java.util.Set;

record Address(String street, String city, String zip, String country) {
    @Override public String toString() { return street + ", " + city + " " + zip + ", " + country; }
}

/** One line of a cart or an order: an immutable snapshot of product, quantity and unit price. */
record Item(String productId, String name, int quantity, BigDecimal unitPrice) {
    Item {
        if (quantity <= 0) throw new IllegalArgumentException("quantity must be positive");
    }
    BigDecimal lineTotal() { return unitPrice.multiply(BigDecimal.valueOf(quantity)); }
}

/** A mutable working area. Owns its items (they die when the cart is cleared). */
final class ShoppingCart {
    private final Map<String, Item> items = new LinkedHashMap<>();

    synchronized void addItem(Product p, int qty) {
        items.merge(p.id(), new Item(p.id(), p.name(), qty, p.price()),
                (old, add) -> new Item(old.productId(), old.name(), old.quantity() + add.quantity(),
p.price()));
    }

    synchronized void updateQuantity(String productId, int qty) {
        if (qty == 0) { items.remove(productId); return; }
        Item old = items.get(productId);
        if (old == null) throw new IllegalArgumentException("not in cart: " + productId);
        items.put(productId, new Item(old.productId(), old.name(), qty, old.unitPrice()));
    }

    synchronized void removeItem(String productId) { items.remove(productId); }

    /** A copy, so an order never shares its lines with the cart. */
    synchronized List<Item> snapshot() { return List.copyOf(items.values()); }

    synchronized boolean isEmpty() { return items.isEmpty(); }

    synchronized void clear() { items.clear(); }

    synchronized void mergeFrom(ShoppingCart other) {
        for (Item i : other.snapshot()) items.merge(i.productId(), i,
                (a, b) -> new Item(a.productId(), a.name(), a.quantity() + b.quantity(),
a.unitPrice()));
    }
}

/** The order state machine. Each state lists the states it may move to. */
enum OrderStatus {
    PENDING, UNSHIPPED, SHIPPED, COMPLETED, CANCELED, REFUND_APPLIED;

    private Set<OrderStatus> next() {
```

```
        return switch (this) {
            case PENDING -> EnumSet.of(UNSHIPPED, CANCELED);      // waiting for payment
            case UNSHIPPED -> EnumSet.of(SHIPPED, CANCELED);      // paid, not yet handed to carrier
            case SHIPPED -> EnumSet.of(COMPLETED);                // too late to cancel
            case CANCELED -> EnumSet.of(REFUND_APPLIED);          // money goes back if it was taken
            case COMPLETED, REFUND_APPLIED -> EnumSet.noneOf(OrderStatus.class);
        };
    }

    boolean canMoveTo(OrderStatus target) { return next().contains(target); }
}

record OrderLog(OrderStatus status, String note) { }

/** A fixed copy of the purchased lines, plus a status that only moves along legal edges. */
final class Order {
    private final String orderNumber;
    private final String memberId;
    private final List<Item> items;              // copied from the cart, never shared
    private final Address shippingAddress;
    private final List<OrderLog> log = new ArrayList<>();
    private OrderStatus status = OrderStatus.PENDING;
    private Payment payment;
    private Shipment shipment;

    Order(String orderNumber, String memberId, List<Item> items, Address shippingAddress) {
        this.orderNumber = orderNumber;
        this.memberId = memberId;
        this.items = List.copyOf(items);
        this.shippingAddress = shippingAddress;
        log.add(new OrderLog(status, "order placed"));
    }

    String orderNumber() { return orderNumber; }
    String memberId() { return memberId; }
    List<Item> items() { return items; }
    Address shippingAddress() { return shippingAddress; }
    synchronized OrderStatus status() { return status; }
    synchronized Payment payment() { return payment; }
    synchronized Shipment shipment() { return shipment; }
    synchronized List<OrderLog> log() { return Collections.unmodifiableList(new ArrayList<>(log)); }

    BigDecimal total() {
        return items.stream().map(Item::lineTotal).reduce(Money.ZERO, BigDecimal::add);
    }

    synchronized void attachPayment(Payment p) { this.payment = p; }
    synchronized void attachShipment(Shipment s) { this.shipment = s; }

    /** Requirement 8 lives in exactly one place. */
    synchronized boolean canCancel() { return status.canMoveTo(OrderStatus.CANCELED); }

    synchronized void moveTo(OrderStatus target, String note) {
        if (!status.canMoveTo(target))
            throw new IllegalStateException(orderNumber + ": " + status + " -> " + target + " is not
allowed");
        status = target;
        log.add(new OrderLog(target, note));
    }
}
```

```java
// File: lld/ch19/amazon/OrderService.java
package lld.ch19.amazon;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.List;
import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.CopyOnWriteArrayList;
import java.util.concurrent.atomic.AtomicInteger;
import java.util.function.Function;

/** Observer: anything that wants to hear about order and shipment changes. */
interface OrderEventListener {
    void onOrderStatus(Order order, OrderStatus status);
    void onShipmentUpdate(Order order, ShipmentLog entry);
}

/** One way of delivering a message (email, SMS, push...). */
interface NotificationChannel {
    void send(Account to, String message);
}

final class EmailChannel implements NotificationChannel {
    @Override public void send(Account to, String message) {
        System.out.println("    [email -> " + to.email() + "] " + message);
    }
}

final class SmsChannel implements NotificationChannel {
    @Override public void send(Account to, String message) {
        System.out.println("    [sms   -> " + to.phone() + "] " + message);
    }
}

/** Requirement 9: tell the member whenever order or shipping status changes. */
final class NotificationService implements OrderEventListener {
    private final Function<String, Account> accountLookup;
    private final List<NotificationChannel> channels;

    NotificationService(Function<String, Account> accountLookup, List<NotificationChannel> channels) {
        this.accountLookup = accountLookup;
        this.channels = List.copyOf(channels);
    }

    @Override public void onOrderStatus(Order order, OrderStatus status) {
        Account to = accountLookup.apply(order.memberId());
        String msg = "Order " + order.orderNumber() + " is now " + status;
        channels.forEach(c -> c.send(to, msg));
    }

    @Override public void onShipmentUpdate(Order order, ShipmentLog entry) {
        Account to = accountLookup.apply(order.memberId());
        channels.get(0).send(to, "Shipment for " + order.orderNumber() + ": " + entry.status() + " @ "
+ entry.location());
    }
}

/** Application service (facade) that turns a cart into an order and drives the order state machine.
*/
final class OrderService {
    private final Catalog catalog;
```

```
    private final Map<String, Order> orders = new ConcurrentHashMap<>();
    private final List<OrderEventListener> listeners = new CopyOnWriteArrayList<>();
    private final AtomicInteger orderSeq = new AtomicInteger(1000);
    private final AtomicInteger shipSeq = new AtomicInteger(1);

    OrderService(Catalog catalog) { this.catalog = catalog; }

    void subscribe(OrderEventListener l) { listeners.add(l); }

    /**
     * Checkout. Stock is reserved when the order is placed (not when paid), all-or-nothing,
     * so two buyers can never both take the last copy.
     */
    Order placeOrder(Member member, Address address, Function<BigDecimal, Payment> paymentMethod) {
        if (!member.canBuy()) throw new IllegalStateException("account " + member.account().status());
        List<Item> items = member.cart().snapshot();
        if (items.isEmpty()) throw new IllegalStateException("cart is empty");

        List<Item> reserved = new ArrayList<>();
        for (Item i : items) {
            Product p = catalog.find(i.productId()).orElseThrow();
            if (!p.reserve(i.quantity())) {
                reserved.forEach(r ->
catalog.find(r.productId()).orElseThrow().release(r.quantity()));
                throw new IllegalStateException("out of stock: " + i.name());
            }
            reserved.add(i);
        }

        Order order = new Order("ORD-" + orderSeq.incrementAndGet(), member.id(), items, address);
        orders.put(order.orderNumber(), order);

        Payment payment = paymentMethod.apply(order.total());
        order.attachPayment(payment);
        switch (payment.process()) {
            case COMPLETED -> { transition(order, OrderStatus.UNSHIPPED, "paid by " +
payment.describe()); member.cart().clear(); }
            case PENDING -> { publish(order, OrderStatus.PENDING); member.cart().clear(); }
            default -> {                                  // declined: give the stock back, keep the
cart
                releaseStock(order);
                transition(order, OrderStatus.CANCELED, "payment declined");
            }
        }
        return order;
    }

    /** Called when a bank transfer finally clears. */
    void onPaymentSettled(String orderNumber) {
        Order o = get(orderNumber);
        o.payment().markSettled();
        transition(o, OrderStatus.UNSHIPPED, "transfer cleared");
    }

    /** Requirement 8: cancel only if not shipped yet. */
    boolean cancel(String orderNumber) {
        Order o = get(orderNumber);
        synchronized (o) {                       // check-then-act must be atomic against ship()
            if (!o.canCancel()) return false;
            releaseStock(o);
            transition(o, OrderStatus.CANCELED, "cancelled by member");
        }
```

```
        if (o.payment().refund() == PaymentStatus.REFUNDED)
            transition(o, OrderStatus.REFUND_APPLIED, "refunded " + Money.fmt(o.total()));
        return true;
    }

    void ship(String orderNumber, String method) {
        Order o = get(orderNumber);
        Shipment s = new Shipment("SHP-" + shipSeq.getAndIncrement(), method);
        synchronized (o) {
            transition(o, OrderStatus.SHIPPED, "handed to " + method);
            o.attachShipment(s);
        }
        updateShipment(orderNumber, ShipmentStatus.SHIPPED, "sort centre");
    }

    void updateShipment(String orderNumber, ShipmentStatus status, String location) {
        Order o = get(orderNumber);
        o.shipment().addLog(status, location);
        ShipmentLog entry = o.shipment().current();
        listeners.forEach(l -> l.onShipmentUpdate(o, entry));
        if (status == ShipmentStatus.DELIVERED) transition(o, OrderStatus.COMPLETED, "delivered");
    }

    Order get(String orderNumber) {
        Order o = orders.get(orderNumber);
        if (o == null) throw new IllegalArgumentException("no order " + orderNumber);
        return o;
    }

    private void releaseStock(Order o) {
        o.items().forEach(i -> catalog.find(i.productId()).orElseThrow().release(i.quantity()));
    }

    private void transition(Order o, OrderStatus target, String note) {
        o.moveTo(target, note);
        publish(o, target);
    }

    private void publish(Order o, OrderStatus s) { listeners.forEach(l -> l.onOrderStatus(o, s)); }
}
```

```java
// File: lld/ch19/amazon/Payments.java
package lld.ch19.amazon;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.List;

enum PaymentStatus { UNPAID, PENDING, COMPLETED, DECLINED, CANCELLED, REFUNDED }

/**
 * Strategy for taking money. Each subclass behaves differently: a card clears at
 * once, a bank transfer comes back PENDING and only clears later.
 */
abstract sealed class Payment permits CreditCardPayment, BankTransferPayment {
    private final BigDecimal amount;
    protected PaymentStatus status = PaymentStatus.UNPAID;

    Payment(BigDecimal amount) { this.amount = Money.of(amount); }

    BigDecimal amount() { return amount; }
    synchronized PaymentStatus status() { return status; }

    /** Try to take the money. Returns the new status. */
    abstract PaymentStatus process();

    abstract String describe();

    synchronized void markSettled() {
        if (status == PaymentStatus.PENDING) status = PaymentStatus.COMPLETED;
    }

    synchronized PaymentStatus refund() {
        if (status == PaymentStatus.COMPLETED) status = PaymentStatus.REFUNDED;
        else if (status == PaymentStatus.PENDING || status == PaymentStatus.UNPAID) status =
PaymentStatus.CANCELLED;
        return status;
    }
}

final class CreditCardPayment extends Payment {
    private final String last4;             // never store the full card number
    private final BigDecimal creditLimit;

    CreditCardPayment(BigDecimal amount, String last4, BigDecimal creditLimit) {
        super(amount);
        this.last4 = last4;
        this.creditLimit = creditLimit;
    }

    @Override synchronized PaymentStatus process() {
        status = amount().compareTo(creditLimit) <= 0 ? PaymentStatus.COMPLETED :
PaymentStatus.DECLINED;
        return status;
    }

    @Override String describe() { return "card ****" + last4; }
}

final class BankTransferPayment extends Payment {
    private final String bankName;

    BankTransferPayment(BigDecimal amount, String bankName) {
```

```
        super(amount);
        this.bankName = bankName;
    }

    @Override synchronized PaymentStatus process() {
        status = PaymentStatus.PENDING;      // clears later, when the bank confirms
        return status;
    }

    @Override String describe() { return "bank transfer via " + bankName; }
}

enum ShipmentStatus { PENDING, SHIPPED, DELIVERED, ON_HOLD }

record ShipmentLog(ShipmentStatus status, String location) { }

/** Keeps a log, not one status field: the last entry is "now", the rest is history. */
final class Shipment {
    private final String shipmentNumber;
    private final String method;
    private final List<ShipmentLog> log = new ArrayList<>();

    Shipment(String shipmentNumber, String method) {
        this.shipmentNumber = shipmentNumber;
        this.method = method;
        log.add(new ShipmentLog(ShipmentStatus.PENDING, "warehouse"));
    }

    String shipmentNumber() { return shipmentNumber; }
    String method() { return method; }
    synchronized void addLog(ShipmentStatus s, String where) { log.add(new ShipmentLog(s, where)); }
    synchronized ShipmentLog current() { return log.get(log.size() - 1); }
    synchronized List<ShipmentLog> history() { return List.copyOf(log); }
}
```

Output:

```
== 1. Search ==
name 'code'      -> [Clean Code ($32.50)]
name 'usb cable' -> [USB-C Cable ($9.99)]
category 'books' -> [Clean Code ($32.50), Refactoring ($41.00), Design Patterns (last copy) ($55.00)]

== 2. Guest fills a cart, then registers (cart carries over) ==
  cart: 1 x Clean Code = $32.50
  cart: 2 x USB-C Cable = $19.98

== 3. Checkout with a credit card ==
    [email -> alice@mail.test] Order ORD-1001 is now UNSHIPPED
    [sms   -> +1-555-0101] Order ORD-1001 is now UNSHIPPED
  ORD-1001 total $52.48, status UNSHIPPED, stock of Clean Code now 4

== 4. Ship, track, deliver ==
    [email -> alice@mail.test] Order ORD-1001 is now SHIPPED
    [sms   -> +1-555-0101] Order ORD-1001 is now SHIPPED
    [email -> alice@mail.test] Shipment for ORD-1001: SHIPPED @ sort centre
    [email -> alice@mail.test] Shipment for ORD-1001: SHIPPED @ Portland hub
    [email -> alice@mail.test] Shipment for ORD-1001: DELIVERED @ front door
    [email -> alice@mail.test] Order ORD-1001 is now COMPLETED
    [sms   -> +1-555-0101] Order ORD-1001 is now COMPLETED
  tracking now: ShipmentLog[status=DELIVERED, location=front door], order COMPLETED
  cancel after delivery? false

== 5. Bank transfer (pending) then cancel before shipping -> refund ==
    [email -> alice@mail.test] Order ORD-1002 is now PENDING
    [sms   -> +1-555-0101] Order ORD-1002 is now PENDING
  ORD-1002 status PENDING, payment PENDING
    [email -> alice@mail.test] Order ORD-1002 is now UNSHIPPED
    [sms   -> +1-555-0101] Order ORD-1002 is now UNSHIPPED
    [email -> alice@mail.test] Order ORD-1002 is now CANCELED
    [sms   -> +1-555-0101] Order ORD-1002 is now CANCELED
    [email -> alice@mail.test] Order ORD-1002 is now REFUND_APPLIED
    [sms   -> +1-555-0101] Order ORD-1002 is now REFUND_APPLIED
  cancel before shipping? true -> REFUND_APPLIED, payment REFUNDED, Refactoring stock back to 2
  order log: [PENDING, UNSHIPPED, CANCELED, REFUND_APPLIED]

== 6. Illegal transition is rejected by the state machine ==
  ORD-1002: REFUND_APPLIED -> SHIPPED is not allowed

== 7. Review ==
  Clean Code average rating = 4.5

== 8. Blocked member cannot buy ==
  rejected: account BLOCKED

== 9. Ten members race for the last copy ==
  winners=1 losers=9 stock left=0
```

Reading the output. Step 1 answers both searches from indexes (a name search matches every query word, so `usb cable` finds "USB-C Cable"). Step 2 shows the guest's cart surviving registration. Step 3 prints two notifications (email + SMS) because both channels are registered. Step 5 walks the full cancel path `PENDING → UNSHIPPED → CANCELED → REFUND_APPLIED`, and the stock returns to 2. Step 6 proves the state machine refuses `REFUND_APPLIED → SHIPPED`. Step 9 starts ten threads at the same instant (a `CountDownLatch` — a gate that keeps threads waiting until a counter reaches zero — releases them together) and exactly one wins the last copy.

Concurrency & edge cases

Risk	What could go wrong	How the code prevents it
Last unit, two buyers	Both see `availableCount == 1`, both decrement → stock −1	`Product.reserve` is `synchronized`: check and decrement are one step. Step 9 proves it.
Multi-line cart, one line out of stock	Lines 1–2 reserved, line 3 fails → stock leaked	`placeOrder` remembers what it reserved and releases it all on failure (all-or-nothing).
Cancel and ship at the same time	Order shipped *and* stock released	`cancel` and `ship` both lock the `Order` object around check-then-move.
Payment declined	Stock stays reserved forever	Declined path releases stock and moves the order to `CANCELED`; the cart is kept so the member can retry.
Bank transfer never clears	Stock held forever	Mention a timeout job that cancels `PENDING` orders after N hours.
Price changes after adding to cart	Customer charged a price they never saw	`Item` stores the unit price at add time; production systems re-validate prices at checkout and show changes.
Blocked member	Places orders anyway	`Member.canBuy()` checks `AccountStatus.ACTIVE` first (step 8).
Deadlock (two threads each waiting forever for a lock the other holds)	Two locks taken in different orders	Each lock (product, order, cart) is held briefly and never while waiting for another object's lock in the opposite order.

Real-world example

On Amazon's "Your Orders" page, the *Cancel items* button is offered only while an order has not yet entered the shipping process — the same edge our state machine removes after `SHIPPED`. Amazon also treated the shopping cart as something that must *always* accept writes: its 2007 Dynamo paper (the key-value store — a database that simply maps a key to a value — behind services such as the cart) chose **availability** (staying usable even when some machines fail; see System Design Fundamentals) so that "add to cart" works even during failures, and merges conflicting cart versions later. Our in-memory design shows the object model; at Amazon's scale the cart, catalog and orders live in separate services.

Extensions & follow-up questions

- **"Add a new payment type, e.g. gift cards or UPI (India's instant bank-to-bank mobile payment system)."** Add a `GiftCardPayment extends Payment` (and list it in `permits`). `OrderService` is untouched, because it only calls `process()`.
- **"How do you prevent two shoppers from both buying the final unit?"** Reserve at order time inside a lock (`Product.reserve`). In a distributed system (one spread over many machines): a conditional database update (`UPDATE ... SET stock = stock - 1 WHERE id = ? AND stock >= 1`) or **optimistic locking** with a version column (no lock is held; each row carries a version number, and an update succeeds only if the version is still the one you read — otherwise you retry).
- **"Abandoned orders hold stock."** Give each reservation an expiry. A scheduled job cancels `PENDING` orders older than, say, 30 minutes and releases stock.
- **"Returns after delivery."** Add `COMPLETED → RETURN_REQUESTED → RETURNED → REFUND_APPLIED` edges; the enum change is local.
- **"Multiple warehouses."** Replace `Product.availableCount` with an `Inventory` per warehouse; checkout picks a warehouse per line (a Strategy: nearest, cheapest shipping).

- **"Split shipments."** One `Order` owns many `Shipment`s, each with its own item subset and log.
- **"Search at scale."** The in-memory `Catalog` becomes a search index (for example Elasticsearch/OpenSearch, popular search-engine servers) fed by product change events; the `Search` interface stays.
- **"Seller payouts, recommendations, wish lists."** Separate modules listening to order events (Observer again).

In the interview

30–45 minute plan

1. (3 min) Clarify: guest vs member, payment types, cancel rule, notifications, what is out of scope.
2. (5 min) Actors and use cases; name the four actors.
3. (10 min) Class diagram. **Say the LSP sentence out loud**: "Ordering is on `Member`, not `Customer`, otherwise every guest inherits an ordering method it is not allowed to use."
4. (5 min) The order state machine; show that "cancel after ship" has no edge.
5. (10 min) Code `placeOrder` (reserve all-or-nothing, copy items, pay, notify) and `cancel`.
6. (5 min) Concurrency: last-item race; abandoned-order timeout; extensions.

> **30-second answer:** "Catalog with name and category indexes; a cart per customer; ordering only on `Member`; checkout reserves stock for every line atomically, copies the lines into an immutable order, then takes payment through a `Payment` strategy. The order moves through a state machine that forbids cancelling after shipping, and every status change is published to a notification observer."

Follow-ups with answers

- *Why is `Payment` abstract rather than an enum?* Because the methods behave differently (clears now vs pending). New methods are new classes — Open/Closed.
- *Why does `Order` copy items?* So later cart edits or price changes cannot rewrite history.
- *Where would you put tax and discounts?* A `PricingRule` chain (Chain of Responsibility — a pattern where a request is passed along a line of handlers, each doing its part — or a list of Strategies) applied when building the order; the order stores the final line prices.
- *How do you make checkout idempotent (safe to repeat: doing it twice has the same effect as doing it once) if the user double-clicks?* Accept a client-generated idempotency key (a unique id the browser sends with the request); if an order with that key exists, return it instead of creating a second one.

Self-check

1. **Q:** Why can't `placeOrder` live on `Customer`? **A:** `Guest extends Customer` and guests must not order; inheriting a method you must refuse violates LSP.
2. **Q:** What does the filled diamond between `Order` and `OrderLog` mean? **A:** Composition — the order owns its log entries; they are created and destroyed with it.
3. **Q:** Name two things wrong with "check stock, take payment, then decrement stock". **A:** A race window lets two buyers pass the check; and a payment can succeed for goods that are no longer there.
4. **Q:** Which status edges allow a cancel? **A:** `PENDING → CANCELED` and `UNSHIPPED → CANCELED`.
5. **Q:** Why `compareTo` and not `equals` on `BigDecimal`? **A:** `equals` also compares scale, so `2.0` and `2.00` are "not equal".
6. **Q:** What pattern lets you add push notifications without touching `OrderService`? **A:** Observer (listener) plus a new `NotificationChannel` implementation.

TL;DR

- Guests browse and fill carts; **only members order** (LSP).
- Checkout = **reserve all lines atomically → copy into an immutable order → pay (Strategy) → publish (Observer)**.
- `OrderStatus` is a **state machine**; "cancel after ship" is impossible because the edge does not exist.
- Money in `BigDecimal`; histories in logs; stock guarded by a lock.

Design an Online Stock Brokerage System

This problem has two halves, and interviews ask both:

- **Part A — the brokerage** (what the course designs): members, money in and out, watchlists, the four order types, lots and positions, statements.
- **Part B — the exchange's matching engine** (a machine-coding problem — write a working program in about 90 minutes — originally from Geektrust, an Indian coding-assessment platform, kept in kumaransg/LLD under a project named "Navi"; we call it "the Navi statement" below): an **order book** (the list of all waiting buy and sell orders for one stock) that matches buy and sell orders by **price-time priority**, implemented and tested.

A **broker** is the company you trade *through* (it holds your cash and shares and forwards your orders). An **exchange** is the market where buy and sell orders from all brokers *meet* and are matched. Keep the two apart in your head; the design keeps them apart in code.

Problem & clarifying questions

An online brokerage lets people buy and sell stocks without a human middleman. Members track and carry out their transactions, see charts of how the stocks they hold are performing, and get an alert when a stock moves by an amount they chose. The platform must keep those transactions secure. Because computers replace the phone call to a broker, each trade is quicker and cheaper, and members get fast access to real-time (updated the moment they change) prices, market trends and stock reports.

Question	Assumed answer
Which order types?	Market, limit, stop-loss, stop-limit (explained in the table under Key flows).
Can one order fill at several prices?	Yes — an order can be filled in parts.
Do we track separate purchases of the same stock?	Yes — each purchase is a **lot**, because tax depends on which lot you sell.
How does money come in and out?	Check, wire (a direct bank-to-bank transfer), or electronic bank transfer, each with a per-transfer limit.
Reports?	Quarterly statements and a yearly tax statement.
Notifications?	When an order executes.
Margin, short selling, options, dividends and stock splits, real-time streaming quotes, the settlement delay, the wash-sale tax rule?	Out of scope — a deliberate decision, say so.

For reference, those out-of-scope terms: **margin** is trading with money borrowed from the broker; **short selling** is selling borrowed shares, hoping to buy them back cheaper; **options** are contracts giving the right to buy or sell later at a fixed price; **dividends** are profits a company pays to its shareholders; a **stock split** turns each share into several cheaper ones; **settlement** is the actual swap of cash and shares, which happens a day or so after the trade; the **wash-sale rule** is a US tax rule that blocks claiming a loss if you rebuy the same stock within 30 days.

Requirements

Functional (brokerage)

1. Any member can buy and sell stocks.
2. A member can have **multiple watchlists** (named lists of stocks to keep an eye on), each with multiple stock quotes (a quote is a stock's latest price).
3. Orders can be **market, limit, stop-loss, stop-limit**.
4. A member can hold **multiple lots** of the same stock; the system keeps them apart.
5. The system generates **quarterly** updates and **yearly tax** statements.
6. Money goes in and out of the account in three ways: **check, wire, or electronic bank transfer**.
7. A notification is sent whenever an order executes.

Functional (matching engine, Navi variant)

8. Orders arrive as `<id> <time> <symbol> <buy/sell> <price> <qty>`, e.g. `#1 09:45 BAC sell 240.12 100`. They are read from a text file whose path is given as the program's argument; trades are printed to the console.
9. On every new order, match against the opposite side. Lower sell prices and higher buy prices go first; at the same price, **earlier orders go first** (price-time priority).
10. A trade happens when buy price ≥ sell price. In the Navi statement the trade is recorded at the **sell order's price**.
11. Print each trade as `<buy_id> <price> <qty> <sell_id>`; unmatched quantity stays in the book.

Non-functional

- Money is never over-committed: ten $1,000 orders cannot all pass a $1,000 balance check.
- Matching is deterministic (no randomness or timing luck): same input, same trades.
- The engine is safe under concurrent submissions and conserves quantity (shares bought = shares sold).

Use cases

Three actors:

- **Admin** — administrative jobs such as blocking or unblocking members.
- **Member** — searches the stock inventory, buys and sells, manages watchlists.
- **System** — sends order notifications and periodically fetches quotes from the exchange.

Top use cases: register / cancel membership; add, remove or edit a watchlist; search stocks by symbol; place an order; cancel an order; deposit or withdraw money.

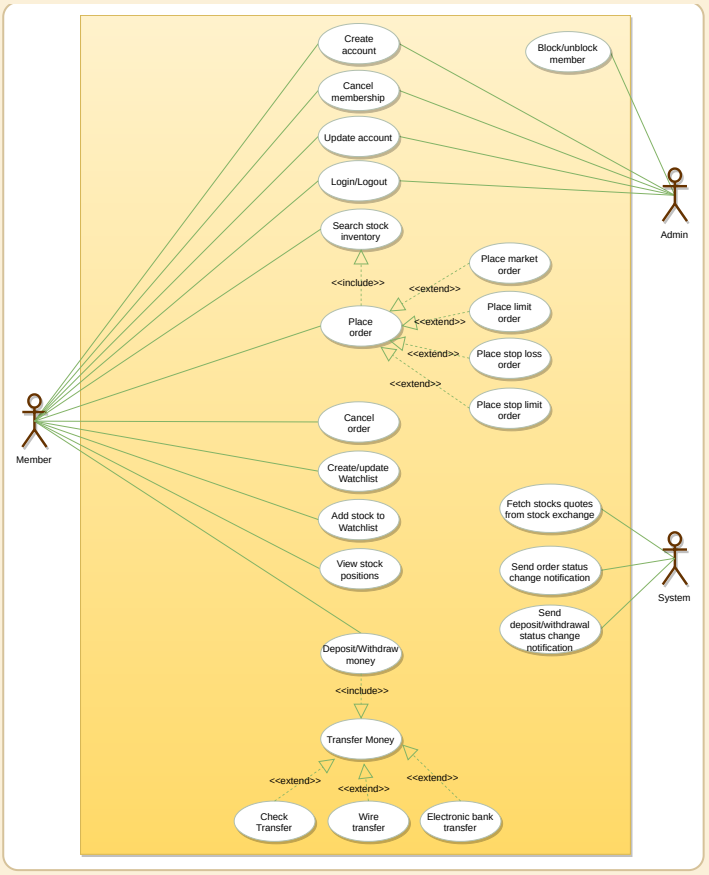


Figure — Use-case diagram for the stock brokerage: Admin, Member and System (course diagram)

Core entities

Class	Responsibility
`MoneyTransfer` → `CheckTransfer`, `WireTransfer`, `ElectronicBankTransfer`	Move money in or out. They differ only in name and clearing days (5 / 1 / 3 in our code); the per-transfer limit (100,000) is a single rule, so it lives only in the base class.
`Notifier` / `EmailNotifier`	Delivers "your order filled" messages. The course draws one base with two delivery ways (email and SMS); an `SmsNotifier` would be a second implementation.
`Order` (abstract)	Shared by all order types: member, symbol, side (buy or sell), quantity, time-in-force (how long the order stays live), status, list of fills. Declares the **two questions**.
`MarketOrder`, `LimitOrder`, `StopLossOrder`, `StopLimitOrder`	Each answers "may I trade now?" and "what is my worst price?". `StopLimitOrder extends StopLossOrder`.
`OrderPart`	One fill: quantity, price, day. A 100-share order might be filled as 70 @ 10.00 plus 30 @ 10.05.
`Stock`, `StockInventory`	Symbol + latest price; the inventory is the **single source of truth** (the one place whose value counts if copies disagree) for prices.
`StockExchange`	Holds open orders; on each new order or price tick tries to fill them; calls back through `OrderListener`.
`OrderListener`	Interface the exchange calls (`onFill`, `onCancelled`). The exchange never learns what a member is.
`StockLot`	One purchase at one price.
`StockPosition`	All lots of one symbol; sells from the oldest lot first (FIFO — first in, first out) unless a lot is named. Returns a `LotSale` per lot touched.
`LotSale`	Shares taken from one lot, buy price, sell price → realised gain (profit actually locked in by selling).
`Watchlist`	A named set of **symbols** (not `Stock` objects, so it never goes stale).
`Trade`	One fill as the member sees it (sells carry their `LotSale`s).
`Statement` → `QuarterlyStatement`, `TaxStatement`	Base class filters trades by day range; subclasses write their own lines.
`Member`	Cash, **reserved funds**, positions, watchlists, trades. Implements `OrderListener`. (The course's `Account` also stores name, address, e-mail and phone; we keep only what the demo uses.)
`Admin`	Blocks / unblocks members.
(engine) `OrderBook`, `MatchingEngine`, `Trade`, `ExecutionPricePolicy`	Price levels with FIFO queues per symbol; routing and locking; who sets the trade price.

Class diagram

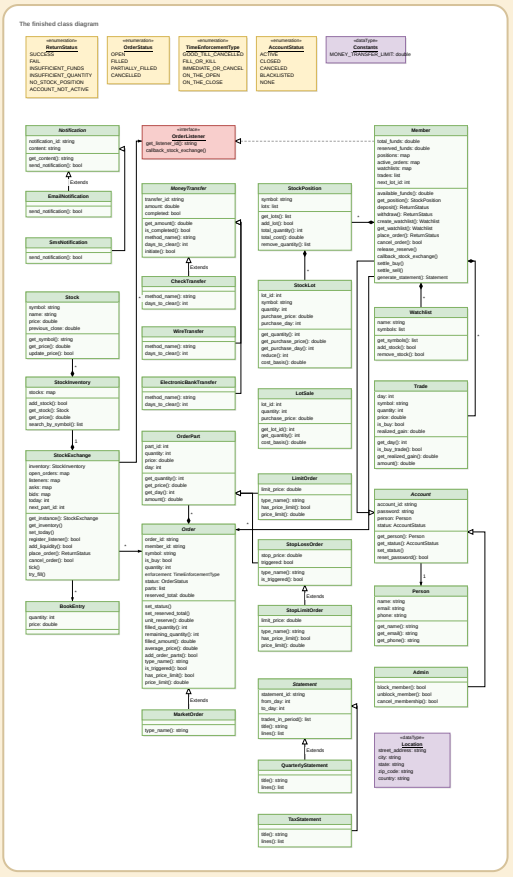


Figure — The finished class diagram of the online stock brokerage (course diagram)

The course builds this model in six stages: money transfers and notifications → the order hierarchy → stock, inventory and exchange → lots and positions → watchlists, trades and statements → members and admins. In the figure above, `Member` extends `Account` and implements `OrderListener` (dashed arrow): the exchange reports fills through that interface and never needs the `Member` type. The filled diamonds from `Member` mean each position, watchlist and trade is owned by exactly one member; the plain arrow to `Order` says the exchange shares those orders.

The course singles out stage 2 — the order hierarchy — as the part that earns the most credit in an interview:

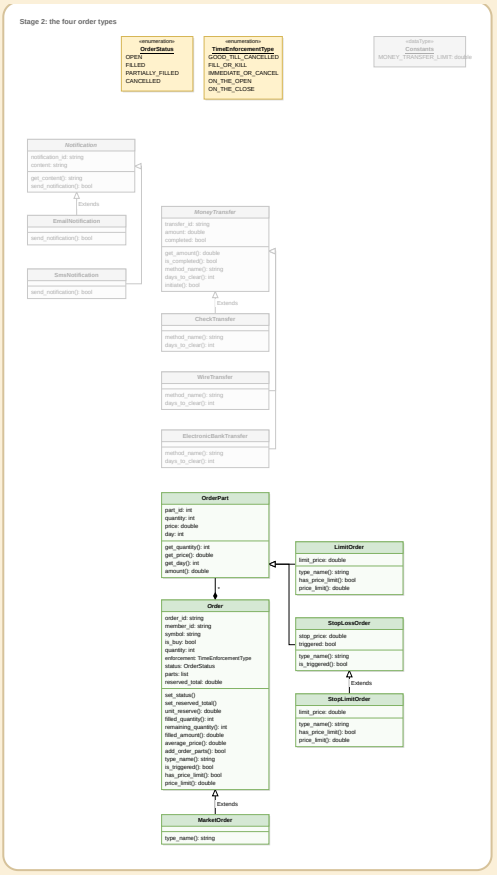
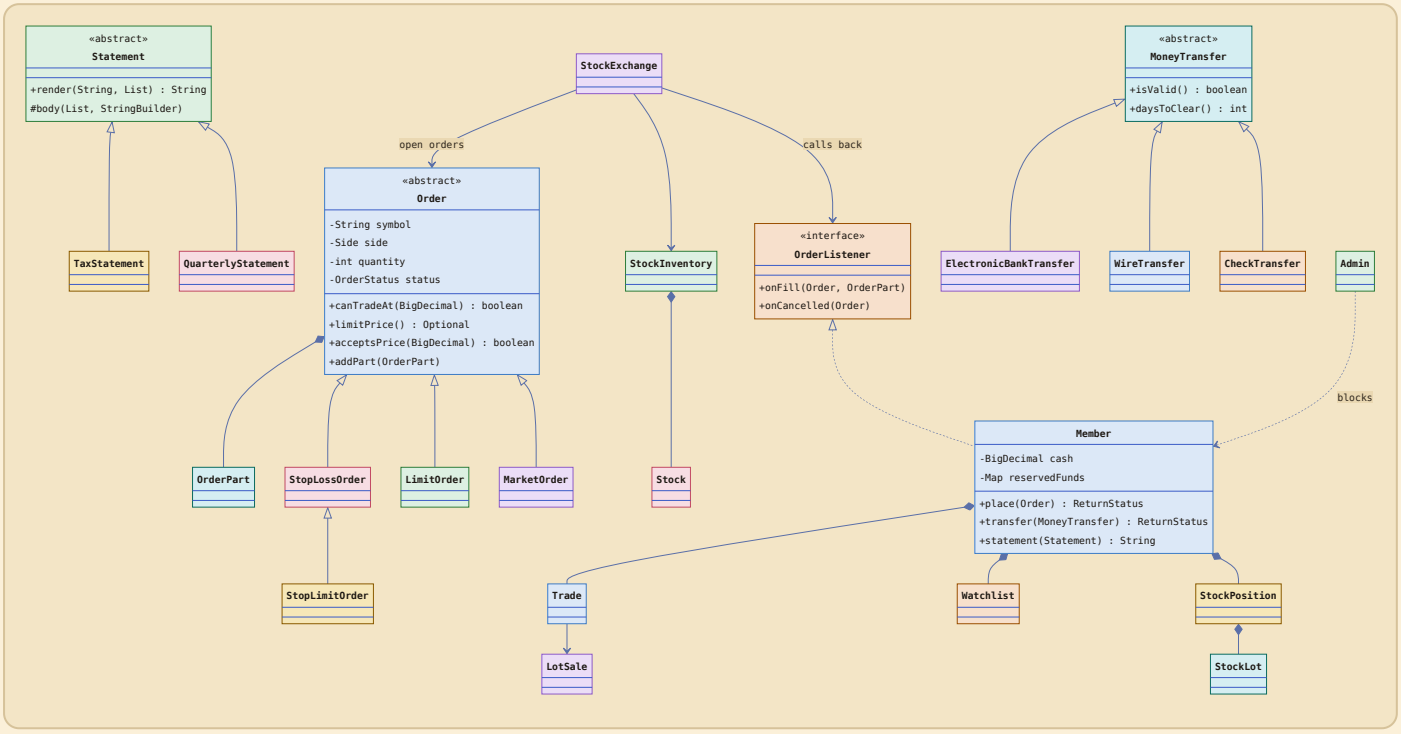
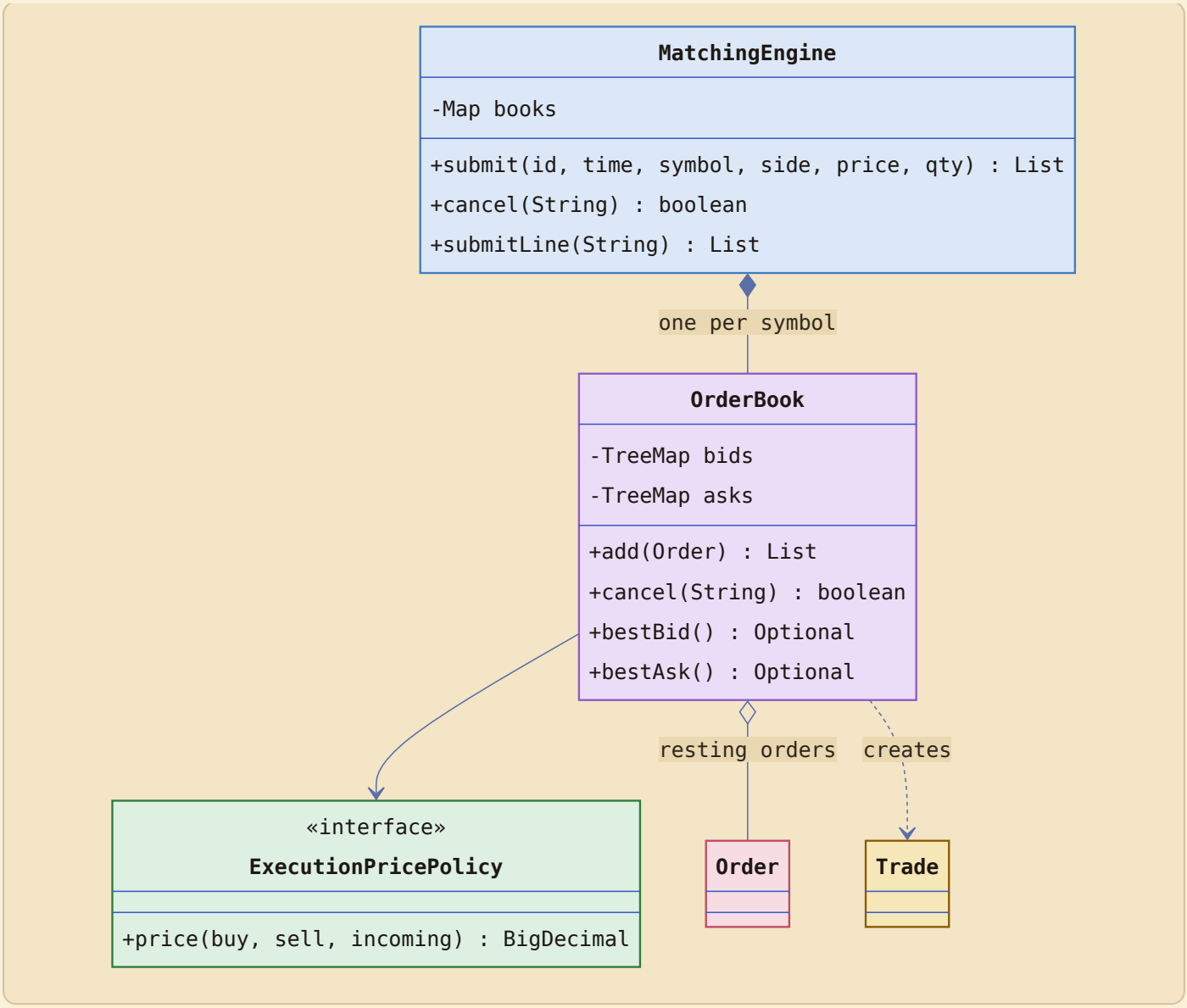


Figure — Stage 2: one abstract Order and four small subclasses that differ only in two answers (course diagram)

Our brokerage code:



And the matching engine (Part B):



Key flows

Placing a buy order. The course's activity diagram: pick a stock, pick the order type (market, limit, stop-loss, stop-limit), enter quantity, price limit and time enforcement; the system checks there is enough money, then deducts it (we *reserve* it) and sends the order to the exchange; when the exchange acknowledges, the member sees "order placed". The member can then update the order.

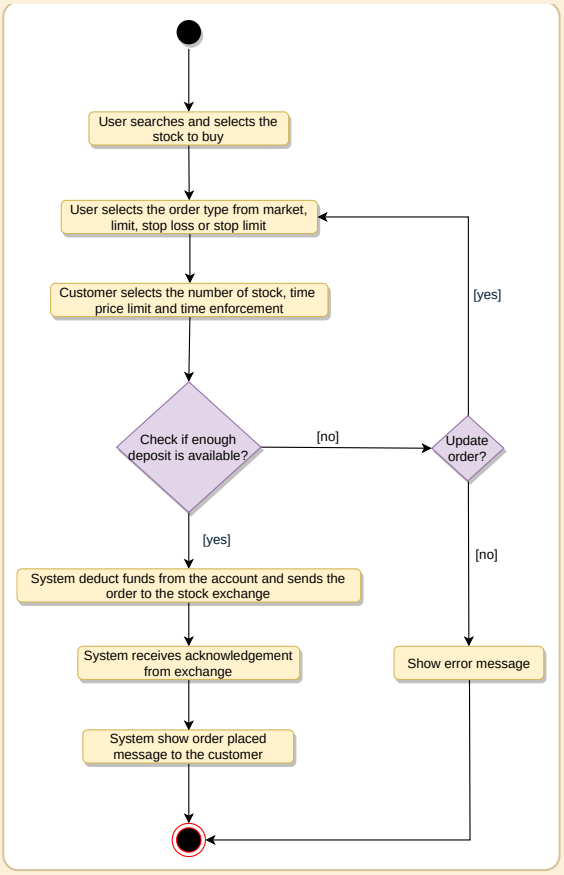


Figure — Activity diagram for placing a buy order (course diagram)

Placing a sell order is the mirror image, with one extra step: the member picks **which lots** to sell, and the check is "do I own enough shares?" rather than "do I have enough money?".

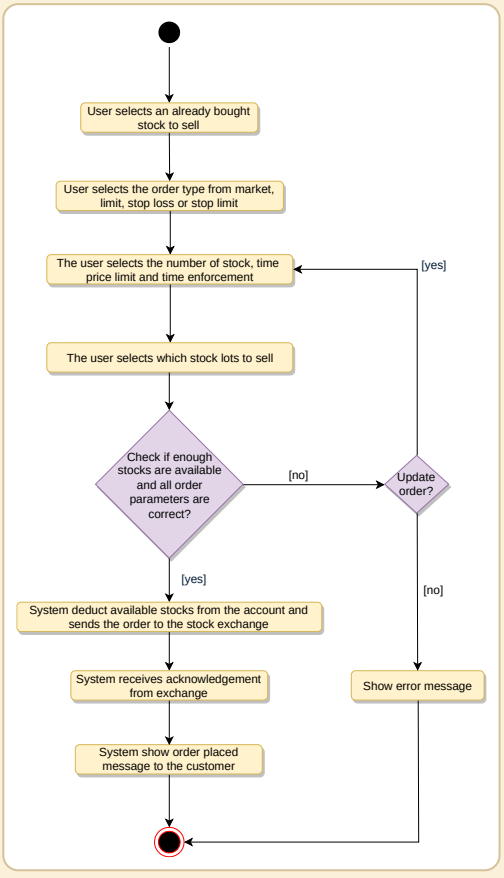


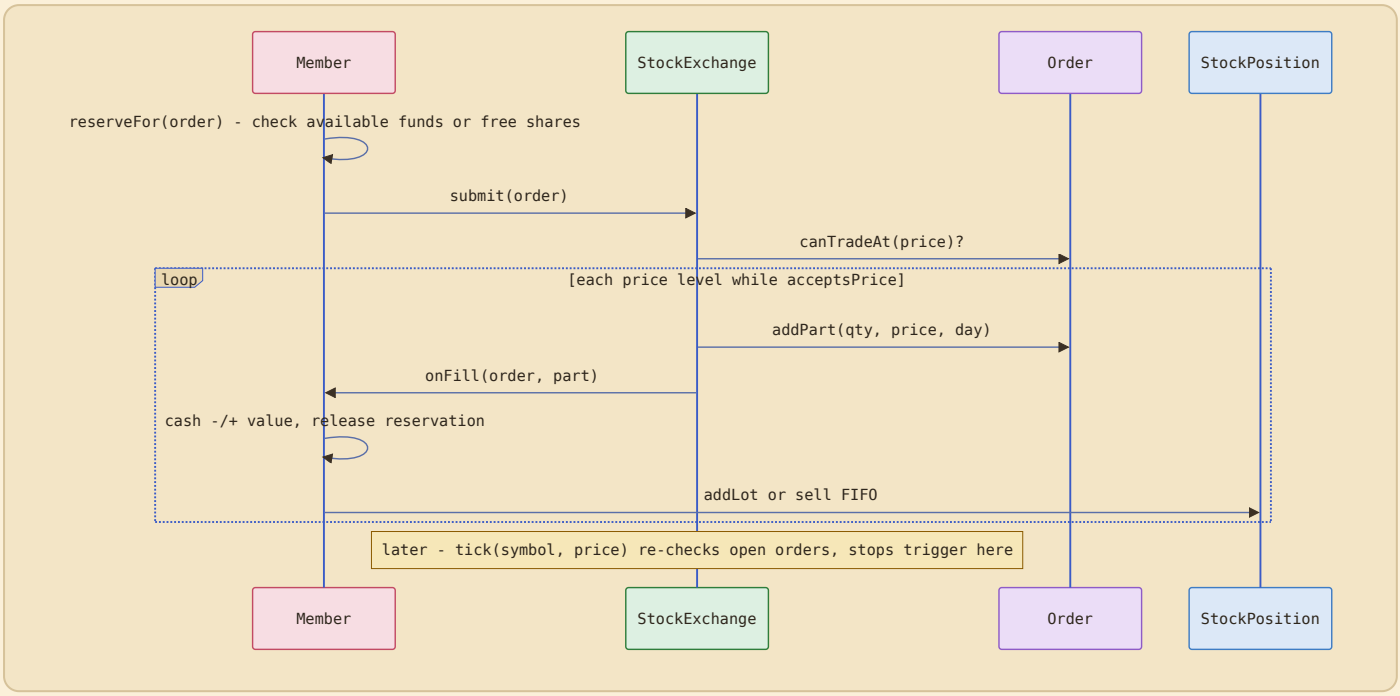
Figure — Activity diagram for placing a sell order, including choosing lots (course diagram)

The four order types in one table. They share everything except two answers:

Type	Q1: may it trade at this market price?	Q2: worst acceptable price	Plain words
Market	Always	None	"Buy/sell now at whatever the market gives."
Limit	Always	The limit	"Buy at 9.50 or less / sell at 10.90 or more."
Stop-loss (stop)	Only after the price **reaches the stop**, then forever	None (becomes a market order)	"If it falls to 10.00, get me out."
Stop-limit	Only after the stop is reached	The limit	"If it rises to 305, buy — but pay no more than 306."

A SELL stop triggers when price ≤ stop; a BUY stop triggers when price ≥ stop. Once triggered it stays triggered (the code "latches" a boolean).

Fill → money and lots move (Observer callback).

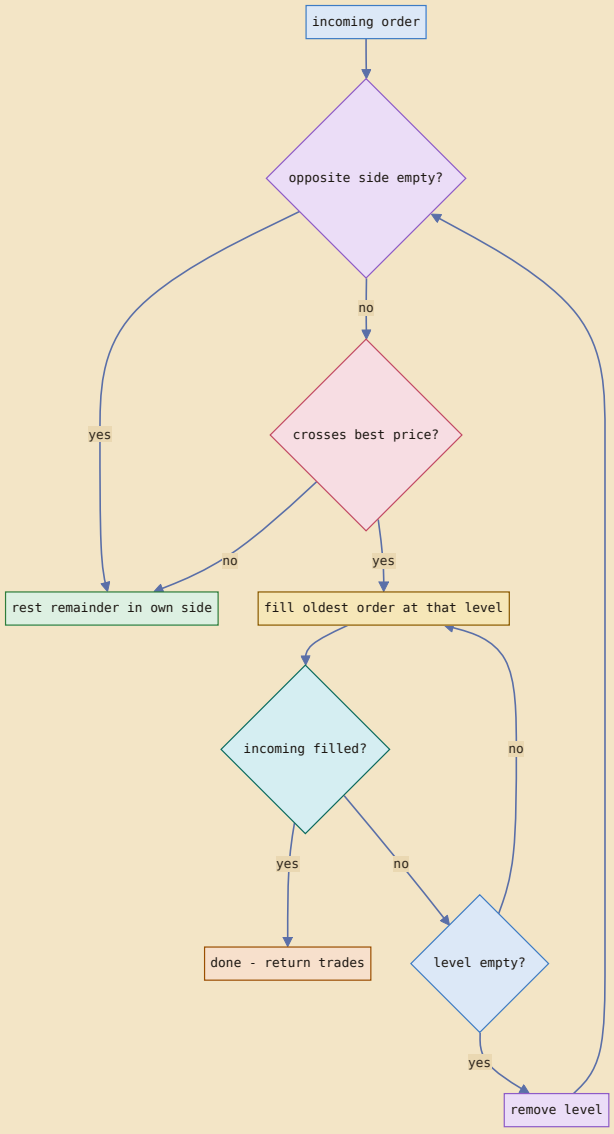


Matching engine: price-time priority. Each symbol has an `OrderBook` with two sides. Each side is a `TreeMap` (a map that keeps its keys sorted) from price to a FIFO queue (`ArrayDeque`, Java's fast double-ended queue). **Bids** (buy orders) are sorted highest-first, **asks** (sell orders) lowest-first. For an incoming order:

1. Look at the best price on the opposite side.
2. If it does not **cross** (buy price < best ask, or sell price > best bid), stop.
3. Otherwise fill against that level's queue, **oldest order first**, for `min(incoming remaining, resting remaining)`.
4. Remove fully filled resting orders; drop empty price levels; repeat from step 1.
5. If the incoming order still has quantity, add it to the back of its own price level's queue.

Walk-through of the sample input:

Order	Book before (asks / bids)	What happens
#1 sell 240.12 × 100	empty	rests on asks
#2 sell 237.45 × 90	asks 240.12×100	rests; best ask now 237.45
#3 buy 238.10 × 110	asks 237.45×90, 240.12×100	crosses 237.45 → trade `#3 237.45 90 #2`; 20 left; 240.12 > 238.10 so stop; rests 20 @ 238.10
#4 buy 237.80 × 10	best ask 240.12	no cross; rests
#5 buy 237.80 × 40	same	no cross; rests **behind #4** at 237.80
#6 sell 236.00 × 50	bids 238.10×20, 237.80×(10, 40)	fills #3 for 20, then #4 for 10, then #5 for 20 — all at 236.00 (the sell price)



Design decisions & patterns used

Decision	Why	Alternative rejected
One abstract `Order` + two abstract questions	The four types differ only in "may I trade?" and "worst price?". Each subclass is a few lines. The course stresses this is just a type hierarchy (plain polymorphism — calling the same method on different subclasses and each running its own version), not a named pattern: every subclass answers for itself whether the current price lets it trade, and no order ever turns into another kind — so it is not the State pattern either. (The only named patterns the course lists are Observer and Singleton — a class that allows exactly one instance; the Template Method and Strategy labels in this table are our additions.)	Four unrelated classes with copied fields; or one class with a `type` field and `switch`es.
`StopLimitOrder extends StopLossOrder`	A stop-limit is a stop that becomes a *limit* instead of a *market* order: it inherits the trigger, overrides only `limitPrice()`.	Duplicating the trigger logic.
`OrderPart` per fill	One order can fill at several prices/days; each fill needs its own price and quantity (and becomes its own lot).	Storing only an average price (loses tax information).
Observer: `OrderListener`	The exchange tells whoever placed the order; it does not import `Member`. Exchange and account code change independently.	The exchange updating member balances directly.
Reserved funds / reserved shares	Without it, a member with $1,000 could place ten $1,000 orders and all would pass the check. Money is reserved at placement and released as fills or cancellations arrive. Money and shares actually move only in `onFill`, which the **exchange** calls (the course's `callbackStockExchange`) — so the moment an order is *placed* and the moment it is *paid for* are different.	Checking the balance only at fill time (over-commitment).
Lots, FIFO by default	Tax depends on *which* shares you sold; each `LotSale` records it.	Merging all purchases into one average-cost position.
Template Method: `Statement`	Filtering by day range is shared; only the lines differ. `render` is `final`; subclasses fill in `body`.	Two statement classes each re-implementing the filter.
`MoneyTransfer` base class holds the limit rule	One rule, three delivery methods.	Copying the limit check into each subclass.
Singleton?	The course makes `StockExchange` a Singleton (one gateway to the real exchange). We pass one instance into `Member` instead (**dependency injection**: an object is handed the things it needs from outside rather than creating or looking them up itself) — same "only one" in practice, but testable with a fake exchange.	A global `getInstance()` that tests cannot replace.
Engine: `TreeMap<price, ArrayDeque>` per side	Best price in O(log n) (Big-O notation: how the work grows with n items — here, very slowly); FIFO inside a level gives time priority for free.	One `PriorityQueue` (a structure that always hands back the smallest item first) sorted by (price, time) — works, but cancel is O(n) (time

Decision	Why	Alternative rejected
		grows in step with the number of orders) and levels are not visible.
Engine: lock per symbol	Orders for different symbols match in parallel; same-symbol orders are strictly serialised (handled one at a time, in order), so "time priority" is well defined.	One global lock (slower) or no lock (corrupt book).
Engine: `ExecutionPricePolicy` (Strategy)	The Navi statement prices trades at the **sell** order's price; real continuous-trading exchanges use the **resting** (already-in-book) order's price. Both are one-line policies.	Hard-coding one rule.

Java implementation

Part A — Brokerage (`lld.ch19.brokerage`)

Files: `Orders.java` (enums, money helper, `OrderPart`, the order hierarchy), `Market.java` (stock, inventory, listener, simulated exchange), `Holdings.java` (lots, positions, watchlists, trades, statements), `Accounts.java` (transfers, notifier, member, admin), `Main.java`.

The simulated exchange offers a fixed depth of shares per price level (70 shares at each level, 5 cents apart, 3 levels) so fills come back in parts, like a real book. Part B builds a real matching engine.

```java
// File: lld/ch19/brokerage/Accounts.java
package lld.ch19.brokerage;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;
import java.util.Optional;

enum AccountStatus { ACTIVE, CLOSED, CANCELED, BLACKLISTED, NONE }

enum ReturnStatus { SUCCESS, FAIL, INSUFFICIENT_FUNDS, INSUFFICIENT_QUANTITY, NO_STOCK_POSITION,
ACCOUNT_NOT_ACTIVE }

enum Direction { DEPOSIT, WITHDRAW }

/** Three ways to move money. They differ only in name and clearing time; the limit rule lives here
once. */
abstract class MoneyTransfer {
    static final BigDecimal TRANSFER_LIMIT = Money.of("100000");
    private final BigDecimal amount;
    private final Direction direction;
    MoneyTransfer(BigDecimal amount, Direction direction) { this.amount = amount; this.direction =
direction; }
    BigDecimal amount() { return amount; }
    Direction direction() { return direction; }
    boolean isValid() { return amount.signum() > 0 && amount.compareTo(TRANSFER_LIMIT) <= 0; }
    abstract String methodName();
    abstract int daysToClear();
}

final class CheckTransfer extends MoneyTransfer {
    CheckTransfer(BigDecimal a, Direction d) { super(a, d); }
    @Override String methodName() { return "check"; }
    @Override int daysToClear() { return 5; }
}

final class WireTransfer extends MoneyTransfer {
    WireTransfer(BigDecimal a, Direction d) { super(a, d); }
    @Override String methodName() { return "wire"; }
    @Override int daysToClear() { return 1; }
}

final class ElectronicBankTransfer extends MoneyTransfer {
    ElectronicBankTransfer(BigDecimal a, Direction d) { super(a, d); }
    @Override String methodName() { return "electronic bank transfer"; }
    @Override int daysToClear() { return 3; }
}

interface Notifier { void send(String to, String message); }

final class EmailNotifier implements Notifier {
    @Override public void send(String to, String message) { System.out.println("    [email " + to + "]
" + message); }
}

/**
 * A member. Implements OrderListener so the exchange can call back without knowing
 * what a member is. Holds reserved funds so ten $1,000 orders cannot all pass a $1,000 check.
 * Lock order: the member never calls the exchange while holding its own lock.
```

```
 */
final class Member implements OrderListener {
    private final String id;
    private final String name;
    private final String email;
    private final StockExchange exchange;
    private final StockInventory inventory;
    private final Notifier notifier;
    private volatile AccountStatus status = AccountStatus.ACTIVE;

    private BigDecimal cash = Money.of("0");
    private final Map<String, BigDecimal> reservedFunds = new HashMap<>();   // buy orderId -> money
still reserved
    private final Map<String, Order> pendingSells = new HashMap<>();            // sell orderId ->
open sell order
    private final Map<String, StockPosition> positions = new LinkedHashMap<>();
    private final Map<String, Watchlist> watchlists = new LinkedHashMap<>();
    private final List<Trade> trades = new ArrayList<>();
    private int lotSeq = 1;

    Member(String id, String name, String email, StockExchange ex, StockInventory inv, Notifier n) {
        this.id = id; this.name = name; this.email = email;
        this.exchange = ex; this.inventory = inv; this.notifier = n;
        ex.register(id, this);
    }

    String id() { return id; }
    String name() { return name; }
    void setStatus(AccountStatus s) { status = s; }

    // ---- money ----
    synchronized ReturnStatus transfer(MoneyTransfer t) {
        if (!t.isValid()) return ReturnStatus.FAIL;
        if (t.direction() == Direction.WITHDRAW) {
            if (t.amount().compareTo(availableFunds()) > 0) return ReturnStatus.INSUFFICIENT_FUNDS;
            cash = cash.subtract(t.amount());
        } else {
            cash = cash.add(t.amount());
        }
        return ReturnStatus.SUCCESS;
    }

    synchronized BigDecimal cash() { return cash; }

    synchronized BigDecimal reserved() {
        BigDecimal r = Money.of("0");
        for (Map.Entry<String, BigDecimal> e : reservedFunds.entrySet()) r = r.add(e.getValue());
        return r;
    }

    synchronized BigDecimal availableFunds() { return cash.subtract(reserved()); }

    // ---- orders ----
    ReturnStatus place(Order o) {
        ReturnStatus check = reserveFor(o);                // under our lock
        if (check == ReturnStatus.SUCCESS) exchange.submit(o);   // outside our lock
        return check;
    }

    private synchronized ReturnStatus reserveFor(Order o) {
        if (status != AccountStatus.ACTIVE) return ReturnStatus.ACCOUNT_NOT_ACTIVE;
        if (o.side() == Side.BUY) {
```

```
            BigDecimal perShare = o.limitPrice().orElse(inventory.priceOf(o.symbol()));
            BigDecimal need = Money.times(perShare, o.quantity());
            if (need.compareTo(availableFunds()) > 0) return ReturnStatus.INSUFFICIENT_FUNDS;
            reservedFunds.put(o.id(), need);           // released part by part as fills arrive
            return ReturnStatus.SUCCESS;
        }
        StockPosition pos = positions.get(o.symbol());
        if (pos == null) return ReturnStatus.NO_STOCK_POSITION;
        int promised = pendingSells.values().stream()              // shares already promised to
other
                .filter(p -> p.symbol().equals(o.symbol()))             // open sells of this symbol
                .mapToInt(Order::remaining).sum();
        if (o.quantity() > pos.quantity() - promised) return ReturnStatus.INSUFFICIENT_QUANTITY;
        pendingSells.put(o.id(), o);
        return ReturnStatus.SUCCESS;
    }

    ReturnStatus cancel(Order o) { return exchange.cancel(o) ? ReturnStatus.SUCCESS :
ReturnStatus.FAIL; }

    /** Called by the exchange: this is where money and shares actually move. */
    @Override public synchronized void onFill(Order o, OrderPart part) {
        BigDecimal value = Money.times(part.price(), part.quantity());
        List<LotSale> sales = List.of();
        if (o.side() == Side.BUY) {
            cash = cash.subtract(value);
            BigDecimal left = reservedFunds.getOrDefault(o.id(), Money.of("0")).subtract(value);
            if (o.remaining() == 0 || left.signum() <= 0) reservedFunds.remove(o.id());
            else reservedFunds.put(o.id(), left);
            positions.computeIfAbsent(o.symbol(), StockPosition::new)
                     .addLot(new StockLot("L" + lotSeq++, part.quantity(), part.price(), part.day()));
        } else {
            cash = cash.add(value);
            if (o.remaining() == 0) pendingSells.remove(o.id());
            sales = positions.get(o.symbol()).sell(part.quantity(), part.price(), Optional.empty());
            if (positions.get(o.symbol()).quantity() == 0) positions.remove(o.symbol());
        }
        trades.add(new Trade(part.day(), o.symbol(), o.side(), part.quantity(), part.price(), sales));
        notifier.send(email, "filled " + part.quantity() + " " + o.symbol() + " @ " + part.price()
                + " (" + o.id() + " " + o.status() + ")");
    }

    @Override public synchronized void onCancelled(Order o) {
        reservedFunds.remove(o.id());
        pendingSells.remove(o.id());
        notifier.send(email, o.id() + " cancelled, " + o.remaining() + " shares unfilled");
    }

    // ---- watchlists, positions, reports ----
    synchronized Watchlist watchlist(String name) { return watchlists.computeIfAbsent(name,
Watchlist::new); }
    synchronized List<Watchlist> watchlists() { return List.copyOf(watchlists.values()); }
    synchronized Map<String, List<StockLot>> holdings() {
        Map<String, List<StockLot>> m = new LinkedHashMap<>();
        positions.forEach((s, p) -> m.put(s, p.lots()));
        return m;
    }
    synchronized String statement(Statement s) { return s.render(name, List.copyOf(trades)); }
}

final class Admin {
    void block(Member m) { m.setStatus(AccountStatus.BLACKLISTED); }
```

```
    void unblock(Member m) { m.setStatus(AccountStatus.ACTIVE); }
}
```

```java
// File: lld/ch19/brokerage/Holdings.java
package lld.ch19.brokerage;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.Iterator;
import java.util.LinkedHashSet;
import java.util.List;
import java.util.Optional;
import java.util.Set;

/** One purchase at one price. Kept apart from other purchases because tax depends on it. */
final class StockLot {
    private final String lotId;
    private final BigDecimal buyPrice;
    private final int day;
    private int quantity;

    StockLot(String lotId, int quantity, BigDecimal buyPrice, int day) {
        this.lotId = lotId; this.quantity = quantity; this.buyPrice = buyPrice; this.day = day;
    }
    String lotId() { return lotId; }
    int quantity() { return quantity; }
    BigDecimal buyPrice() { return buyPrice; }
    int day() { return day; }
    void take(int q) { quantity -= q; }
    @Override public String toString() { return lotId + ":" + quantity + "@" + buyPrice; }
}

/** What came out of one lot in one sale - exactly what a tax statement needs. */
record LotSale(String lotId, int quantity, BigDecimal buyPrice, BigDecimal sellPrice) {
    BigDecimal gain() { return Money.times(sellPrice.subtract(buyPrice), quantity); }
}

/** All lots of one symbol. Sells from the named lot first, then oldest first (FIFO). */
final class StockPosition {
    private final String symbol;
    private final List<StockLot> lots = new ArrayList<>();

    StockPosition(String symbol) { this.symbol = symbol; }

    String symbol() { return symbol; }
    void addLot(StockLot lot) { lots.add(lot); }
    int quantity() { return lots.stream().mapToInt(StockLot::quantity).sum(); }
    List<StockLot> lots() { return List.copyOf(lots); }

    List<LotSale> sell(int qty, BigDecimal price, Optional<String> preferredLot) {
        if (qty > quantity()) throw new IllegalStateException("not enough shares");
        List<LotSale> out = new ArrayList<>();
        List<StockLot> order = new ArrayList<>(lots);
        preferredLot.flatMap(id -> lots.stream().filter(l -> l.lotId().equals(id)).findFirst())
                    .ifPresent(l -> { order.remove(l); order.add(0, l); });
        int left = qty;
        for (StockLot lot : order) {
            if (left == 0) break;
            int take = Math.min(left, lot.quantity());
            if (take == 0) continue;
            lot.take(take);
            out.add(new LotSale(lot.lotId(), take, lot.buyPrice(), price));
            left -= take;
        }
        for (Iterator<StockLot> it = lots.iterator(); it.hasNext(); ) if (it.next().quantity() == 0)
```

```
it.remove();
        return out;
    }
}

/** A watchlist holds symbols, not Stock objects, so it never goes stale. */
final class Watchlist {
    private final String name;
    private final Set<String> symbols = new LinkedHashSet<>();
    Watchlist(String name) { this.name = name; }
    String name() { return name; }
    void add(String symbol) { symbols.add(symbol); }
    void remove(String symbol) { symbols.remove(symbol); }
    Set<String> symbols() { return Set.copyOf(symbols); }
    String render(StockInventory inv) {
        StringBuilder sb = new StringBuilder(name + ":");
        symbols.forEach(s -> sb.append(' ').append(s).append('=').append(inv.priceOf(s)));
        return sb.toString();
    }
}

/** One executed fill as the member sees it. Sells carry the lots they consumed. */
record Trade(int day, String symbol, Side side, int quantity, BigDecimal price, List<LotSale> sales) {
}

/** Template Method: the base class filters trades by day range, subclasses write the lines. */
abstract class Statement {
    private final int fromDay;
    private final int toDay;
    Statement(int fromDay, int toDay) { this.fromDay = fromDay; this.toDay = toDay; }

    final String render(String memberName, List<Trade> all) {
        List<Trade> inRange = all.stream().filter(t -> t.day() >= fromDay && t.day() <=
toDay).toList();
        StringBuilder sb = new StringBuilder(title() + " for " + memberName + " (days " + fromDay + "-
" + toDay + ")\n");
        body(inRange, sb);
        return sb.toString();
    }
    abstract String title();
    abstract void body(List<Trade> trades, StringBuilder sb);
}

final class QuarterlyStatement extends Statement {
    QuarterlyStatement(int from, int to) { super(from, to); }
    @Override String title() { return "Quarterly statement"; }
    @Override void body(List<Trade> trades, StringBuilder sb) {
        for (Trade t : trades)
            sb.append(String.format("  day %d %-4s %-4s %4d @ %s%n", t.day(), t.side(), t.symbol(),
t.quantity(), t.price()));
    }
}

final class TaxStatement extends Statement {
    TaxStatement(int from, int to) { super(from, to); }
    @Override String title() { return "Tax statement"; }
    @Override void body(List<Trade> trades, StringBuilder sb) {
        BigDecimal total = Money.of("0");
        for (Trade t : trades) for (LotSale s : t.sales()) {
            sb.append(String.format("  %s %s: %d sh bought @ %s sold @ %s gain %s%n",
                    t.symbol(), s.lotId(), s.quantity(), s.buyPrice(), s.sellPrice(),
Money.fmt(s.gain())));
```

```
            total = total.add(s.gain());
        }
        sb.append("  realised gain: ").append(Money.fmt(total)).append('\n');
    }
}
```

```java
// File: lld/ch19/brokerage/Main.java
package lld.ch19.brokerage;

import java.math.BigDecimal;

public class Main {
    public static void main(String[] args) {
        StockInventory inv = new StockInventory();
        inv.list("AAPL", Money.of("10.00"));
        inv.list("MSFT", Money.of("300.00"));
        inv.list("AMZN", Money.of("120.00"));
        // 70 shares offered per price level, levels 5 cents apart, 3 levels deep
        StockExchange ex = new StockExchange(inv, 70, new BigDecimal("0.05"), 3);
        Member alice = new Member("M1", "Alice", "alice@mail.test", ex, inv, new EmailNotifier());
        Admin admin = new Admin();

        System.out.println("== 1. Fund the account ==");
        System.out.println("  wire 5,000        -> " + alice.transfer(new
WireTransfer(Money.of("5000"), Direction.DEPOSIT)));
        System.out.println("  check 200,000     -> " + alice.transfer(new
CheckTransfer(Money.of("200000"), Direction.DEPOSIT))
                + " (over the 100,000 limit)");
        System.out.println("  cash = " + Money.fmt(alice.cash()));

        System.out.println("\n== 2. Watchlists and search ==");
        alice.watchlist("tech").add("AAPL");
        alice.watchlist("tech").add("MSFT");
        alice.watchlist("retail").add("AMZN");
        alice.watchlists().forEach(w -> System.out.println("  " + w.render(inv)));
        System.out.println("  search 'A' -> " + inv.search("A"));

        System.out.println("\n== 3. Market buy 100 AAPL fills in two parts (two lots) ==");
        Order m1 = new MarketOrder("O1", "M1", "AAPL", Side.BUY, 100);
        System.out.println("  place -> " + alice.place(m1));
        System.out.println("  " + m1 + " parts=" + m1.parts());

        System.out.println("\n== 4. Next day: limit buy 50 AAPL @ 9.50 waits, then fills when price
drops ==");
        ex.setDay(2);
        Order l1 = new LimitOrder("O2", "M1", "AAPL", Side.BUY, 50, Money.of("9.50"),
TimeInForce.GOOD_TILL_CANCELLED);
        System.out.println("  place -> " + alice.place(l1) + ", " + l1 + ", reserved " +
Money.fmt(alice.reserved()));
        ex.tick("AAPL", Money.of("9.40"));
        System.out.println("  after tick 9.40: " + l1 + ", reserved " + Money.fmt(alice.reserved()));
        System.out.println("  holdings " + alice.holdings());

        System.out.println("\n== 5. Day 3: price rises, sell 120 AAPL (FIFO across lots) ==");
        ex.setDay(3);
        ex.tick("AAPL", Money.of("11.00"));
        Order s1 = new LimitOrder("O3", "M1", "AAPL", Side.SELL, 120, Money.of("10.90"),
TimeInForce.GOOD_TILL_CANCELLED);
        System.out.println("  place -> " + alice.place(s1) + ", " + s1);
        System.out.println("  holdings " + alice.holdings());

        System.out.println("\n== 6. Protect the rest with a stop-loss @ 10.00, then the price falls
==");
        ex.setDay(4);
        Order stop = new StopLossOrder("O4", "M1", "AAPL", Side.SELL, 30, Money.of("10.00"));
        System.out.println("  place -> " + alice.place(stop) + ", " + stop);
        ex.tick("AAPL", Money.of("10.50"));
        System.out.println("  tick 10.50: " + stop);
```

```
        ex.tick("AAPL", Money.of("9.95"));
        System.out.println("  tick 9.95: " + stop + " parts=" + stop.parts());

        System.out.println("\n== 7. Stop-limit buy triggers but its limit is not met ==");
        Order sl = new StopLimitOrder("O5", "M1", "MSFT", Side.BUY, 5, Money.of("305.00"),
Money.of("306.00"));
        System.out.println("  place -> " + alice.place(sl) + ", " + sl);
        ex.tick("MSFT", Money.of("310.00"));
        System.out.println("  tick 310 (stop hit, above limit): " + sl);
        ex.tick("MSFT", Money.of("305.50"));
        System.out.println("  tick 305.50 (within limit): " + sl);

        System.out.println("\n== 8. Reserved funds block over-spending, cancel releases them ==");
        System.out.println("  cash " + Money.fmt(alice.cash()) + ", available " +
Money.fmt(alice.availableFunds()));
        Order big1 = new LimitOrder("O6", "M1", "AMZN", Side.BUY, 20, Money.of("100.00"),
TimeInForce.GOOD_TILL_CANCELLED);
        Order big2 = new LimitOrder("O7", "M1", "AMZN", Side.BUY, 20, Money.of("100.00"),
TimeInForce.GOOD_TILL_CANCELLED);
        System.out.println("  O6 buy 20 AMZN @ 100 -> " + alice.place(big1) + ", available " +
Money.fmt(alice.availableFunds()));
        System.out.println("  O7 buy 20 AMZN @ 100 -> " + alice.place(big2));
        System.out.println("  cancel O6 -> " + alice.cancel(big1) + ", available " +
Money.fmt(alice.availableFunds()));
        System.out.println("  sell 10 AMZN (none owned) -> "
                + alice.place(new MarketOrder("O8", "M1", "AMZN", Side.SELL, 10)));

        System.out.println("\n== 9. Blocked account cannot trade ==");
        admin.block(alice);
        System.out.println("  -> " + alice.place(new MarketOrder("O9", "M1", "AAPL", Side.BUY, 1)));
        admin.unblock(alice);

        System.out.println("\n== 10. Reports ==");
        System.out.print(alice.statement(new QuarterlyStatement(1, 90)));
        System.out.print(alice.statement(new TaxStatement(1, 365)));
    }
}
```

```java
// File: lld/ch19/brokerage/Market.java
package lld.ch19.brokerage;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.List;
import java.util.Map;
import java.util.Optional;
import java.util.concurrent.ConcurrentHashMap;

/** A symbol and its latest price. */
final class Stock {
    private final String symbol;
    private volatile BigDecimal price;
    Stock(String symbol, BigDecimal price) { this.symbol = symbol; this.price = price; }
    String symbol() { return symbol; }
    BigDecimal price() { return price; }
    void setPrice(BigDecimal p) { price = p; }
}

/** The one place prices are read from, so no two parts of the system disagree. */
final class StockInventory {
    private final Map<String, Stock> stocks = new ConcurrentHashMap<>();
    void list(String symbol, BigDecimal price) { stocks.put(symbol, new Stock(symbol, price)); }
    Optional<Stock> find(String symbol) { return Optional.ofNullable(stocks.get(symbol)); }
    BigDecimal priceOf(String symbol) {
        return find(symbol).orElseThrow(() -> new IllegalArgumentException("unknown " +
symbol)).price();
    }
    List<String> search(String prefix) {
        return stocks.keySet().stream().filter(s ->
s.startsWith(prefix.toUpperCase())).sorted().toList();
    }
}

/** Observer: how the exchange calls back without knowing what a Member is. */
interface OrderListener {
    void onFill(Order order, OrderPart part);
    void onCancelled(Order order);
}

/**
 * A simulated exchange. At each price level it offers a fixed depth of shares
 * (e.g. 70 @ price, 70 @ price+0.05 ...) so that fills come back in parts, like a
 * real order book. (A real matching engine is built in lld.ch19.stockexchange.)
 * All methods are synchronized: one lock orders every event on the exchange.
 */
final class StockExchange {
    private final StockInventory inventory;
    private final int depthPerLevel;
    private final BigDecimal tickSize;
    private final int levels;
    private final Map<String, List<Order>> openOrders = new ConcurrentHashMap<>();
    private final Map<String, OrderListener> listeners = new ConcurrentHashMap<>();
    private int day = 1;

    StockExchange(StockInventory inventory, int depthPerLevel, BigDecimal tickSize, int levels) {
        this.inventory = inventory;
        this.depthPerLevel = depthPerLevel;
        this.tickSize = tickSize;
        this.levels = levels;
    }
```

```
    synchronized void register(String memberId, OrderListener l) { listeners.put(memberId, l); }
    synchronized void setDay(int d) { day = d; }
    synchronized int day() { return day; }

    synchronized void submit(Order o) {
        openOrders.computeIfAbsent(o.symbol(), k -> new ArrayList<>()).add(o);
        tryFill(o, inventory.priceOf(o.symbol()));
        prune(o.symbol());
    }

    /** A new price arrives: every open order on that symbol gets another chance (stops trigger here).
*/
    synchronized void tick(String symbol, BigDecimal newPrice) {
        inventory.find(symbol).orElseThrow().setPrice(newPrice);
        for (Order o : List.copyOf(openOrders.getOrDefault(symbol, List.of()))) tryFill(o, newPrice);
        prune(symbol);
    }

    synchronized boolean cancel(Order o) {
        if (!o.cancel()) return false;
        prune(o.symbol());
        listeners.get(o.memberId()).onCancelled(o);
        return true;
    }

    private void tryFill(Order o, BigDecimal market) {
        if (!o.isOpen() || !o.canTradeAt(market)) return;
        for (int k = 0; k < levels && o.remaining() > 0; k++) {
            BigDecimal step = tickSize.multiply(BigDecimal.valueOf(k));
            BigDecimal levelPrice = o.side() == Side.BUY ? market.add(step) : market.subtract(step);
            if (!o.acceptsPrice(levelPrice)) break;          // limit reached: rest stays open
            OrderPart part = new OrderPart(Math.min(depthPerLevel, o.remaining()), levelPrice, day);
            o.addPart(part);
            listeners.get(o.memberId()).onFill(o, part);
        }
    }

    private void prune(String symbol) {
        openOrders.getOrDefault(symbol, new ArrayList<>()).removeIf(o -> !o.isOpen());
    }
}
```

```java
// File: lld/ch19/brokerage/Orders.java
package lld.ch19.brokerage;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.util.ArrayList;
import java.util.List;
import java.util.Optional;

enum Side { BUY, SELL }

enum OrderStatus { OPEN, PARTIALLY_FILLED, FILLED, CANCELLED }

/** Time in force: how long an order may stay alive. Modelled, only GTC used in the demo. */
enum TimeInForce { GOOD_TILL_CANCELLED, FILL_OR_KILL, IMMEDIATE_OR_CANCEL, ON_THE_OPEN, ON_THE_CLOSE }

final class Money {
    private Money() { }
    static BigDecimal of(String s) { return new BigDecimal(s).setScale(2, RoundingMode.HALF_EVEN); }
    static BigDecimal times(BigDecimal price, int qty) {
        return price.multiply(BigDecimal.valueOf(qty)).setScale(2, RoundingMode.HALF_EVEN);
    }
    static String fmt(BigDecimal b) { return "$" + b.setScale(2,
RoundingMode.HALF_EVEN).toPlainString(); }
}

/** One fill. A 100-share order might be filled as 70 @ 10.00 plus 30 @ 10.05. */
record OrderPart(int quantity, BigDecimal price, int day) { }

/**
 * Everything the four order types share. They differ in exactly two answers:
 * (1) may I trade at this market price? (2) what is the worst price I accept?
 */
abstract class Order {
    private final String id;
    private final String memberId;
    private final String symbol;
    private final Side side;
    private final int quantity;
    private final TimeInForce timeInForce;
    private final List<OrderPart> parts = new ArrayList<>();
    private OrderStatus status = OrderStatus.OPEN;

    Order(String id, String memberId, String symbol, Side side, int quantity, TimeInForce tif) {
        if (quantity <= 0) throw new IllegalArgumentException("quantity must be positive");
        this.id = id;
        this.memberId = memberId;
        this.symbol = symbol;
        this.side = side;
        this.quantity = quantity;
        this.timeInForce = tif;
    }

    /** Question 1. Market and limit orders always say yes; stop orders wait for the trigger. */
    abstract boolean canTradeAt(BigDecimal marketPrice);

    /** Question 2. Empty means "any price" (market behaviour). */
    abstract Optional<BigDecimal> limitPrice();

    abstract String type();

    /** Is a fill at this price acceptable? BUY: price <= limit. SELL: price >= limit. */
```

```
    final boolean acceptsPrice(BigDecimal price) {
        return limitPrice().map(l -> side == Side.BUY ? price.compareTo(l) <= 0 : price.compareTo(l)
>= 0)
                           .orElse(true);
    }

    final synchronized void addPart(OrderPart p) {
        if (p.quantity() > remaining()) throw new IllegalStateException("overfill");
        parts.add(p);
        status = remaining() == 0 ? OrderStatus.FILLED : OrderStatus.PARTIALLY_FILLED;
    }

    final synchronized boolean cancel() {
        if (status == OrderStatus.FILLED || status == OrderStatus.CANCELLED) return false;
        status = OrderStatus.CANCELLED;
        return true;
    }

    final synchronized int filled() { return parts.stream().mapToInt(OrderPart::quantity).sum(); }
    final synchronized int remaining() { return quantity - filled(); }
    final synchronized OrderStatus status() { return status; }
    final synchronized boolean isOpen() { return status == OrderStatus.OPEN || status ==
OrderStatus.PARTIALLY_FILLED; }
    final synchronized List<OrderPart> parts() { return List.copyOf(parts); }

    final String id() { return id; }
    final String memberId() { return memberId; }
    final String symbol() { return symbol; }
    final Side side() { return side; }
    final int quantity() { return quantity; }
    final TimeInForce timeInForce() { return timeInForce; }

    @Override public String toString() {
        return id + " " + type() + " " + side + " " + quantity + " " + symbol + " [" + status() + "]";
    }
}

final class MarketOrder extends Order {
    MarketOrder(String id, String memberId, String symbol, Side side, int qty) {
        super(id, memberId, symbol, side, qty, TimeInForce.GOOD_TILL_CANCELLED);
    }
    @Override boolean canTradeAt(BigDecimal p) { return true; }
    @Override Optional<BigDecimal> limitPrice() { return Optional.empty(); }
    @Override String type() { return "MARKET"; }
}

final class LimitOrder extends Order {
    private final BigDecimal limit;
    LimitOrder(String id, String memberId, String symbol, Side side, int qty, BigDecimal limit,
TimeInForce tif) {
        super(id, memberId, symbol, side, qty, tif);
        this.limit = limit;
    }
    @Override boolean canTradeAt(BigDecimal p) { return true; }
    @Override Optional<BigDecimal> limitPrice() { return Optional.of(limit); }
    @Override String type() { return "LIMIT@" + limit; }
}

/**
 * Stop (stop-loss) order: sleeps until the market reaches the stop price, then trades
 * like a market order. SELL stop fires when price <= stop, BUY stop when price >= stop.
 * Once triggered it stays triggered.
```

```
 */
class StopLossOrder extends Order {
    private final BigDecimal stopPrice;
    private boolean triggered;

    StopLossOrder(String id, String memberId, String symbol, Side side, int qty, BigDecimal stopPrice)
{
        super(id, memberId, symbol, side, qty, TimeInForce.GOOD_TILL_CANCELLED);
        this.stopPrice = stopPrice;
    }

    @Override final synchronized boolean canTradeAt(BigDecimal p) {
        if (!triggered) {
            triggered = side() == Side.SELL ? p.compareTo(stopPrice) <= 0 : p.compareTo(stopPrice) >=
0;
        }
        return triggered;
    }
    @Override Optional<BigDecimal> limitPrice() { return Optional.empty(); }
    @Override String type() { return "STOP@" + stopPrice; }
    BigDecimal stopPrice() { return stopPrice; }
}

/** A stop order that becomes a LIMIT order (not a market order) once triggered. */
final class StopLimitOrder extends StopLossOrder {
    private final BigDecimal limit;
    StopLimitOrder(String id, String memberId, String symbol, Side side, int qty, BigDecimal stop,
BigDecimal limit) {
        super(id, memberId, symbol, side, qty, stop);
        this.limit = limit;
    }
    @Override Optional<BigDecimal> limitPrice() { return Optional.of(limit); }
    @Override String type() { return "STOP@" + stopPrice() + "/LIMIT@" + limit; }
}
```

Output:

```
== 1. Fund the account ==
  wire 5,000        -> SUCCESS
  check 200,000     -> FAIL (over the 100,000 limit)
  cash = $5000.00

== 2. Watchlists and search ==
  tech: AAPL=10.00 MSFT=300.00
  retail: AMZN=120.00
  search 'A' -> [AAPL, AMZN]

== 3. Market buy 100 AAPL fills in two parts (two lots) ==
    [email alice@mail.test] filled 70 AAPL @ 10.00 (O1 PARTIALLY_FILLED)
    [email alice@mail.test] filled 30 AAPL @ 10.05 (O1 FILLED)
  place -> SUCCESS
  O1 MARKET BUY 100 AAPL [FILLED] parts=[OrderPart[quantity=70, price=10.00, day=1],
OrderPart[quantity=30, price=10.05, day=1]]

== 4. Next day: limit buy 50 AAPL @ 9.50 waits, then fills when price drops ==
  place -> SUCCESS, O2 LIMIT@9.50 BUY 50 AAPL [OPEN], reserved $475.00
    [email alice@mail.test] filled 50 AAPL @ 9.40 (O2 FILLED)
  after tick 9.40: O2 LIMIT@9.50 BUY 50 AAPL [FILLED], reserved $0.00
  holdings {AAPL=[L1:70@10.00, L2:30@10.05, L3:50@9.40]}

== 5. Day 3: price rises, sell 120 AAPL (FIFO across lots) ==
    [email alice@mail.test] filled 70 AAPL @ 11.00 (O3 PARTIALLY_FILLED)
    [email alice@mail.test] filled 50 AAPL @ 10.95 (O3 FILLED)
  place -> SUCCESS, O3 LIMIT@10.90 SELL 120 AAPL [FILLED]
  holdings {AAPL=[L3:30@9.40]}

== 6. Protect the rest with a stop-loss @ 10.00, then the price falls ==
  place -> SUCCESS, O4 STOP@10.00 SELL 30 AAPL [OPEN]
  tick 10.50: O4 STOP@10.00 SELL 30 AAPL [OPEN]
    [email alice@mail.test] filled 30 AAPL @ 9.95 (O4 FILLED)
  tick 9.95: O4 STOP@10.00 SELL 30 AAPL [FILLED] parts=[OrderPart[quantity=30, price=9.95, day=4]]

== 7. Stop-limit buy triggers but its limit is not met ==
  place -> SUCCESS, O5 STOP@305.00/LIMIT@306.00 BUY 5 MSFT [OPEN]
  tick 310 (stop hit, above limit): O5 STOP@305.00/LIMIT@306.00 BUY 5 MSFT [OPEN]
    [email alice@mail.test] filled 5 MSFT @ 305.50 (O5 FILLED)
  tick 305.50 (within limit): O5 STOP@305.00/LIMIT@306.00 BUY 5 MSFT [FILLED]

== 8. Reserved funds block over-spending, cancel releases them ==
  cash $3617.00, available $3617.00
  O6 buy 20 AMZN @ 100 -> SUCCESS, available $1617.00
  O7 buy 20 AMZN @ 100 -> INSUFFICIENT_FUNDS
    [email alice@mail.test] O6 cancelled, 20 shares unfilled
  cancel O6 -> SUCCESS, available $3617.00
  sell 10 AMZN (none owned) -> NO_STOCK_POSITION

== 9. Blocked account cannot trade ==
  -> ACCOUNT_NOT_ACTIVE

== 10. Reports ==
Quarterly statement for Alice (days 1-90)
  day 1 BUY  AAPL   70 @ 10.00
  day 1 BUY  AAPL   30 @ 10.05
  day 2 BUY  AAPL   50 @ 9.40
  day 3 SELL AAPL   70 @ 11.00
  day 3 SELL AAPL   50 @ 10.95
  day 4 SELL AAPL   30 @ 9.95
  day 4 BUY  MSFT    5 @ 305.50
Tax statement for Alice (days 1-365)
```

```
  AAPL L1: 70 sh bought @ 10.00 sold @ 11.00 gain $70.00
  AAPL L2: 30 sh bought @ 10.05 sold @ 10.95 gain $27.00
  AAPL L3: 20 sh bought @ 9.40 sold @ 10.95 gain $31.00
  AAPL L3: 30 sh bought @ 9.40 sold @ 9.95 gain $16.50
  realised gain: $144.50
```

Reading the output. Step 1: the 200,000 check fails the 100,000 transfer limit. Step 3 reproduces the course's example: a market order for 100 fills as 70 @ 10.00 and 30 @ 10.05 and creates **two lots**. Notice the course's warning in action: the order reserved $1,000.00 (100 × last price 10.00) but cost $1,001.50 — a market order can spend more than was reserved. Step 4: a limit buy waits until the price tick to 9.40. Step 5 sells 120 shares FIFO across three lots. Step 6: the stop does nothing at 10.50 and fires at 9.95. Step 7: the stop-limit triggers at 310 but its 306 limit blocks the fill until 305.50. Step 8: the reserve blocks a second $2,000 order; cancelling returns the money. Step 10: the tax statement lists the gain **per lot**.

Part B — Matching engine (`lld.ch19.stockexchange`)

Files: `OrderBook.java` (side, order, trade, price policy, book), `MatchingEngine.java` (routing, per-symbol locking, line parser), `MatchingEngineTests.java` (nine tests, standard library only), `Main.java` (runs the sample — or a file given as the first argument — then the tests).

```java
// File: lld/ch19/stockexchange/Main.java
package lld.ch19.stockexchange;

import java.io.IOException;
import java.nio.file.Files;
import java.nio.file.Path;
import java.util.List;

/** Usage: java lld.ch19.stockexchange.Main [orders.txt]  (no file = the sample from the problem
statement). */
public class Main {
    static final List<String> SAMPLE = List.of(
            "#1 09:45 BAC sell 240.12 100",
            "#2 09:46 BAC sell 237.45 90",
            "#3 09:47 BAC buy 238.10 110",
            "#4 09:48 BAC buy 237.80 10",
            "#5 09:49 BAC buy 237.80 40",
            "#6 09:50 BAC sell 236.00 50");

    public static void main(String[] args) throws IOException {
        List<String> lines = args.length > 0 ? Files.readAllLines(Path.of(args[0])) : SAMPLE;

        System.out.println("== Trades (execution price = sell order's price) ==");
        MatchingEngine engine = new MatchingEngine(ExecutionPricePolicy.SELL_PRICE,
System.out::println);
        lines.stream().filter(l -> !l.isBlank()).forEach(engine::submitLine);
        System.out.println("book after: " + engine.book("BAC").snapshot());

        System.out.println("\n== Same input, real-exchange rule (price of the resting order) ==");
        MatchingEngine real = new MatchingEngine(ExecutionPricePolicy.RESTING_PRICE,
System.out::println);
        lines.stream().filter(l -> !l.isBlank()).forEach(real::submitLine);

        System.out.println("\n== Tests ==");
        int failed = MatchingEngineTests.runAll();
        if (failed > 0) System.exit(1);
    }
}
```

```java
// File: lld/ch19/stockexchange/MatchingEngine.java
package lld.ch19.stockexchange;

import java.math.BigDecimal;
import java.time.LocalTime;
import java.util.List;
import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.atomic.AtomicLong;
import java.util.function.Consumer;

/**
 * Routes each order to its symbol's book. One lock per book: orders for different
 * symbols match in parallel, orders for the same symbol are strictly serialised
 * (which is what makes "time priority" well defined).
 */
final class MatchingEngine {
    private final Map<String, OrderBook> books = new ConcurrentHashMap<>();
    private final Map<String, String> symbolOfOrder = new ConcurrentHashMap<>();
    private final AtomicLong sequence = new AtomicLong();
    private final ExecutionPricePolicy policy;
    private final Consumer<Trade> tradeSink;

    MatchingEngine(ExecutionPricePolicy policy, Consumer<Trade> tradeSink) {
        this.policy = policy;
        this.tradeSink = tradeSink;
    }

    List<Trade> submit(String id, LocalTime time, String symbol, Side side, BigDecimal price, int qty)
{
        if (qty <= 0 || price.signum() <= 0) throw new IllegalArgumentException("bad quantity/price: "
+ id);
        if (symbolOfOrder.putIfAbsent(id, symbol) != null) throw new
IllegalArgumentException("duplicate order id " + id);
        OrderBook book = books.computeIfAbsent(symbol, s -> new OrderBook(s, policy));
        List<Trade> trades;
        synchronized (book) {
            // the sequence is taken inside the lock, so arrival order == matching order
            Order o = new Order(id, time, symbol, side, price, qty, sequence.incrementAndGet());
            trades = book.add(o);
        }
        trades.forEach(tradeSink);        // publish outside the lock
        return trades;
    }

    boolean cancel(String orderId) {
        String symbol = symbolOfOrder.get(orderId);
        if (symbol == null) return false;
        OrderBook book = books.get(symbol);
        synchronized (book) { return book.cancel(orderId); }
    }

    OrderBook book(String symbol) { return books.get(symbol); }

    /** Parses "#1 09:45 BAC sell 240.12 100" and submits it. */
    List<Trade> submitLine(String line) {
        String[] f = line.trim().split("\\s+");
        if (f.length != 6) throw new IllegalArgumentException("bad line: " + line);
        return submit(f[0], LocalTime.parse(f[1]), f[2], Side.valueOf(f[3].toUpperCase()),
                new BigDecimal(f[4]), Integer.parseInt(f[5]));
```

```
    }
}
```

```java
// File: lld/ch19/stockexchange/MatchingEngineTests.java
package lld.ch19.stockexchange;

import java.math.BigDecimal;
import java.time.LocalTime;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.concurrent.CountDownLatch;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.TimeUnit;
import java.util.concurrent.atomic.AtomicInteger;

/** Tiny test harness (no JUnit - standard library only). Each test throws AssertionError on failure.
*/
final class MatchingEngineTests {
    private MatchingEngineTests() { }

    interface Test { void run() throws Exception; }

    record Case(String name, Test body) { }

    static int runAll() {
        List<Case> tests = List.of(
                new Case("sample input gives the expected four trades", MatchingEngineTests::sample),
                new Case("better price wins over earlier time", MatchingEngineTests::pricePriority),
                new Case("same price: earlier order fills first", MatchingEngineTests::timePriority),
                new Case("no trade when prices do not cross", MatchingEngineTests::noCross),
                new Case("partial fill leaves the remainder resting",
MatchingEngineTests::partialFill),
                new Case("cancelled order is never matched", MatchingEngineTests::cancel),
                new Case("symbols never match each other", MatchingEngineTests::symbolIsolation),
                new Case("duplicate id and bad quantity rejected", MatchingEngineTests::validation),
                new Case("concurrent submits conserve quantity", MatchingEngineTests::concurrency));
        int failed = 0;
        for (Case t : tests) {
            try {
                t.body().run();
                System.out.println("  PASS  " + t.name());
            } catch (Exception | AssertionError e) {
                failed++;
                System.out.println("  FAIL  " + t.name() + " -> " + e.getMessage());
            }
        }
        System.out.println("  " + (tests.size() - failed) + "/" + tests.size() + " passed");
        return failed;
    }

    // ---------- helpers ----------
    private static MatchingEngine engine(List<Trade> sink) {
        return new MatchingEngine(ExecutionPricePolicy.SELL_PRICE, sink::add);
    }

    private static void eq(Object expected, Object actual) {
        if (!expected.equals(actual)) throw new AssertionError("expected <" + expected + "> but was <"
+ actual + ">");
    }

    private static List<String> str(List<Trade> trades) { return
trades.stream().map(Trade::toString).toList(); }
```

```
    private static final LocalTime T = LocalTime.of(10, 0);

    // ---------- tests ----------
    static void sample() {
        List<Trade> out = new ArrayList<>();
        MatchingEngine e = engine(out);
        Main.SAMPLE.forEach(e::submitLine);
        eq(List.of("#3 237.45 90 #2", "#3 236.00 20 #6", "#4 236.00 10 #6", "#5 236.00 20 #6"),
str(out));
    }

    static void pricePriority() {
        List<Trade> out = new ArrayList<>();
        MatchingEngine e = engine(out);
        e.submit("s1", T, "X", Side.SELL, new BigDecimal("101"), 10);   // earlier but worse
        e.submit("s2", T, "X", Side.SELL, new BigDecimal("100"), 10);   // later but cheaper
        e.submit("b1", T, "X", Side.BUY, new BigDecimal("105"), 10);
        eq(List.of("b1 100 10 s2"), str(out));
    }

    static void timePriority() {
        List<Trade> out = new ArrayList<>();
        MatchingEngine e = engine(out);
        e.submit("b1", T, "X", Side.BUY, new BigDecimal("50"), 5);
        e.submit("b2", T, "X", Side.BUY, new BigDecimal("50"), 5);
        e.submit("s1", T, "X", Side.SELL, new BigDecimal("49"), 7);
        eq(List.of("b1 49 5 s1", "b2 49 2 s1"), str(out));
        eq(3, e.book("X").restingQty(Side.BUY));
    }

    static void noCross() {
        List<Trade> out = new ArrayList<>();
        MatchingEngine e = engine(out);
        e.submit("b1", T, "X", Side.BUY, new BigDecimal("99.99"), 5);
        e.submit("s1", T, "X", Side.SELL, new BigDecimal("100.00"), 5);
        eq(0, out.size());
        eq(5, e.book("X").restingQty(Side.BUY));
        eq(5, e.book("X").restingQty(Side.SELL));
    }

    static void partialFill() {
        List<Trade> out = new ArrayList<>();
        MatchingEngine e = engine(out);
        e.submit("s1", T, "X", Side.SELL, new BigDecimal("10"), 100);
        e.submit("b1", T, "X", Side.BUY, new BigDecimal("10"), 30);
        e.submit("b2", T, "X", Side.BUY, new BigDecimal("10"), 30);
        eq(40, e.book("X").restingQty(Side.SELL));
        eq(0, e.book("X").restingQty(Side.BUY));
    }

    static void cancel() {
        List<Trade> out = new ArrayList<>();
        MatchingEngine e = engine(out);
        e.submit("s1", T, "X", Side.SELL, new BigDecimal("10"), 10);
        e.submit("s2", T, "X", Side.SELL, new BigDecimal("10"), 10);
        eq(true, e.cancel("s1"));
        eq(false, e.cancel("s1"));
        e.submit("b1", T, "X", Side.BUY, new BigDecimal("10"), 10);
        eq(List.of("b1 10 10 s2"), str(out));
    }

    static void symbolIsolation() {
```

```
        List<Trade> out = new ArrayList<>();
        MatchingEngine e = engine(out);
        e.submit("s1", T, "AAA", Side.SELL, new BigDecimal("1"), 10);
        e.submit("b1", T, "BBB", Side.BUY, new BigDecimal("5"), 10);
        eq(0, out.size());
    }

    static void validation() {
        MatchingEngine e = engine(new ArrayList<>());
        e.submit("x1", T, "X", Side.BUY, new BigDecimal("1"), 1);
        try { e.submit("x1", T, "X", Side.BUY, new BigDecimal("1"), 1); throw new AssertionError("dup
accepted"); }
        catch (IllegalArgumentException expected) { /* ok */ }
        try { e.submit("x2", T, "X", Side.BUY, new BigDecimal("1"), 0); throw new AssertionError("qty
0 accepted"); }
        catch (IllegalArgumentException expected) { /* ok */ }
    }

    /** 8 threads x 500 orders on one symbol. Every share bought must equal every share sold. */
    static void concurrency() throws InterruptedException {
        List<Trade> out = Collections.synchronizedList(new ArrayList<>());
        MatchingEngine e = engine(out);
        ExecutorService pool = Executors.newFixedThreadPool(8);
        CountDownLatch go = new CountDownLatch(1);
        AtomicInteger buys = new AtomicInteger();
        AtomicInteger sells = new AtomicInteger();
        for (int t = 0; t < 8; t++) {
            int thread = t;
            pool.submit(() -> {
                go.await();
                for (int i = 0; i < 500; i++) {
                    Side side = (i + thread) % 2 == 0 ? Side.BUY : Side.SELL;
                    int qty = 1 + (i % 7);
                    (side == Side.BUY ? buys : sells).addAndGet(qty);
                    e.submit("t" + thread + "-" + i, T, "X", side, new BigDecimal(100 + (i % 3)),
qty);
                }
                return null;
            });
        }
        go.countDown();
        pool.shutdown();
        if (!pool.awaitTermination(30, TimeUnit.SECONDS)) throw new AssertionError("timeout");
        int traded = out.stream().mapToInt(Trade::quantity).sum();
        eq(buys.get(), traded + e.book("X").restingQty(Side.BUY));
        eq(sells.get(), traded + e.book("X").restingQty(Side.SELL));
        // the book must never be left crossed: best bid < best ask
        OrderBook book = e.book("X");
        if (book.bestBid().isPresent() && book.bestAsk().isPresent()
                && book.bestBid().get().compareTo(book.bestAsk().get()) >= 0)
            throw new AssertionError("book left crossed: " + book.snapshot());
    }
}
```

```java
// File: lld/ch19/stockexchange/OrderBook.java
package lld.ch19.stockexchange;

import java.math.BigDecimal;
import java.time.LocalTime;
import java.util.ArrayDeque;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Deque;
import java.util.HashMap;
import java.util.Iterator;
import java.util.List;
import java.util.Map;
import java.util.NavigableMap;
import java.util.Optional;
import java.util.TreeMap;

enum Side { BUY, SELL }

/** An order. Only remainingQty changes, and only under the book's lock. */
final class Order {
    private final String id;
    private final LocalTime time;
    private final String symbol;
    private final Side side;
    private final BigDecimal price;
    private final long sequence;          // arrival order: breaks ties when times are equal
    private int remainingQty;

    Order(String id, LocalTime time, String symbol, Side side, BigDecimal price, int qty, long
sequence) {
        if (qty <= 0) throw new IllegalArgumentException("quantity must be positive: " + id);
        if (price.signum() <= 0) throw new IllegalArgumentException("price must be positive: " + id);
        this.id = id; this.time = time; this.symbol = symbol; this.side = side;
        this.price = price; this.remainingQty = qty; this.sequence = sequence;
    }

    String id() { return id; }
    LocalTime time() { return time; }
    String symbol() { return symbol; }
    Side side() { return side; }
    BigDecimal price() { return price; }
    long sequence() { return sequence; }
    int remainingQty() { return remainingQty; }
    void fill(int qty) { remainingQty -= qty; }
    boolean isDone() { return remainingQty == 0; }
}

/** Output line: "<buy_order_id> <execution_price> <quantity> <sell_order_id>". */
record Trade(String buyOrderId, BigDecimal price, int quantity, String sellOrderId) {
    @Override public String toString() { return buyOrderId + " " + price.toPlainString() + " " +
quantity + " " + sellOrderId; }
}

/** Who sets the trade price. The Navi problem uses the SELL order's price; real exchanges use the
resting order's. */
interface ExecutionPricePolicy {
    BigDecimal price(Order buy, Order sell, Order incoming);

    ExecutionPricePolicy SELL_PRICE = (buy, sell, incoming) -> sell.price();
    ExecutionPricePolicy RESTING_PRICE = (buy, sell, incoming) -> incoming == buy ? sell.price() :
buy.price();
```

```
}

/**
 * The order book for ONE symbol. Each side is a sorted map of price level -> FIFO queue.
 * Bids: highest price first. Asks: lowest price first. Inside a level: first come, first served.
 * That is price-time priority. Not thread-safe by itself: the engine locks it.
 */
final class OrderBook {
    private final String symbol;
    private final NavigableMap<BigDecimal, Deque<Order>> bids = new TreeMap<>
(Collections.reverseOrder());
    private final NavigableMap<BigDecimal, Deque<Order>> asks = new TreeMap<>();
    private final Map<String, Order> resting = new HashMap<>();
    private final ExecutionPricePolicy pricePolicy;

    OrderBook(String symbol, ExecutionPricePolicy pricePolicy) {
        this.symbol = symbol;
        this.pricePolicy = pricePolicy;
    }

    /** Match the incoming order against the other side, then rest whatever is left. */
    List<Trade> add(Order incoming) {
        List<Trade> trades = new ArrayList<>();
        NavigableMap<BigDecimal, Deque<Order>> opposite = incoming.side() == Side.BUY ? asks : bids;

        while (!incoming.isDone() && !opposite.isEmpty()) {
            Map.Entry<BigDecimal, Deque<Order>> best = opposite.firstEntry();
            if (!crosses(incoming, best.getKey())) break;          // best price does not meet ours:
stop
            Deque<Order> queue = best.getValue();
            Iterator<Order> it = queue.iterator();
            while (!incoming.isDone() && it.hasNext()) {
                Order restingOrder = it.next();                        // oldest first = time priority
                int qty = Math.min(incoming.remainingQty(), restingOrder.remainingQty());
                Order buy = incoming.side() == Side.BUY ? incoming : restingOrder;
                Order sell = incoming.side() == Side.SELL ? incoming : restingOrder;
                trades.add(new Trade(buy.id(), pricePolicy.price(buy, sell, incoming), qty,
sell.id()));
                incoming.fill(qty);
                restingOrder.fill(qty);
                if (restingOrder.isDone()) {
                    it.remove();
                    resting.remove(restingOrder.id());
                }
            }
            if (queue.isEmpty()) opposite.pollFirstEntry();
        }
        if (!incoming.isDone()) rest(incoming);
        return trades;
    }

    boolean cancel(String orderId) {
        Order o = resting.remove(orderId);
        if (o == null) return false;
        NavigableMap<BigDecimal, Deque<Order>> side = o.side() == Side.BUY ? bids : asks;
        Deque<Order> q = side.get(o.price());
        q.remove(o);
        if (q.isEmpty()) side.remove(o.price());
        return true;
    }

    int restingQty(Side side) {
```

```
        return (side == Side.BUY ? bids : asks).values().stream()
                .flatMap(Deque::stream).mapToInt(Order::remainingQty).sum();
    }

    Optional<BigDecimal> bestBid() { return bids.isEmpty() ? Optional.empty() :
Optional.of(bids.firstKey()); }

    Optional<BigDecimal> bestAsk() { return asks.isEmpty() ? Optional.empty() :
Optional.of(asks.firstKey()); }

    String snapshot() {
        return symbol + " bids=" + levels(bids) + " asks=" + levels(asks);
    }

    private static boolean crosses(Order incoming, BigDecimal bestOpposite) {
        return incoming.side() == Side.BUY
                ? incoming.price().compareTo(bestOpposite) >= 0     // buy meets or beats the ask
                : incoming.price().compareTo(bestOpposite) <= 0;    // sell meets or undercuts the bid
    }

    private void rest(Order o) {
        (o.side() == Side.BUY ? bids : asks).computeIfAbsent(o.price(), p -> new ArrayDeque<>
()).addLast(o);
        resting.put(o.id(), o);
    }

    private static String levels(NavigableMap<BigDecimal, Deque<Order>> side) {
        List<String> out = new ArrayList<>();
        side.forEach((p, q) -> out.add(p.toPlainString() + "x" +
q.stream().mapToInt(Order::remainingQty).sum()));
        return out.toString();
    }
}
```

Output:

```
== Trades (execution price = sell order's price) ==
#3 237.45 90 #2
#3 236.00 20 #6
#4 236.00 10 #6
#5 236.00 20 #6
book after: BAC bids=[237.80x20] asks=[240.12x100]

== Same input, real-exchange rule (price of the resting order) ==
#3 237.45 90 #2
#3 238.10 20 #6
#4 237.80 10 #6
#5 237.80 20 #6

== Tests ==
  PASS  sample input gives the expected four trades
  PASS  better price wins over earlier time
  PASS  same price: earlier order fills first
  PASS  no trade when prices do not cross
  PASS  partial fill leaves the remainder resting
  PASS  cancelled order is never matched
  PASS  symbols never match each other
  PASS  duplicate id and bad quantity rejected
  PASS  concurrent submits conserve quantity
  9/9 passed
```

Reading the output. The first block matches the expected output in the problem statement exactly. The second block runs the same orders with the real-exchange rule: #6 (the incoming sell) trades at each *resting* buyer's price (238.10, 237.80), which is how a continuous limit order book normally prices trades. The tests cover price priority, time priority, no-cross, partial fills, cancel, symbol isolation, validation, and a concurrency test (8 threads × 500 orders) that checks quantity is conserved and the book is never left crossed (best bid must stay below best ask).

Concurrency & edge cases

Risk	Where	Prevention
Over-spending with many open orders	Brokerage	`reserveFor` (synchronized on the member) subtracts all reservations from cash before accepting a new buy.
Selling the same shares twice	Brokerage	Open sell orders are tracked; free shares = position − remaining quantity of open sells *of that symbol*.
Deadlock between member and exchange	Brokerage	Lock order is fixed: the member never calls the exchange while holding its own lock (`place` reserves under the lock, then submits outside it). The exchange may call the member (exchange → member), never the reverse.
Market order costs more than reserved	Brokerage	Known gap (shown in step 3). Fix: reserve a buffer (e.g. +5%) or reject prices too far above the last price ("price band").
Time in force not enforced	Brokerage	The enum has five values (GTC, fill-or-kill, immediate-or-cancel, on-the-open, on-the-close) and no code reads them — every order behaves as good-till-cancelled within the demo. A GTC order should survive into the next trading day; FOK must be cancelled if it cannot be filled in full; IOC cancels whatever is left after one pass. All of this belongs in the exchange, next to the fill loop.
Two orders at the same price arrive "together"	Engine	Whichever thread gets the book's lock first is appended to the level's queue first, so lock-acquisition order = time priority. The `sequence` number is taken inside the same lock, so it records exactly that order (useful for logs and audits).
Cancel racing a match	Engine	Cancel takes the same per-symbol lock; an order is either matched or cancelled, never both.
Duplicate id / zero quantity / negative price	Engine	Validated before the id is registered (test `validation`).
Trade publishing slowing matching	Engine	Trades are collected under the lock and published after releasing it.
Floating-point prices	Both	`BigDecimal` everywhere; `TreeMap` keys compare with `compareTo`, so `237.8` and `237.80` land on the same level.

Real-world example

Indian brokers such as Zerodha offer the same stop order types to retail users (individual investors): **SL** (stop-loss *limit*) and **SL-M** (stop-loss *market*) — our `StopLimitOrder` and `StopLossOrder`. (Which segments allow SL-M has changed over time, so check the broker's current rules.) Their orders go to exchanges such as NSE (National Stock Exchange of India), whose continuous-trading order book matches on price-time priority, like our `OrderBook`. At the extreme end of performance, the LMAX exchange (described in Martin Fowler's article "The LMAX Architecture") runs its business logic, including matching, on a **single thread** that reads input events from a ring buffer (a fixed-size array used as a circular queue; LMAX's version is called the Disruptor) — the

same idea as our "one lock per symbol, strictly serialised", taken to its logical end. On the tax side, US rules treat shares as sold first-in-first-out unless you identify specific lots, and FIFO is a common broker default — the reason `StockPosition.sell` takes an optional lot id.

Extensions & follow-up questions

- **"Add a trailing-stop order."** New subclass: it tracks the best price seen and moves its stop by a fixed distance. `canTradeAt` changes; nothing else does.
- **"Implement IOC and FOK."** IOC: after the fill loop, cancel any remainder. FOK: before filling, sum the quantity available at acceptable prices; if short, cancel without filling.
- **"Price alerts (the 'notify when a stock moves by X%' feature)."** A `PriceAlert` observer registered on `StockInventory`/`tick`; fires once when the move crosses the threshold.
- **"Modify an order."** On many exchanges, reducing quantity keeps time priority, but changing price (or increasing quantity) loses it — implement as cancel + new order.
- **"Make the engine faster."** Single writer thread per symbol fed by a queue (no locks on the hot path — the code that runs for every single order), pre-allocated objects, prices as `long` ticks instead of `BigDecimal`.
- **"Market orders in the engine."** Treat as a limit at +∞ (buy) or 0 (sell); never rest the remainder — cancel it.
- **"Settlement."** Trades settle later (for example T+1: one business day after the trade); add `unsettledCash` to the member so sale proceeds are not withdrawable immediately.

In the interview

30–45 minute plan (brokerage)

1. Clarify order types, lots, transfers, reports (3 min).
2. Actors and use cases (3 min).
3. Say the key sentence: *"The four order types differ in exactly two things: whether they may trade yet, and the worst price they accept."* Draw `Order` + four tiny subclasses (8 min).
4. Explain **lots** and why they exist: tax depends on which shares you sold (4 min).
5. Explain **reserved funds** (3 min).
6. Code `canTradeAt` / `limitPrice` for the four types and `StockPosition.sell` (10 min).
7. Observer callback, statements, extensions (5 min).

Machine-coding (Navi) plan, 90 minutes: parse lines → `Order`; `OrderBook` with two `TreeMap`s of `ArrayDeque`; `add()` matching loop; print trades; then tests for the sample, price priority, time priority, partial fill and cancel. Get the sample passing first, then refactor (restructure the code without changing what it does).

Follow-ups with answers

- *Why not four unrelated order classes?* They share member, symbol, quantity, side, status and fills; duplicating that invites bugs. The hierarchy isolates the two real differences.
- *Why is `StockInventory` separate from `StockExchange`?* One place to read prices, so no two components disagree about a price.
- *What data structure for the book, and the complexity?* `TreeMap` of price levels (a red-black tree — a self-balancing sorted tree): finding the best level with `firstEntry` and inserting a new level are both O(log L) for L price levels; FIFO queue per level: append O(1), match O(1) per fill. Cancel: O(log L) to find the level + O(k) in that level (use a linked structure with a handle for O(1)).
- *At what price does a trade execute?* In the Navi statement: the sell order's price. In real continuous trading: the resting order's price — the incoming order may get a better price than it asked for ("price improvement").

Self-check

1. **Q:** A SELL stop-loss at 10.00; the price goes 10.50 → 9.95 → 10.20. When does it trade? **A:** At the 9.95 tick. It stays triggered even when the price goes back up.
2. **Q:** What does a stop-limit add to a stop-loss? **A:** A worst acceptable price after triggering — it becomes a limit order, not a market order.
3. **Q:** Why does one market order create two lots in step 3? **A:** It filled at two prices, and a lot is defined by its purchase price.
4. **Q:** Two bids at 50.00 arrive, b1 then b2; a sell for 7 at 49 arrives. Trades? **A:** b1 gets 5, b2 gets 2 — time priority within the level (test `timePriority`).
5. **Q:** What does "the book is crossed" mean and why is it a bug? **A:** Best bid ≥ best ask; those two orders should already have traded.
6. **Q:** Which pattern produces the tax and quarterly statements? **A:** Template Method — shared filtering in `Statement.render`, specific lines in `body`.

TL;DR

- **Brokerage:** one abstract `Order`, four tiny subclasses answering *may I trade now?* and *worst price?*; fills are `OrderPart`s; purchases are **lots**; money is **reserved** at placement; the exchange calls back via **Observer**; statements are a **Template Method**.
- **Matching engine:** per symbol, two `TreeMap<price, FIFO queue>`; match best price first, oldest first; rest the remainder; one lock per symbol; trade price is a pluggable policy (Navi: sell price; real exchanges: resting price).

Design an Online Auction System

Problem & clarifying questions

An online auction (think eBay auctions) lets a **seller** list an item with a starting price and an end time. **Bidders** offer higher and higher amounts. When time runs out, the highest bidder wins — if the seller's hidden minimum (the **reserve price**) was reached.

Question	Assumed answer
What does a listing contain?	Item name, description, category, starting price, minimum bid increment, optional reserve price, duration.
What is a valid bid?	The first bid must be at least the starting price; later bids must beat the current highest by at least the increment.
Can the seller bid? Can the leader outbid themselves?	No and no.
When does an auction end?	At its end time. A late bid is refused even if the closing job has not run yet.
Who is told what?	The outbid bidder immediately; the winner and seller at close; the seller if unsold.
Proxy (automatic) bidding, anti-sniping (stopping last-second bids from stealing a win), payments?	Extensions — mention them.

Requirements

Functional

1. Users register and log in. (We model registration; login itself is the usual authentication (checking who a user is) service and is not coded here.)
2. Users create listings with item name, description, starting price and duration.
3. Users browse and search listings by name, category and price range.
4. Users place bids on **active** listings.
5. The system keeps the **current highest bid** up to date and notifies bidders.
6. When the duration ends, the auction closes and the **highest bidder wins** (if the reserve is met).

Non-functional

7. Many users bid on the same listing at the same moment; the system must stay consistent (one true highest bid).
8. Extensible: new bid rules, notification channels or search criteria without rewriting the core.

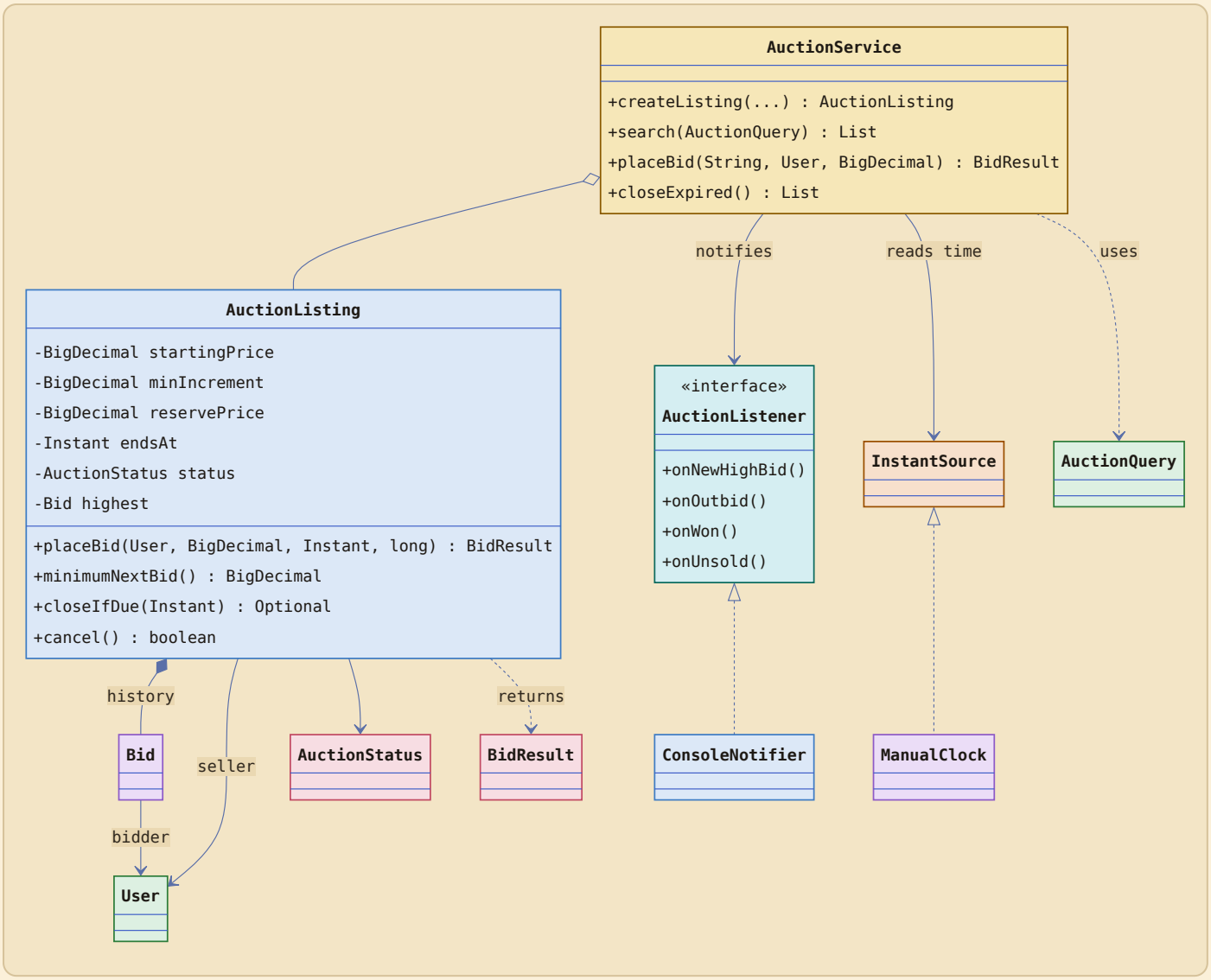
Use cases

Actor	Use cases
Seller	Create listing, cancel listing (only while nobody has bid), see result
Bidder	Search, view listing and bid history, place bid, be told when outbid or when won
System (scheduler)	Activate scheduled auctions, close expired ones, declare winner, send notifications

Core entities

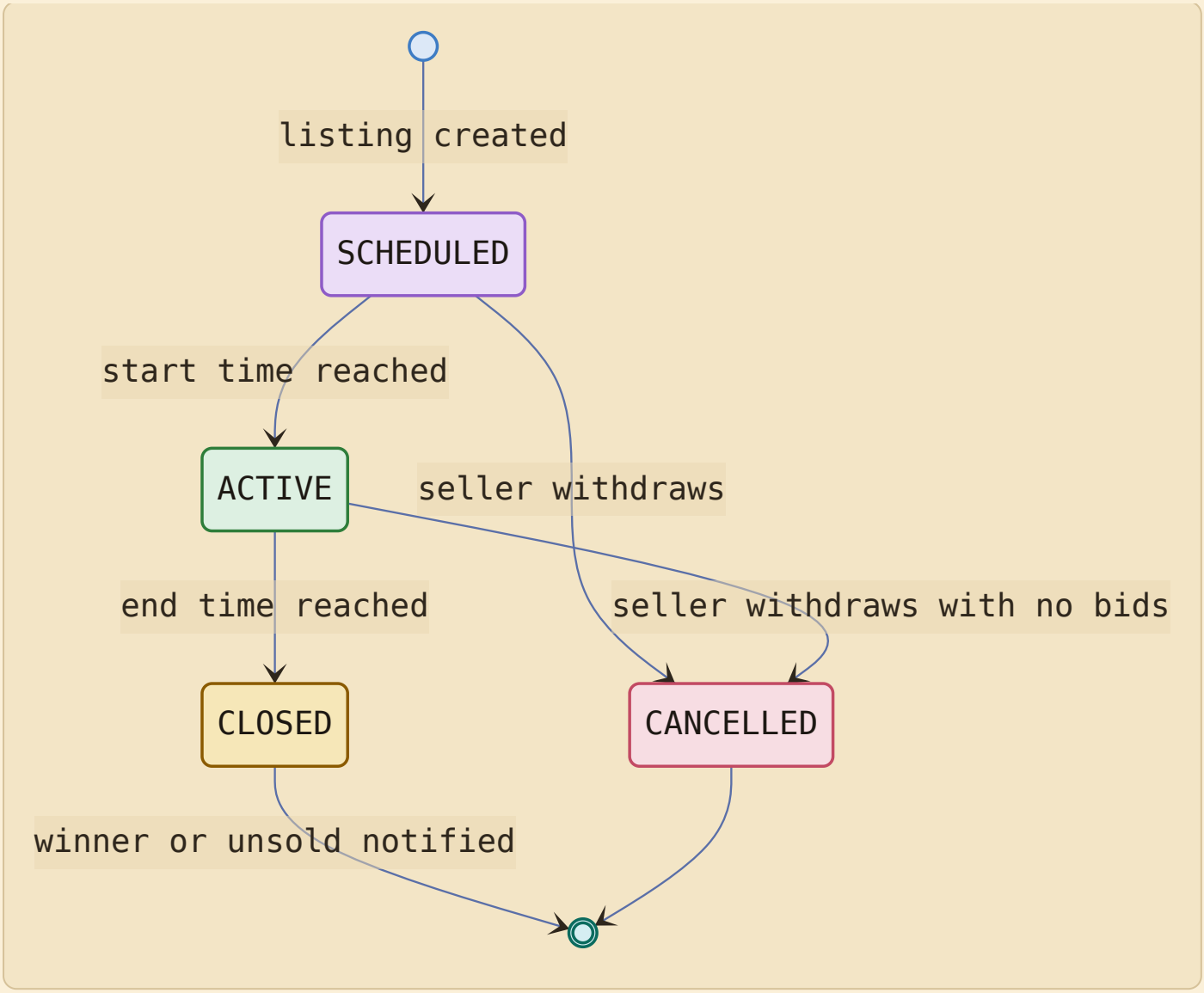
Class	Responsibility
`User`	id, name, email (a `record`).
`Bid`	bidder, amount (`BigDecimal`), time, sequence number. Immutable.
`AuctionStatus`	`SCHEDULED → ACTIVE → CLOSED`, or `CANCELLED`.
`AuctionListing`	Item details, seller, prices, times, bid history, highest bid. **Owns the bidding rules** and guards them with its own lock.
`BidOutcome` / `BidResult`	Why a bid was accepted or refused, the next minimum bid, the accepted bid, who was outbid.
`CloseResult`	Listing, winning bid (if any), whether the reserve was met.
`AuctionQuery`	Fluent (calls chained one after another) search criteria; each criterion is a `Predicate` (a function that answers true/false; Specification pattern).
`AuctionListener` / `ConsoleNotifier`	Observer: new high bid, outbid, won, unsold.
`AuctionService`	Facade: register, create, search, bid, close expired. Calls listeners **outside** listing locks.
`ManualClock`	An `InstantSource` (Java's "what time is it now?" interface) we can move forward in demos and tests.

Class diagram



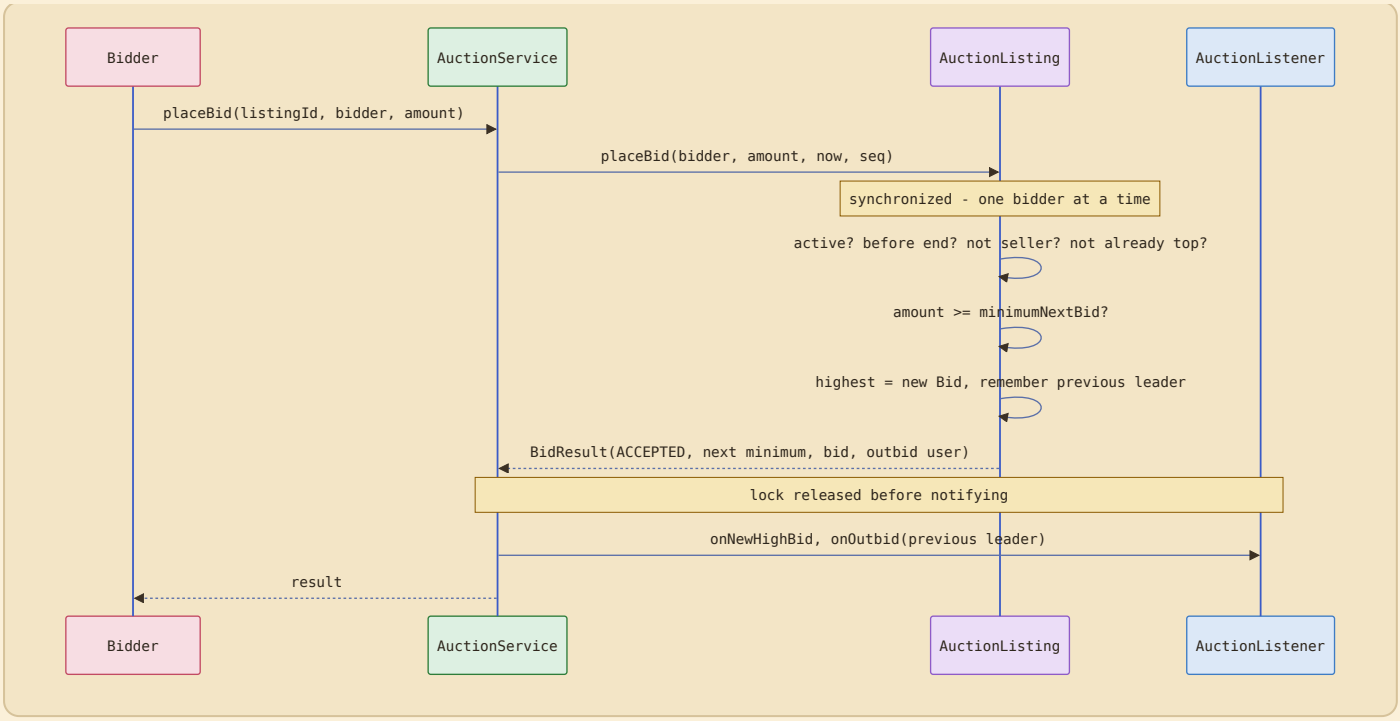
Key flows

Auction lifecycle



`SCHEDULED → ACTIVE` happens **lazily** (only when someone next touches the listing, not at the exact moment): every method first calls `refresh(now)`. No background thread is needed just to flip the state.

Placing a bid (the critical section) — a **critical section** is a piece of code that only one thread may run at a time because it reads and changes shared data.



Closing. A scheduler (in production a `ScheduledExecutorService` — Java's built-in timer that runs a task repeatedly — running every few seconds) calls `closeExpired()`. Each listing's `closeIfDue(now)` atomically moves `ACTIVE → CLOSED` and returns the winner only if the reserve is met. Notifications go out after all locks are released.

Design decisions & patterns used

Decision	Why	Alternative rejected
Bid rules live inside `AuctionListing` and run under its lock (a *monitor*: an object whose methods are all `synchronized`)	"Read highest, compare, replace" must be one atomic step. The lock is per listing, so bids on different auctions never wait for each other.	A global lock in the service (all auctions serialised); or `AtomicReference<Bid>` (a reference that can be swapped atomically) + CAS loop (retry compare-and-set until it succeeds) — valid, but the rule set (seller, leader, end time, increment) is easier to get right under one lock.
Refusals are returned (`BidOutcome`), not thrown	A too-low bid is normal business, not an exceptional error (an **exception** is Java's way of aborting a method to signal that something went wrong). The caller also gets the next minimum to show in the UI.	Throwing an exception for every low bid.
Observer for notifications, called outside the lock	Slow email or push calls must not block other bidders. `BidResult` carries a **snapshot** (a copy frozen at one moment) of the accepted bid taken inside the lock, so listeners never read a newer, unrelated value.	Notifying inside `synchronized` (slow, and risks deadlock if a listener calls back).
Minimum increment	Stops 1-cent "wars" and makes the next minimum predictable.	Any higher amount accepted.
Reserve price	Lets a seller refuse to sell too cheaply; checked only at close.	Rejecting bids below the reserve (would reveal it).
`InstantSource` injected	Time-dependent code becomes testable: the demo moves a `ManualClock` forward instead of sleeping.	Calling `Instant.now()` everywhere.
Specification pattern for search	Each criterion is a `Predicate<AuctionListing>`; they combine with `and`. New criteria = one method.	One `search(name, category, min, max, ...)` method whose parameter list keeps growing.
The service is not a Singleton	awesome-low-level-design's version uses a Singleton `AuctionSystem`; we pass one instance around instead so tests can create fresh ones.	Global state shared by all tests.

Java implementation

Files: `Auction.java` (user, bid, status, outcome records, listing, manual clock), `AuctionService.java` (listener, notifier, query, service), `Main.java`.

```java
// File: lld/ch19/auction/Auction.java
package lld.ch19.auction;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.time.Duration;
import java.time.Instant;
import java.time.InstantSource;
import java.util.ArrayList;
import java.util.List;
import java.util.Optional;

record User(String id, String name, String email) { }

/** One bid. seq breaks ties between bids with the same timestamp. */
record Bid(User bidder, BigDecimal amount, Instant at, long seq) { }

enum AuctionStatus { SCHEDULED, ACTIVE, CLOSED, CANCELLED }

/** Why a bid was refused - returned, not thrown, because refusals are normal business. */
enum BidOutcome { ACCEPTED, AUCTION_NOT_ACTIVE, AUCTION_ENDED, SELLER_CANNOT_BID, TOO_LOW,
ALREADY_HIGHEST }

/** Result of placeBid: what happened, the bid that was accepted, who (if anyone) was just outbid. */
record BidResult(BidOutcome outcome, BigDecimal minimumNextBid, Optional<Bid> accepted, Optional<User>
outbid) { }

/** Result of closing: sold to a winner, or unsold. */
record CloseResult(AuctionListing listing, Optional<Bid> winningBid, boolean reserveMet) { }

/**
 * One auction. All state changes go through synchronized methods, so the
 * "read highest bid, compare, replace" sequence is atomic: two bidders can never
 * both believe they are the highest.
 */
final class AuctionListing {
    private final String id;
    private final String itemName;
    private final String description;
    private final String category;
    private final User seller;
    private final BigDecimal startingPrice;
    private final BigDecimal minIncrement;
    private final BigDecimal reservePrice;          // hidden minimum; ZERO = no reserve
    private final Instant startsAt;
    private final Instant endsAt;
    private AuctionStatus status = AuctionStatus.SCHEDULED;
    private final List<Bid> bids = new ArrayList<>();
    private Bid highest;

    AuctionListing(String id, String itemName, String description, String category, User seller,
                   BigDecimal startingPrice, BigDecimal minIncrement, BigDecimal reservePrice,
                   Instant startsAt, Duration duration) {
        if (startingPrice.signum() <= 0 || minIncrement.signum() <= 0) throw new
IllegalArgumentException("prices must be > 0");
        this.id = id; this.itemName = itemName; this.description = description; this.category =
category;
        this.seller = seller; this.startingPrice = money(startingPrice); this.minIncrement =
money(minIncrement);
        this.reservePrice = money(reservePrice); this.startsAt = startsAt; this.endsAt =
startsAt.plus(duration);
    }
```

```
    /** Lazily moves SCHEDULED -> ACTIVE when the start time has passed. */
    private void refresh(Instant now) {
        if (status == AuctionStatus.SCHEDULED && !now.isBefore(startsAt)) status =
AuctionStatus.ACTIVE;
    }

    synchronized BidResult placeBid(User bidder, BigDecimal rawAmount, Instant now, long seq) {
        refresh(now);
        BigDecimal amount = money(rawAmount);
        if (status != AuctionStatus.ACTIVE) return refuse(BidOutcome.AUCTION_NOT_ACTIVE);
        if (!now.isBefore(endsAt)) return refuse(BidOutcome.AUCTION_ENDED);  // closing job may not
have run yet
        if (bidder.id().equals(seller.id())) return refuse(BidOutcome.SELLER_CANNOT_BID);
        if (highest != null && highest.bidder().id().equals(bidder.id())) return
refuse(BidOutcome.ALREADY_HIGHEST);
        if (amount.compareTo(minimumNextBid()) < 0) return refuse(BidOutcome.TOO_LOW);

        Optional<User> outbid = Optional.ofNullable(highest).map(Bid::bidder);
        highest = new Bid(bidder, amount, now, seq);
        bids.add(highest);
        return new BidResult(BidOutcome.ACCEPTED, minimumNextBid(), Optional.of(highest), outbid);
    }

    /** First bid must reach the starting price; later bids must beat the leader by the increment. */
    synchronized BigDecimal minimumNextBid() {
        return highest == null ? startingPrice : highest.amount().add(minIncrement);
    }

    /** ACTIVE -> CLOSED once time is up. Returns empty if not yet due (or already closed). */
    synchronized Optional<CloseResult> closeIfDue(Instant now) {
        refresh(now);
        if (status != AuctionStatus.ACTIVE || now.isBefore(endsAt)) return Optional.empty();
        status = AuctionStatus.CLOSED;
        boolean reserveMet = highest != null && highest.amount().compareTo(reservePrice) >= 0;
        return Optional.of(new CloseResult(this, reserveMet ? Optional.of(highest) : Optional.empty(),
reserveMet));
    }

    /** A seller may withdraw only while nobody has bid. */
    synchronized boolean cancel() {
        if (status == AuctionStatus.CLOSED || status == AuctionStatus.CANCELLED || highest != null)
return false;
        status = AuctionStatus.CANCELLED;
        return true;
    }

    private BidResult refuse(BidOutcome why) { return new BidResult(why, minimumNextBid(),
Optional.empty(), Optional.empty()); }

    private static BigDecimal money(BigDecimal b) { return b.setScale(2, RoundingMode.HALF_EVEN); }

    String id() { return id; }
    String itemName() { return itemName; }
    String description() { return description; }
    String category() { return category; }
    User seller() { return seller; }
    BigDecimal startingPrice() { return startingPrice; }
    synchronized AuctionStatus status(Instant now) { refresh(now); return status; }
    Instant endsAt() { return endsAt; }
    synchronized Optional<Bid> highestBid() { return Optional.ofNullable(highest); }
    synchronized BigDecimal currentPrice() { return highest == null ? startingPrice :
```

```
highest.amount(); }
    synchronized List<Bid> bidHistory() { return List.copyOf(bids); }

    @Override public String toString() { return id + " '" + itemName + "' now " + currentPrice(); }
}

/** A controllable clock for demos and tests (production code passes InstantSource.system()). */
final class ManualClock implements InstantSource {
    private Instant now;
    ManualClock(Instant start) { now = start; }
    @Override public synchronized Instant instant() { return now; }
    synchronized void advance(Duration d) { now = now.plus(d); }
}
```

```java
// File: lld/ch19/auction/AuctionService.java
package lld.ch19.auction;

import java.math.BigDecimal;
import java.time.Duration;
import java.time.InstantSource;
import java.util.ArrayList;
import java.util.Comparator;
import java.util.List;
import java.util.Locale;
import java.util.Map;
import java.util.Optional;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.CopyOnWriteArrayList;
import java.util.concurrent.atomic.AtomicInteger;
import java.util.concurrent.atomic.AtomicLong;
import java.util.function.Predicate;

/** Observer: who wants to hear about bids and results. */
interface AuctionListener {
    default void onNewHighBid(AuctionListing a, Bid bid) { }
    default void onOutbid(AuctionListing a, User outbid, BigDecimal newPrice) { }
    default void onWon(AuctionListing a, Bid winning) { }
    default void onUnsold(AuctionListing a, boolean hadBids) { }
}

final class ConsoleNotifier implements AuctionListener {
    private volatile boolean muted;
    void mute(boolean m) { muted = m; }
    private void say(String to, String msg) { if (!muted) System.out.println("    [notify " + to + "]
" + msg); }

    @Override public void onOutbid(AuctionListing a, User u, BigDecimal p) {
        say(u.name(), "you were outbid on " + a.itemName() + ", now " + p);
    }
    @Override public void onWon(AuctionListing a, Bid w) {
        say(w.bidder().name(), "you WON " + a.itemName() + " for " + w.amount());
        say(a.seller().name(), a.itemName() + " sold to " + w.bidder().name());
    }
    @Override public void onUnsold(AuctionListing a, boolean hadBids) {
        say(a.seller().name(), a.itemName() + " unsold" + (hadBids ? " (reserve not met)" : " (no
bids)"));
    }
}

/** Search criteria, built up fluently. Each criterion is a Predicate (Specification pattern). */
final class AuctionQuery {
    private final List<Predicate<AuctionListing>> parts = new ArrayList<>();
    AuctionQuery nameContains(String s) {
        parts.add(a -> a.itemName().toLowerCase(Locale.ROOT).contains(s.toLowerCase(Locale.ROOT)));
return this;
    }
    AuctionQuery category(String c) { parts.add(a -> a.category().equalsIgnoreCase(c)); return this; }
    AuctionQuery priceBetween(BigDecimal lo, BigDecimal hi) {
        parts.add(a -> a.currentPrice().compareTo(lo) >= 0 && a.currentPrice().compareTo(hi) <= 0);
return this;
    }
    Predicate<AuctionListing> toPredicate() { return parts.stream().reduce(x -> true, Predicate::and);
}
}

/** Facade over listings, bidding and closing. Listeners are called OUTSIDE listing locks. */
```

```
final class AuctionService {
    private final InstantSource clock;
    private final Map<String, User> users = new ConcurrentHashMap<>();
    private final Map<String, AuctionListing> listings = new ConcurrentHashMap<>();
    private final List<AuctionListener> listeners = new CopyOnWriteArrayList<>();
    private final AtomicInteger listingSeq = new AtomicInteger();
    private final AtomicInteger userSeq = new AtomicInteger();
    private final AtomicLong bidSeq = new AtomicLong();

    AuctionService(InstantSource clock) { this.clock = clock; }

    void subscribe(AuctionListener l) { listeners.add(l); }

    User register(String name, String email) {
        User u = new User("U" + userSeq.incrementAndGet(), name, email);
        if (users.putIfAbsent(u.id(), u) != null) throw new IllegalStateException("id clash");
        return u;
    }

    AuctionListing createListing(User seller, String item, String description, String category,
                                 BigDecimal startingPrice, BigDecimal increment, BigDecimal reserve,
Duration duration) {
        AuctionListing a = new AuctionListing("A" + listingSeq.incrementAndGet(), item, description,
category, seller,
                startingPrice, increment, reserve, clock.instant(), duration);
        listings.put(a.id(), a);
        return a;
    }

    List<AuctionListing> search(AuctionQuery q) {
        return listings.values().stream()
                .filter(a -> a.status(clock.instant()) == AuctionStatus.ACTIVE)
                .filter(q.toPredicate())
                .sorted(Comparator.comparing(AuctionListing::endsAt))
                .toList();
    }

    BidResult placeBid(String listingId, User bidder, BigDecimal amount) {
        AuctionListing a = listings.get(listingId);
        if (a == null) throw new IllegalArgumentException("no listing " + listingId);
        BidResult r = a.placeBid(bidder, amount, clock.instant(), bidSeq.incrementAndGet());
        r.accepted().ifPresent(top -> {                 // snapshot taken inside the lock
            listeners.forEach(l -> l.onNewHighBid(a, top));
            r.outbid().ifPresent(u -> listeners.forEach(l -> l.onOutbid(a, u, top.amount())));
        });
        return r;
    }

    /** Run periodically (e.g. by a ScheduledExecutorService). Closes every auction whose time is up.
*/
    List<CloseResult> closeExpired() {
        List<CloseResult> closed = new ArrayList<>();
        for (AuctionListing a : listings.values()) {
            Optional<CloseResult> r = a.closeIfDue(clock.instant());
            r.ifPresent(closed::add);
        }
        for (CloseResult r : closed) {
            if (r.winningBid().isPresent()) listeners.forEach(l -> l.onWon(r.listing(),
r.winningBid().get()));
            else listeners.forEach(l -> l.onUnsold(r.listing(),
r.listing().highestBid().isPresent()));
        }
```

```
        return closed;
    }
}
```

```java
// File: lld/ch19/auction/Main.java
package lld.ch19.auction;

import java.math.BigDecimal;
import java.time.Duration;
import java.time.Instant;
import java.util.ArrayList;
import java.util.List;
import java.util.concurrent.CountDownLatch;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.TimeUnit;
import java.util.concurrent.atomic.AtomicInteger;

public class Main {
    public static void main(String[] args) throws Exception {
        ManualClock clock = new ManualClock(Instant.parse("2026-09-28T10:00:00Z"));
        AuctionService svc = new AuctionService(clock);
        ConsoleNotifier notifier = new ConsoleNotifier();
        svc.subscribe(notifier);

        User sam = svc.register("Sam", "sam@mail.test");
        User ana = svc.register("Ana", "ana@mail.test");
        User bob = svc.register("Bob", "bob@mail.test");

        System.out.println("== 1. Create listings ==");
        AuctionListing guitar = svc.createListing(sam, "Vintage Guitar", "1965 hollow body", "Music",
                bd("100"), bd("10"), bd("0"), Duration.ofHours(2));
        AuctionListing watch = svc.createListing(sam, "Pocket Watch", "Silver, working",
"Collectibles",
                bd("50"), bd("5"), bd("500"), Duration.ofHours(1));
        System.out.println("  " + guitar + ", " + watch);

        System.out.println("\n== 2. Search (name 'watch', price 0-100) ==");
        System.out.println("  " + svc.search(new
AuctionQuery().nameContains("watch").priceBetween(bd("0"), bd("100"))));

        System.out.println("\n== 3. Bidding rules ==");
        show("Ana bids 90 (below start)", svc.placeBid(guitar.id(), ana, bd("90")));
        show("Ana bids 100", svc.placeBid(guitar.id(), ana, bd("100")));
        show("Ana bids 120 (already top)", svc.placeBid(guitar.id(), ana, bd("120")));
        show("Bob bids 105 (increment is 10)", svc.placeBid(guitar.id(), bob, bd("105")));
        show("Bob bids 110", svc.placeBid(guitar.id(), bob, bd("110")));
        show("Sam bids on own item", svc.placeBid(guitar.id(), sam, bd("500")));
        show("Ana bids 150", svc.placeBid(guitar.id(), ana, bd("150")));
        show("Bob bids 60 on the watch", svc.placeBid(watch.id(), bob, bd("60")));

        System.out.println("\n== 4. One hour later: the watch closes, reserve (500) not met ==");
        clock.advance(Duration.ofHours(1));
        svc.closeExpired().forEach(r -> System.out.println("  closed " + r.listing().id() + "
reserveMet=" + r.reserveMet()));

        System.out.println("\n== 5. 50 threads bid on the guitar at once ==");
        List<User> crowd = new ArrayList<>();
        for (int i = 0; i < 50; i++) crowd.add(svc.register("bidder" + i, "b" + i + "@mail.test"));
        notifier.mute(true);                // ~50 outbid messages would drown the output
        AtomicInteger accepted = new AtomicInteger();
        CountDownLatch go = new CountDownLatch(1);
        ExecutorService pool = Executors.newFixedThreadPool(16);
        for (int i = 0; i < crowd.size(); i++) {
            User u = crowd.get(i);
            BigDecimal amount = bd(String.valueOf(160 + i * 10));
```

```
            pool.submit(() -> {
                go.await();
                if (svc.placeBid(guitar.id(), u, amount).outcome() == BidOutcome.ACCEPTED)
accepted.incrementAndGet();
                return null;
            });
        }
        go.countDown();
        pool.shutdown();
        pool.awaitTermination(10, TimeUnit.SECONDS);
        notifier.mute(false);
        Bid top = guitar.highestBid().orElseThrow();
        boolean increasing = true;
        List<Bid> history = guitar.bidHistory();
        for (int i = 1; i < history.size(); i++)
            increasing &= history.get(i).amount().compareTo(history.get(i - 1).amount().add(bd("10")))
>= 0;
        System.out.println("  accepted=" + accepted + " (varies run to run), highest=" + top.amount()
+ " by " + top.bidder().name()
                + ", history strictly increasing by >= increment: " + increasing);

        System.out.println("\n== 6. Time is up: the guitar closes and the winner is told ==");
        clock.advance(Duration.ofHours(1));
        show("late bid", svc.placeBid(guitar.id(), ana, bd("5000")));
        svc.closeExpired();
        System.out.println("  status " + guitar.status(clock.instant()) + ", final price " +
guitar.currentPrice());
    }

    private static BigDecimal bd(String s) { return new BigDecimal(s); }

    private static void show(String what, BidResult r) {
        System.out.println("  " + what + " -> " + r.outcome() + " (next minimum " + r.minimumNextBid()
+ ")");
    }
}
```

Output:

```
== 1. Create listings ==
  A1 'Vintage Guitar' now 100.00, A2 'Pocket Watch' now 50.00

== 2. Search (name 'watch', price 0-100) ==
  [A2 'Pocket Watch' now 50.00]

== 3. Bidding rules ==
  Ana bids 90 (below start) -> TOO_LOW (next minimum 100.00)
  Ana bids 100 -> ACCEPTED (next minimum 110.00)
  Ana bids 120 (already top) -> ALREADY_HIGHEST (next minimum 110.00)
  Bob bids 105 (increment is 10) -> TOO_LOW (next minimum 110.00)
    [notify Ana] you were outbid on Vintage Guitar, now 110.00
  Bob bids 110 -> ACCEPTED (next minimum 120.00)
  Sam bids on own item -> SELLER_CANNOT_BID (next minimum 120.00)
    [notify Bob] you were outbid on Vintage Guitar, now 150.00
  Ana bids 150 -> ACCEPTED (next minimum 160.00)
  Bob bids 60 on the watch -> ACCEPTED (next minimum 65.00)

== 4. One hour later: the watch closes, reserve (500) not met ==
    [notify Sam] Pocket Watch unsold (reserve not met)
  closed A2 reserveMet=false

== 5. 50 threads bid on the guitar at once ==
  accepted=28 (varies run to run), highest=650.00 by bidder49, history strictly increasing by >=
increment: true

== 6. Time is up: the guitar closes and the winner is told ==
  late bid -> AUCTION_ENDED (next minimum 660.00)
    [notify bidder49] you WON Vintage Guitar for 650.00
    [notify Sam] Vintage Guitar sold to bidder49
  status CLOSED, final price 650.00
```

Reading the output. Step 3 exercises every rule: below start, leader re-bidding, below increment, seller bidding. The outbid notices print *before* the bidder's own result line because the service notifies before returning. Step 4 closes the watch as unsold because 60 < the 500 reserve. Step 5 fires 50 concurrent bids: how many are accepted depends on thread timing (a bid that arrives after a higher one is refused as too low), but the final highest is always the maximum, 650.00, and the history is always strictly increasing by at least the increment — the invariant (a rule that must always be true) the lock protects. Step 6 refuses a bid after the end time even though the closing job has not yet run.

Concurrency & edge cases

Case	Handling
Two bidders submit the same amount at the same time	The first one through the lock wins; the second is now below `minimumNextBid()` and gets `TOO_LOW`.
Bid arrives exactly at the end time	`!now.isBefore(endsAt)` → `AUCTION_ENDED`. The end instant is exclusive.
Close job runs late	Bids after `endsAt` are refused anyway, so a late job cannot change the winner.
Close and bid race	Both are `synchronized` on the listing; the bid either lands before the close (and counts) or after (and is refused).
Notification listener is slow or throws	Called outside the lock; in production, publish to a queue so failures are retried without affecting bidding.
Seller cancels after bids	Refused (`cancel()` returns false) — protects bidders.
Money precision	Amounts normalised to scale 2 with `BigDecimal`; compared with `compareTo`.
Very hot auction (thousands of bids per second)	One lock per listing is fine for one JVM (Java Virtual Machine — one running Java process); distributed: a single-writer per auction (partition bids by auction id — split them into separate streams so all bids for one auction go to the same place, e.g. a Kafka partition, Kafka being a widely used event-streaming system), or a conditional update `UPDATE ... SET high = ? WHERE id = ? AND high < ?`.

Real-world example

eBay auctions use **proxy bidding**: you enter the most you are willing to pay, and eBay bids on your behalf only as much as needed to stay ahead, by the current increment. eBay also supports hidden **reserve prices**, and its auctions end at a fixed time, which is why "sniping" (bidding in the last seconds) is common there. Some other auction sites use a **soft close** — a late bid extends the end time — precisely to defeat sniping.

Extensions & follow-up questions

- **"Add proxy (automatic) bidding."** Each bidder may register a `maxAmount`. After any accepted bid, inside the same lock, let the strongest proxy respond with `min(its max, competitor + increment)`. Ties go to the earlier proxy (use `Bid.seq`).
- **"Stop sniping."** Soft close: in `placeBid`, if an accepted bid arrives within the last N minutes, set `endsAt = endsAt + N` (make `endsAt` non-final, change under the same lock).
- **"Buy It Now."** A fixed price that closes the auction immediately if paid before the first bid.
- **"Dutch / sealed-bid / Vickrey auctions."** (Dutch: the price starts high and drops until someone accepts; sealed-bid: bidders cannot see each other's bids.) Extract an `AuctionRules` strategy: `validate(bid)`, `winner(bids)`, `price(bids)`. Vickrey (second-price sealed bid): highest bidder wins but pays the second-highest bid.
- **"Payment after winning."** Winner has 48 hours to pay (new state `AWAITING_PAYMENT`); if not, offer to the second-highest bidder.
- **"Scale the close job."** Keep a priority queue (or a sorted index) by `endsAt` so the scheduler only touches auctions that are due.

In the interview

30–45 minute plan

1. Clarify the bid rules (start price, increment, reserve, seller/leader restrictions) — 4 min.
2. Entities and the lifecycle state machine — 8 min.
3. Code `placeBid` under the listing lock and **say why the lock is per listing** — 12 min.
4. Closing + notifications outside the lock — 6 min.
5. Concurrency discussion and extensions (proxy bidding, soft close) — 10 min.

> **30-second answer:** "Each listing owns its bid rules and its highest bid, and `placeBid` is synchronized on the listing, so compare-and-replace is atomic and different auctions never contend. Refusals return a reason and the next minimum. The service notifies outbid users and, at close, the winner and seller, always after releasing the lock. Time comes from an injected clock, so closing and late bids are testable."

Follow-ups with answers

- *Why not `AtomicReference` + CAS?* Possible for the amount alone, but the rule also checks seller, leader and time. A tiny critical section under one lock is simpler and still fast because the lock is per listing.
- *How do you close auctions on time across servers?* A scheduler that owns a time-ordered index of end times; the close itself is idempotent (`closeIfDue` returns empty the second time), so running it twice is harmless.
- *How do you show a live price to thousands of viewers?* Push `onNewHighBid` events to a pub/sub channel (publish/subscribe: senders post messages to a topic and every subscriber receives them) and WebSockets (a connection the browser keeps open so the server can push data at any time); viewers read a cache (a fast saved copy of data — see System Design Fundamentals), not the lock-holding object.

Self-check

1. **Q:** What two conditions make a bid `TOO_LOW`? **A:** Below the starting price when there is no bid, or below highest + increment otherwise.
2. **Q:** Why are listeners called after `placeBid` returns from the listing? **A:** To keep the lock short and avoid deadlock or slowdown from notification code.
3. **Q:** A bid arrives at 12:00:00.000 and the auction ends at 12:00:00.000. Accepted? **A:** No — the end time is exclusive.
4. **Q:** Where does the reserve price matter? **A:** Only at close: if the highest bid is below it, the item is unsold.
5. **Q:** What is the Specification pattern here? **A:** Search criteria as composable `Predicate`s combined with `and`.

TL;DR

> - One **lock per listing**; all bid rules inside it; refusals are values, not exceptions.
> - Lifecycle `SCHEDULED → ACTIVE → CLOSED | CANCELLED`; activation is lazy, closing is idempotent.
> - **Observer** notifications after the lock; **injected clock** for testable time; **Specification** for search.

Design a Music Streaming Service (Spotify)

Problem & clarifying questions

A music streaming service lets users search a huge catalog, build playlists, and play songs with the usual controls (play, pause, skip, previous, seek, shuffle, repeat). Free and paid plans behave differently. The service suggests new music.

Question	Assumed answer
What can users search?	Songs, albums, artists.
Playlists?	Users create, reorder and edit their own playlists.
Playback controls?	Play, pause, resume, skip, previous, seek; shuffle on/off; repeat off / all / one.
Plans?	FREE (ads, limited skips per hour, no downloads) and PREMIUM (no ads, unlimited skips, offline).
Recommendations?	A stub (a simple placeholder) based on listening history; real ones are an ML (machine learning — software that learns patterns from data) system.
Actual audio streaming, DRM (copy protection), CDNs (content delivery networks — servers near users that hold copies of files)?	Out of scope for LLD (low-level design: classes and their code) — the player models *what* plays, not the bytes.

Requirements

Functional

1. Browse and search songs, albums and artists.
2. Create and manage playlists.
3. User accounts and authentication (we model users and plans; login itself is the usual auth service).
4. Playback controls: play, pause, skip, seek (plus previous, shuffle, repeat).
5. Recommendations based on preferences and listening history.
6. Subscriptions: FREE vs PREMIUM with different limits.

Non-functional

7. Many users stream at once; the shared catalog is read-mostly and thread-safe (works correctly when many threads use it at once).
8. Scales to a large catalog and user base (indexes, not scans, in production).
9. Extensible: social sharing, offline mode, new play orders, new recommenders.

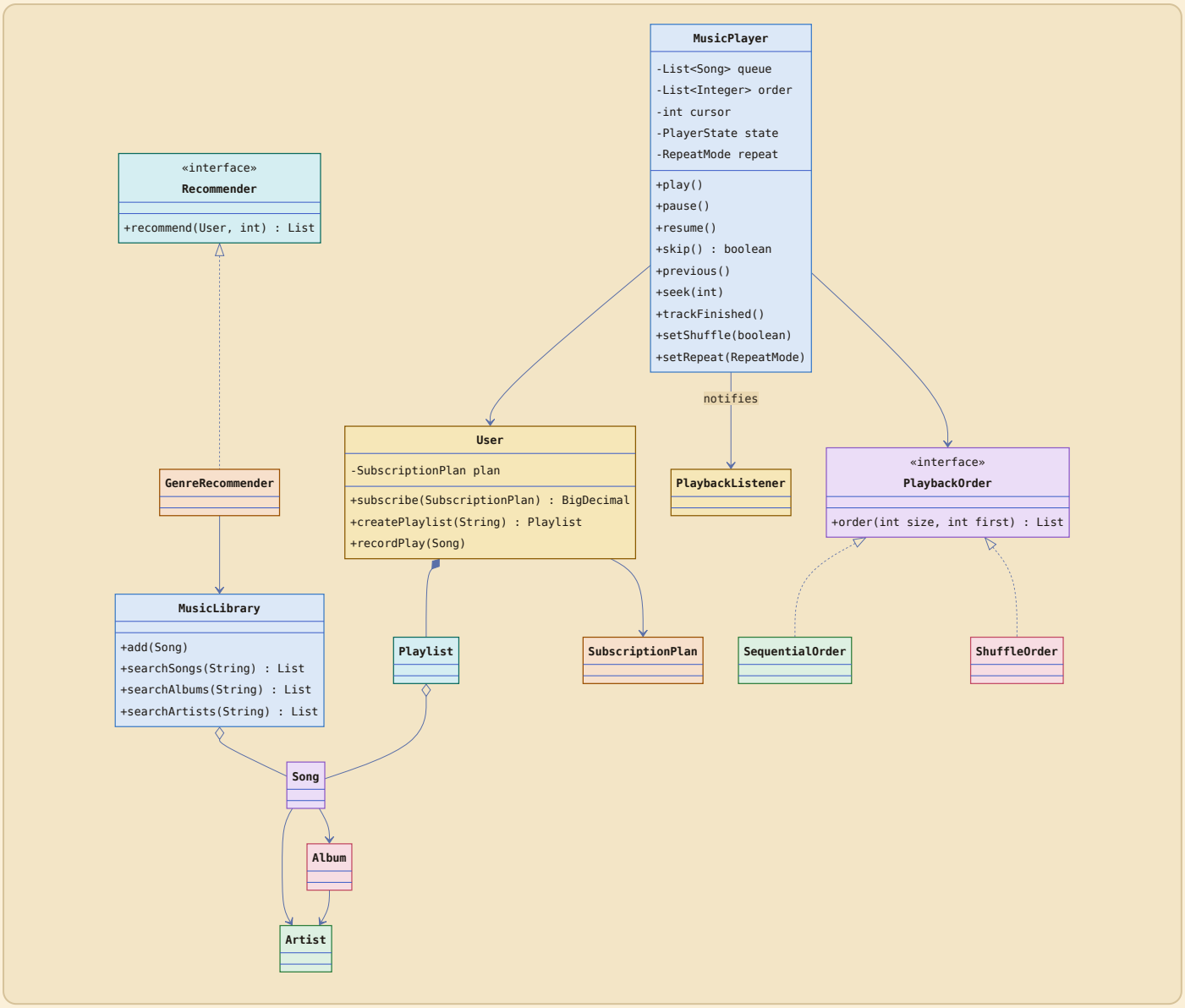
Use cases

Actor	Use cases
Listener	Search, play/pause/skip/seek, shuffle/repeat, create and edit playlists, upgrade plan
Artist / label (content admin)	Add songs and albums to the catalog
System	Record plays (history, royalties — money owed to artists per play), insert ads for free users, recommend songs

Core entities

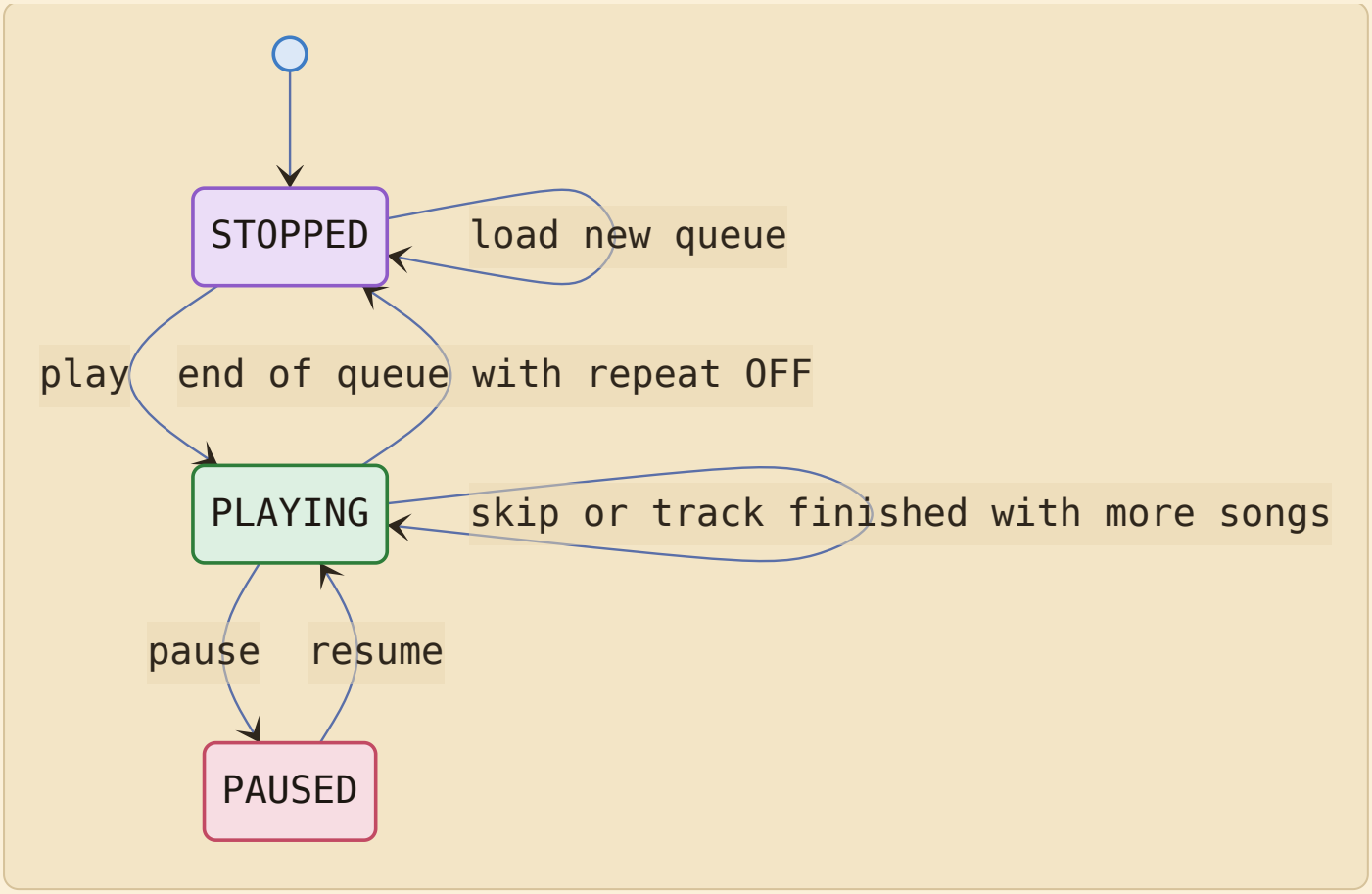
Class	Responsibility
`Artist`, `Album`, `Song`, `Genre`	Immutable catalog records.
`MusicLibrary`	Thread-safe catalog with search by song title, album title, artist name.
`SubscriptionPlan`	Enum that **carries the rules**: monthly price, skips per hour, ads, offline downloads.
`User`	Name, current plan, playlists, listening history.
`Playlist`	Ordered, user-owned song list: add, remove, move.
`PlaybackOrder` → `SequentialOrder`, `ShuffleOrder`	Strategy: produces the play order (a permutation — every song exactly once, in some order) starting from the current song.
`RepeatMode`, `PlayerState`	`OFF / ALL / ONE`; `STOPPED / PLAYING / PAUSED`.
`MusicPlayer`	One user's session: queue, play order, cursor (where we are in the play order), position, state, skip window, ad counter.
`PlaybackListener`	Observer on "song started" (history, royalties, recommendations).
`Recommender` → `GenreRecommender`	Strategy: suggest unplayed songs from the user's favourite genres.

Class diagram

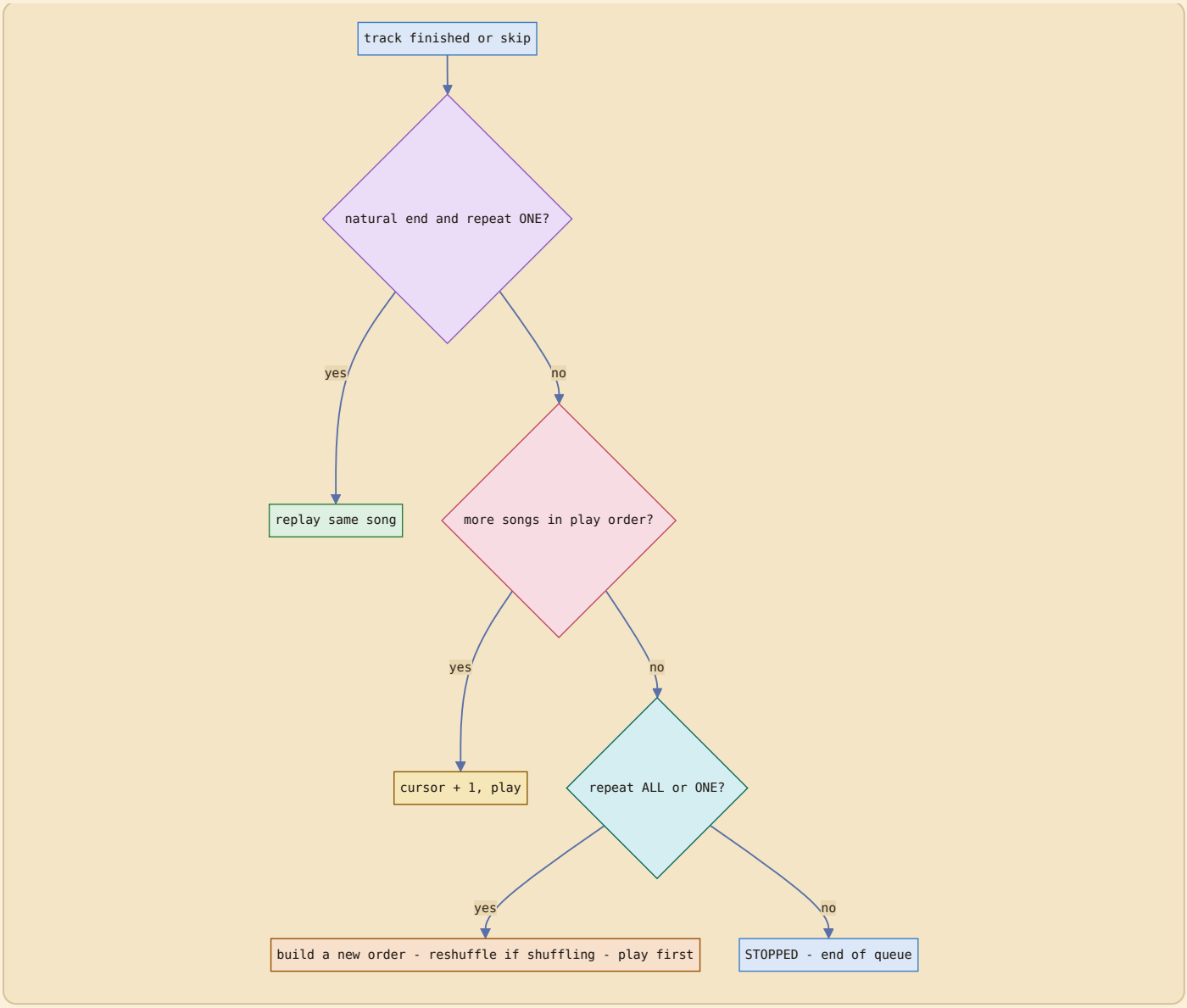


Key flows

Player states



What happens when a song ends or the user presses Next



Skip first checks the plan's rolling (always the last 60 minutes, not a fixed clock hour) one-hour limit (a deque — a queue you can add to or remove from at both ends — of skip timestamps; drop those older than an hour, compare the size with `skipsPerHour`). Pressing Next on repeat-ONE moves on — like real players, repeat-one only affects songs that end naturally.

Shuffle keeps the current song. Turning shuffle on builds a new permutation whose first element is the song playing now, and shuffles the rest with `Collections.shuffle(list, random)`. Turning it off rebuilds the sequential order starting from the current song. The queue itself never changes — only the **order** of indexes into it — so switching back and forth loses nothing.

Design decisions & patterns used

Decision	Why	Alternative rejected
Strategy for play order (`PlaybackOrder`)	Shuffle, sequential, "smart shuffle" (spread the same artist apart) are interchangeable algorithms. The player just asks for an order.	`if (shuffle) ... else ...` inside `skip`, `previous`, `wrap`.
Order as a list of indexes into an unchanged queue	Turning shuffle off/on never loses the original order; `upNext` is a simple walk.	Physically shuffling the queue list (original order lost).
Seeded `Random`	Repeatable demos and tests. Production uses an unseeded (or per-session) seed.	`Math.random()` (not testable).
Plan rules inside `SubscriptionPlan` enum	`user.plan().skipsPerHour` and `.ads` — no `if (plan == FREE)` scattered around; a new plan is one enum constant (or a class if rules get complex).	Boolean flags on `User`.
Observer for plays (`PlaybackListener`)	History, royalty counting and recommendation features all react to "song started" without the player knowing them.	Player calling each subsystem directly.
Strategy for recommendations	`Recommender` can be genre-based today, collaborative filtering ("people who liked what you like also liked…") tomorrow.	Hard-coding one algorithm in `User`.
One `MusicPlayer` per user session, all methods `synchronized`	Commands can arrive from two devices (phone + desktop "connect"). The shared `MusicLibrary` uses concurrent maps.	A single shared player object.
State as an enum, not the State pattern	Three states with trivial behaviour; the GoF ("Gang of Four", the authors of the classic 1994 *Design Patterns* book) State pattern (one class per state) pays off only when each state has substantial, different behaviour.	Three state classes for four one-line methods.
Singletons?	awesome-low-level-design makes `MusicLibrary`, `UserManager` and `MusicRecommender` Singletons. We inject single instances instead — same sharing, easier tests.	Global `getInstance()`.

Java implementation

Files: `Catalog.java` (genre, artist, album, song, library, plans, user, playlist), `Player.java` (play-order strategies, repeat/state enums, listener, player, recommender), `Main.java`.

```java
// File: lld/ch19/music/Catalog.java
package lld.ch19.music;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Locale;
import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.CopyOnWriteArrayList;
import java.util.function.Function;

enum Genre { ROCK, POP, JAZZ, CLASSICAL, HIPHOP }

record Artist(String id, String name) { }

record Album(String id, String title, Artist artist, int year) { }

record Song(String id, String title, Artist artist, Album album, Genre genre, int durationSec) {
    @Override public String toString() { return title + " - " + artist.name(); }
}

/** The shared, read-mostly catalog. Thread-safe collections because every user reads it at once. */
final class MusicLibrary {
    private final Map<String, Song> songs = new ConcurrentHashMap<>();
    private final Map<String, Album> albums = new ConcurrentHashMap<>();
    private final Map<String, Artist> artists = new ConcurrentHashMap<>();

    void add(Song s) {
        songs.put(s.id(), s);
        albums.putIfAbsent(s.album().id(), s.album());
        artists.putIfAbsent(s.artist().id(), s.artist());
    }

    Song song(String id) { return songs.get(id); }
    List<Song> allSongs() { return sorted(songs.values(), Song::id); }

    List<Song> searchSongs(String q) { return match(songs.values(), q, Song::title); }
    List<Album> searchAlbums(String q) { return match(albums.values(), q, Album::title); }
    List<Artist> searchArtists(String q) { return match(artists.values(), q, Artist::name); }
    List<Song> byArtist(Artist a) { return songs.values().stream().filter(s ->
s.artist().equals(a)).sorted(
            (x, y) -> x.id().compareTo(y.id())).toList(); }

    private static <T> List<T> match(Iterable<T> items, String q, Function<T, String> field) {
        String needle = q.toLowerCase(Locale.ROOT);
        List<T> out = new ArrayList<>();
        for (T t : items) if (field.apply(t).toLowerCase(Locale.ROOT).contains(needle)) out.add(t);
        out.sort((a, b) -> field.apply(a).compareTo(field.apply(b)));
        return out;
    }

    private static <T> List<T> sorted(Iterable<T> items, Function<T, String> key) {
        List<T> out = new ArrayList<>();
        items.forEach(out::add);
        out.sort((a, b) -> key.apply(a).compareTo(key.apply(b)));
        return out;
    }
}

/** What each plan allows. The enum carries the rules, so no if (plan == FREE) scattered around. */
```

```
enum SubscriptionPlan {
    FREE(new BigDecimal("0.00"), 6, true, false),
    PREMIUM(new BigDecimal("10.99"), Integer.MAX_VALUE, false, true);

    final BigDecimal monthlyPrice;
    final int skipsPerHour;
    final boolean ads;
    final boolean offlineDownloads;

    SubscriptionPlan(BigDecimal price, int skipsPerHour, boolean ads, boolean offline) {
        this.monthlyPrice = price; this.skipsPerHour = skipsPerHour; this.ads = ads;
this.offlineDownloads = offline;
    }
}

final class User {
    private final String id;
    private final String name;
    private volatile SubscriptionPlan plan = SubscriptionPlan.FREE;
    private final List<Playlist> playlists = new CopyOnWriteArrayList<>();
    private final List<Song> history = new CopyOnWriteArrayList<>();

    User(String id, String name) { this.id = id; this.name = name; }
    String id() { return id; }
    String name() { return name; }
    SubscriptionPlan plan() { return plan; }

    /** Returns the amount to charge now. */
    BigDecimal subscribe(SubscriptionPlan p) { plan = p; return p.monthlyPrice; }

    Playlist createPlaylist(String name) { Playlist p = new Playlist(name, this); playlists.add(p);
return p; }
    List<Playlist> playlists() { return List.copyOf(playlists); }
    void recordPlay(Song s) { history.add(s); }
    List<Song> history() { return Collections.unmodifiableList(history); }
}

/** An ordered, user-owned list of songs. Duplicates are allowed, as in real apps. */
final class Playlist {
    private final String name;
    private final User owner;
    private final List<Song> songs = new ArrayList<>();

    Playlist(String name, User owner) { this.name = name; this.owner = owner; }
    String name() { return name; }
    User owner() { return owner; }
    synchronized Playlist add(Song s) { songs.add(s); return this; }
    synchronized void remove(Song s) { songs.remove(s); }
    synchronized void move(int from, int to) { songs.add(to, songs.remove(from)); }
    synchronized List<Song> songs() { return List.copyOf(songs); }
}
```

```java
// File: lld/ch19/music/Main.java
package lld.ch19.music;

import java.time.Instant;
import java.time.InstantSource;
import java.util.List;

public class Main {
    public static void main(String[] args) {
        MusicLibrary lib = new MusicLibrary();
        Artist queen = new Artist("AR1", "Queen");
        Artist miles = new Artist("AR2", "Miles Davis");
        Artist adele = new Artist("AR3", "Adele");
        Album opera = new Album("AL1", "A Night at the Opera", queen, 1975);
        Album blue = new Album("AL2", "Kind of Blue", miles, 1959);
        Album twentyFive = new Album("AL3", "25", adele, 2015);
        lib.add(new Song("S1", "Bohemian Rhapsody", queen, opera, Genre.ROCK, 354));
        lib.add(new Song("S2", "Love of My Life", queen, opera, Genre.ROCK, 219));
        lib.add(new Song("S3", "So What", miles, blue, Genre.JAZZ, 562));
        lib.add(new Song("S4", "Blue in Green", miles, blue, Genre.JAZZ, 337));
        lib.add(new Song("S5", "Hello", adele, twentyFive, Genre.POP, 295));
        lib.add(new Song("S6", "Death on Two Legs", queen, opera, Genre.ROCK, 223));
        lib.add(new Song("S7", "Freddie Freeloader", miles, blue, Genre.JAZZ, 589));
        lib.add(new Song("S8", "All Blues", miles, blue, Genre.JAZZ, 693));
        lib.add(new Song("S9", "When We Were Young", adele, twentyFive, Genre.POP, 291));
        lib.add(new Song("S10", "You're My Best Friend", queen, opera, Genre.ROCK, 172));

        System.out.println("== 1. Search ==");
        System.out.println("  songs 'blue'    -> " + lib.searchSongs("blue"));
        System.out.println("  albums 'opera'  -> " +
lib.searchAlbums("opera").stream().map(Album::title).toList());
        System.out.println("  artists 'da'    -> " +
lib.searchArtists("da").stream().map(Artist::name).toList());

        System.out.println("\n== 2. Playlist ==");
        User mia = new User("U1", "Mia");
        Playlist mix = mia.createPlaylist("Road trip")

.add(lib.song("S1")).add(lib.song("S3")).add(lib.song("S5")).add(lib.song("S2")).add(lib.song("S4"));
        mix.move(4, 1);
        System.out.println("  " + mix.name() + ": " + mix.songs());

        System.out.println("\n== 3. Play in order, pause, seek, repeat-one ==");
        MusicPlayer player = new MusicPlayer(mia, InstantSource.fixed(Instant.parse("2026-09-
28T08:00:00Z")));
        player.addListener((u, s) -> u.recordPlay(s));          // history is just another listener
        player.load(mix.songs());
        player.play();
        player.seek(120);
        player.pause();
        System.out.println("  state " + player.state() + " at " + player.position() + "s");
        player.resume();
        player.setRepeat(RepeatMode.ONE);
        player.trackFinished();                                  // replays the same song
        player.setRepeat(RepeatMode.OFF);
        player.trackFinished();

        System.out.println("\n== 4. Shuffle keeps the current song, reorders the rest ==");
        player.setShuffle(true);
        System.out.println("  now " + player.current() + ", up next " + player.upNext(4));

        System.out.println("\n== 5. FREE plan: skip limit (6/hour) and ads, repeat-all wraps the
```

```
shuffled queue ==");
        player.setRepeat(RepeatMode.ALL);
        for (int i = 0; i < 7; i++) player.skip();

        System.out.println("\n== 6. Upgrade to PREMIUM: no ads, unlimited skips ==");
        System.out.println("  charged $" + mia.subscribe(SubscriptionPlan.PREMIUM));
        player.setShuffle(false);
        player.load(List.of(lib.song("S6"), lib.song("S7")));
        player.setRepeat(RepeatMode.ALL);
        player.play();
        player.skip();
        player.skip();                                          // wraps to the start
        player.setRepeat(RepeatMode.OFF);
        player.trackFinished();
        player.trackFinished();                                 // end of queue

        System.out.println("\n== 7. Recommendations from listening history ==");
        System.out.println("  plays so far: " + mia.history().size());
        System.out.println("  recommended: " + new GenreRecommender(lib).recommend(mia, 3));
    }
}
```

```java
// File: lld/ch19/music/Player.java
package lld.ch19.music;

import java.time.Duration;
import java.time.Instant;
import java.time.InstantSource;
import java.util.ArrayDeque;
import java.util.ArrayList;
import java.util.Collections;
import java.util.Deque;
import java.util.EnumMap;
import java.util.List;
import java.util.Map;
import java.util.Random;
import java.util.stream.IntStream;

/** Strategy: in what order should the queue be played? */
interface PlaybackOrder {
    /** A permutation of 0..size-1 that starts with {@code first}. */
    List<Integer> order(int size, int first);
}

final class SequentialOrder implements PlaybackOrder {
    @Override public List<Integer> order(int size, int first) {
        List<Integer> out = new ArrayList<>(IntStream.range(0, size).boxed().toList());
        Collections.rotate(out, -first);        // start at the current track, keep the rest in order
        return out;
    }
}

final class ShuffleOrder implements PlaybackOrder {
    private final Random random;
    ShuffleOrder(long seed) { this.random = new Random(seed); }   // seeded: repeatable demo and tests
    @Override public List<Integer> order(int size, int first) {
        List<Integer> rest = new ArrayList<>(IntStream.range(0, size).filter(i -> i !=
first).boxed().toList());
        Collections.shuffle(rest, random);
        rest.add(0, first);                     // the song you are hearing keeps playing
        return rest;
    }
}

enum RepeatMode { OFF, ALL, ONE }

enum PlayerState { STOPPED, PLAYING, PAUSED }

/** Observer hook: history, royalties, recommendations all listen to plays. */
interface PlaybackListener { void onStart(User user, Song song); }

/**
 * One user's player. Not shared between users, but a phone and a web tab could send
 * commands at once, so every public method is synchronized.
 */
final class MusicPlayer {
    private final User user;
    private final InstantSource clock;
    private final List<PlaybackListener> listeners = new ArrayList<>();
    private final Deque<Instant> recentSkips = new ArrayDeque<>();
    private List<Song> queue = List.of();
    private List<Integer> order = List.of();
    private int cursor;                         // index into order
    private int positionSec;
```

```
    private int songsSinceAd;
    private PlayerState state = PlayerState.STOPPED;
    private RepeatMode repeat = RepeatMode.OFF;
    private PlaybackOrder playbackOrder = new SequentialOrder();

    MusicPlayer(User user, InstantSource clock) { this.user = user; this.clock = clock; }

    synchronized void addListener(PlaybackListener l) { listeners.add(l); }

    synchronized void load(List<Song> songs) {
        queue = List.copyOf(songs);
        order = playbackOrder.order(queue.size(), 0);
        cursor = 0;
        state = PlayerState.STOPPED;
    }

    synchronized void play() {
        if (queue.isEmpty()) throw new IllegalStateException("queue is empty");
        start();
    }

    synchronized void pause() { if (state == PlayerState.PLAYING) state = PlayerState.PAUSED; }

    synchronized void resume() { if (state == PlayerState.PAUSED) state = PlayerState.PLAYING; }

    synchronized void seek(int sec) {
        positionSec = Math.max(0, Math.min(sec, current().durationSec()));
    }

    /** User pressed "next". Free users have a skip limit per rolling hour. */
    synchronized boolean skip() {
        if (state == PlayerState.STOPPED) return false;          // nothing to skip
        Instant now = clock.instant();
        while (!recentSkips.isEmpty() &&
recentSkips.peekFirst().isBefore(now.minus(Duration.ofHours(1)))) recentSkips.pollFirst();
        if (recentSkips.size() >= user.plan().skipsPerHour) {
            System.out.println("  [player] skip limit reached on " + user.plan() + " plan");
            return false;
        }
        recentSkips.addLast(now);
        advance(false);
        return true;
    }

    /** The current song ended naturally. Repeat-one replays it; skip does not honour repeat-one. */
    synchronized void trackFinished() { advance(true); }

    synchronized void previous() {
        if (positionSec > 3 || cursor == 0) { positionSec = 0; return; }   // like most players:
restart first
        cursor--;
        start();
    }

    synchronized void setShuffle(boolean on) {
        int now = order.isEmpty() ? 0 : order.get(cursor);
        playbackOrder = on ? new ShuffleOrder(42) : new SequentialOrder();
        order = playbackOrder.order(queue.size(), now);
        cursor = 0;
    }

    synchronized void setRepeat(RepeatMode m) { repeat = m; }
```

```
    synchronized Song current() { return queue.get(order.get(cursor)); }
    synchronized PlayerState state() { return state; }
    synchronized int position() { return positionSec; }

    /** The next few songs in play order (what the "Up next" panel shows). */
    synchronized List<Song> upNext(int n) {
        List<Song> out = new ArrayList<>();
        for (int i = cursor + 1; i < order.size() && out.size() < n; i++)
out.add(queue.get(order.get(i)));
        return out;
    }

    private void advance(boolean natural) {
        if (natural && repeat == RepeatMode.ONE) { start(); return; }
        if (cursor + 1 < order.size()) { cursor++; start(); return; }
        if (repeat == RepeatMode.ALL || repeat == RepeatMode.ONE) {   // wrap around (reshuffle if
shuffling)
            order = playbackOrder.order(queue.size(), order.get(0));
            cursor = 0;
            start();
            return;
        }
        state = PlayerState.STOPPED;
        System.out.println("  [player] end of queue");
    }

    private void start() {
        if (user.plan().ads && ++songsSinceAd > 3) {
            System.out.println("  [player] (30-second ad for FREE plan)");
            songsSinceAd = 1;
        }
        positionSec = 0;
        state = PlayerState.PLAYING;
        Song s = current();
        System.out.println("  [player] > " + s);
        listeners.forEach(l -> l.onStart(user, s));
    }
}

/** Strategy for "what should this user hear next?" - a simple stub based on favourite genre. */
interface Recommender { List<Song> recommend(User user, int n); }

final class GenreRecommender implements Recommender {
    private final MusicLibrary library;
    GenreRecommender(MusicLibrary library) { this.library = library; }

    @Override public List<Song> recommend(User user, int n) {
        Map<Genre, Integer> counts = new EnumMap<>(Genre.class);
        user.history().forEach(s -> counts.merge(s.genre(), 1, Integer::sum));
        List<Genre> favourite = counts.entrySet().stream()
                .sorted((a, b) -> b.getValue() - a.getValue()).map(Map.Entry::getKey).toList();
        List<Song> heard = user.history();
        List<Song> out = new ArrayList<>();
        for (Genre g : favourite)
            for (Song s : library.allSongs())
                if (out.size() < n && s.genre() == g && !heard.contains(s)) out.add(s);
        return out;
    }
}
```

Output:

```
== 1. Search ==
  songs 'blue'    -> [All Blues - Miles Davis, Blue in Green - Miles Davis]
  albums 'opera'  -> [A Night at the Opera]
  artists 'da'    -> [Miles Davis]

== 2. Playlist ==
  Road trip: [Bohemian Rhapsody - Queen, Blue in Green - Miles Davis, So What - Miles Davis, Hello -
Adele, Love of My Life - Queen]

== 3. Play in order, pause, seek, repeat-one ==
  [player] > Bohemian Rhapsody - Queen
  state PAUSED at 120s
  [player] > Bohemian Rhapsody - Queen
  [player] > Blue in Green - Miles Davis

== 4. Shuffle keeps the current song, reorders the rest ==
  now Blue in Green - Miles Davis, up next [Love of My Life - Queen, So What - Miles Davis, Bohemian
Rhapsody - Queen, Hello - Adele]

== 5. FREE plan: skip limit (6/hour) and ads, repeat-all wraps the shuffled queue ==
  [player] (30-second ad for FREE plan)
  [player] > Love of My Life - Queen
  [player] > So What - Miles Davis
  [player] > Bohemian Rhapsody - Queen
  [player] (30-second ad for FREE plan)
  [player] > Hello - Adele
  [player] > Blue in Green - Miles Davis
  [player] > Hello - Adele
  [player] skip limit reached on FREE plan

== 6. Upgrade to PREMIUM: no ads, unlimited skips ==
  charged $10.99
  [player] > Death on Two Legs - Queen
  [player] > Freddie Freeloader - Miles Davis
  [player] > Death on Two Legs - Queen
  [player] > Freddie Freeloader - Miles Davis
  [player] end of queue

== 7. Recommendations from listening history ==
  plays so far: 13
  recommended: [You're My Best Friend - Queen, All Blues - Miles Davis, When We Were Young - Adele]
```

Reading the output. Step 3: repeat-ONE replays "Bohemian Rhapsody" when it ends naturally. Step 4: shuffle keeps "Blue in Green" playing and reorders the rest. Step 5: on the FREE plan an ad plays after every three songs, repeat-ALL wraps and reshuffles the queue, and the seventh skip within the hour is refused. Step 6: PREMIUM has no ads; with repeat-ALL the second skip wraps to the start; with repeat OFF the queue ends. Step 7: the recommender suggests songs the user has **not** heard, from the genres they played most.

Concurrency & edge cases

Case	Handling
Two devices send "next" at the same instant	Player methods are `synchronized`; the two skips apply one after the other.
Catalog updated while users search	`ConcurrentHashMap` (a thread-safe hash map) for songs/albums/artists; search iterates a weakly consistent view (it may or may not show changes made during the loop, but never breaks).
Playlist edited while it is playing	`load` copies the playlist (`List.copyOf`), so the running queue is a snapshot; a real app would sync edits into "up next".
Seek past the end or before 0	Clamped to `[0, duration]`.
Previous after 3 seconds	Restarts the current song (common player behaviour); within 3 seconds goes back one song.
Skip when stopped	Refused (and does not count against the limit).
Plan changes mid-session	`plan` is `volatile` (a Java keyword that makes every thread see the latest written value); the next skip or ad check sees it.
Empty queue	`play()` throws `IllegalStateException`.

Real-world example

Spotify's free tier has historically limited skips (for example, six per hour on mobile) and inserts ads between songs, while Premium removes both and allows downloads — the rules our `SubscriptionPlan` enum carries. Spotify engineers also wrote (2014) that users complained a truly random shuffle "felt" non-random when the same artist came up twice in a row, so they changed the algorithm to spread songs by the same artist across the list. That is exactly the kind of change the `PlaybackOrder` strategy absorbs: a new `ArtistSpreadShuffle` class, no player changes. Personalised playlists such as Discover Weekly are the industrial version of our `Recommender` stub.

Extensions & follow-up questions

- **"Add 'play next' and 'add to queue'."** Keep a separate *user queue* that is consumed before the context (playlist/album) order.
- **"Offline mode."** `SubscriptionPlan.offlineDownloads`; a `DownloadManager` stores encrypted files; the player chooses a local or remote `AudioSource` (Strategy).
- **"Social sharing / collaborative playlists."** Playlist gets a collaborator set and an edit log; listeners (Observer) push changes to other users.
- **"Crossfade, gapless playback."** Player behaviour around `trackFinished`; a `TransitionStrategy`.
- **"Royalties."** A `PlaybackListener` that records a play once the user has listened for a threshold time.
- **"Search at scale."** Inverted index (a map from each word to the items containing it) / search service with prefix (typeahead — suggestions as you type) and fuzzy matching (tolerating typos); the `MusicLibrary` interface stays.
- **"Smart shuffle."** New `PlaybackOrder` that spaces out songs by the same artist.

In the interview

30–45 minute plan

1. Clarify controls, plans and scope (skip audio bytes) — 4 min.
2. Catalog entities + library search — 6 min.

3. User, playlist, subscription plan — 5 min.
4. **Player**: queue + order + cursor; state machine; shuffle/repeat as strategy/enum — 15 min.
5. Observer for history; recommender strategy; concurrency; extensions — 10 min.

> **30-second answer:** "Immutable catalog records in a thread-safe library. Each user has a player session holding the queue, a play order produced by a `PlaybackOrder` strategy, a cursor and a repeat mode. Shuffle rebuilds the order around the current song. Plan limits live in the plan enum. A playback listener records history, which feeds a pluggable recommender."

Follow-ups with answers

- *How does shuffle avoid repeating a song before all have played?* It is a permutation (each index once), not "pick a random next song".
- *Where do ads fit?* The player asks the plan whether ads apply and inserts one every N songs; in production an ad service picks which ad.
- *How would you implement "previous" in shuffle mode?* Walk back along the same order list (the cursor), so previous is the song you actually heard before.

Self-check

1. **Q:** Why store the play order as indexes rather than reordering the queue? **A:** So switching shuffle off restores the original order and nothing is lost.
2. **Q:** What does repeat-ONE do when the user presses Next? **A:** Moves to the next song; it only replays on natural end.
3. **Q:** Which pattern lets you add a smarter shuffle without touching the player? **A:** Strategy (`PlaybackOrder`).
4. **Q:** How is "6 skips per hour" enforced? **A:** A deque of skip times; drop entries older than one hour; refuse if the size has reached the limit.
5. **Q:** Why is the player's state an enum rather than the State pattern? **A:** The states have almost no distinct behaviour; the pattern would add classes without benefit.

TL;DR

- Catalog = immutable records in a concurrent library.
- Player = **queue + order (Strategy) + cursor + repeat mode + state**; shuffle keeps the current song.
- Plan rules in the **enum**; plays published via **Observer**; recommendations via **Strategy**.

Design an Online Food Delivery Service (Swiggy) + FoodKart

This problem also comes in two shapes:

- **Part A — Swiggy-style delivery platform** (the awesome-low-level-design problem): restaurants with menus, a cart, orders with tracking states, payments, and **assigning a delivery rider** by a pluggable strategy. We add **kitchen capacity** (a restaurant can only cook so many items at once), the key twist in a commonly reported Flipkart food-ordering variant (it is not part of the FoodKart statement below).
- **Part B — FoodKart** (Flipkart machine-coding round, from kumaransg/LLD): a smaller, command-driven problem — users and pincodes (Indian postal codes), one-dish restaurants, ratings, `show_restaurant` sorted by rating or price, `place_order` — plus a **restaurant selection strategy** (highest rating / lowest price) filtered by remaining capacity.

Problem & clarifying questions

Question	Assumed answer
Can one cart mix restaurants?	No — one restaurant per cart (as in Swiggy/Zomato).
Who confirms an order?	The restaurant; it may refuse if its kitchen is full.
When is capacity freed?	When the food is ready (handed over), not when it is delivered.
How is a rider chosen?	By a strategy: nearest to the restaurant, or fair share (fewest deliveries today).
What if no rider is free?	The order waits in a queue; the next freed rider takes the oldest waiting order.
Can a customer cancel?	Only before the kitchen starts cooking (PENDING or CONFIRMED).
Payment types?	UPI, card, cash on delivery (COD: nothing to refund).
Live GPS, ETA (estimated time of arrival) prediction, batching two orders per rider, surge fees (higher fees at busy times)?	Extensions.

Requirements

Functional (Part A)

1. Customers browse restaurants, view menus and place orders.
2. Restaurants manage menus, prices and availability.
3. Delivery agents accept and fulfil orders.
4. Customers track order status and get notified of changes.
5. Several payment methods.
6. Real-time notifications to customers, restaurants and delivery agents.

Functional (Part B — FoodKart)

7. `register_user(name, gender, phone, pincode)` — phone is unique.
8. `login_user(phone)` — sets the context; logging in someone else logs out the previous user.
9. `register_restaurant(name, pincodes "A/B", dish, price, quantity)` — in the context of the logged-in user (the owner). A restaurant serves **one** dish to **many** pincodes.
10. `update_quantity(name, add)`, `update_location(name, pincodes)` — owner only.
11. `rate_restaurant(name, 1..5, optional comment)`; rating = average of all ratings.
12. `show_restaurant(rating|price)` — **serviceable** restaurants (deliver to my pincode **and** quantity > 0), sorted **descending** by the chosen field.
13. `place_order(name, qty)`; bonus: order history.
14. *(Variant)* place an order without naming a restaurant: pick one by a **selection strategy**, among those that deliver here **and** have capacity for the whole quantity.

Non-functional

- Handle many concurrent orders without double-booking kitchen capacity or riders.
- Keep data consistent (no negative stock, no rider on two orders).
- In-memory only for the machine-coding round; demoable driver (a small `Main` program that feeds in commands and prints results); graceful failures.

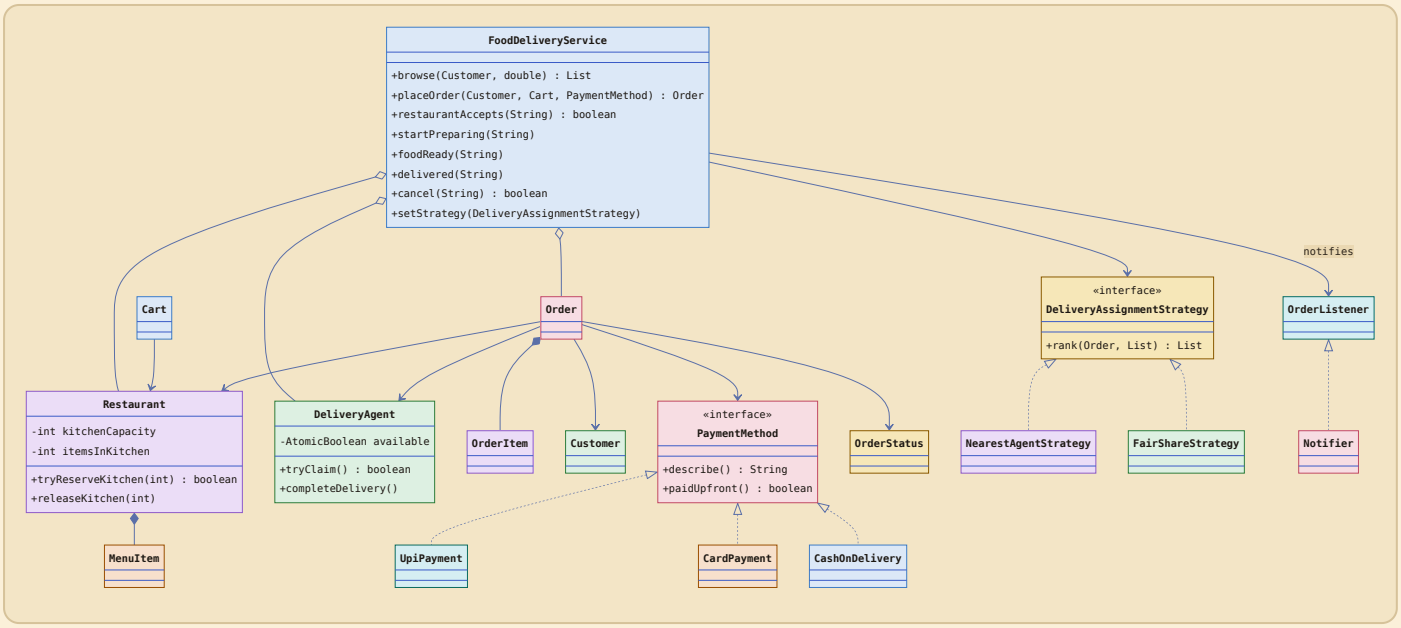
Use cases

Actor	Use cases
Customer	Browse nearby restaurants, build cart, pay, place order, track, cancel, rate
Restaurant	Manage menu/prices/availability, accept or refuse order, mark preparing, mark ready
Delivery agent	Get assigned, pick up, deliver
System	Compute fees, assign riders, queue orders when no rider is free, notify everyone

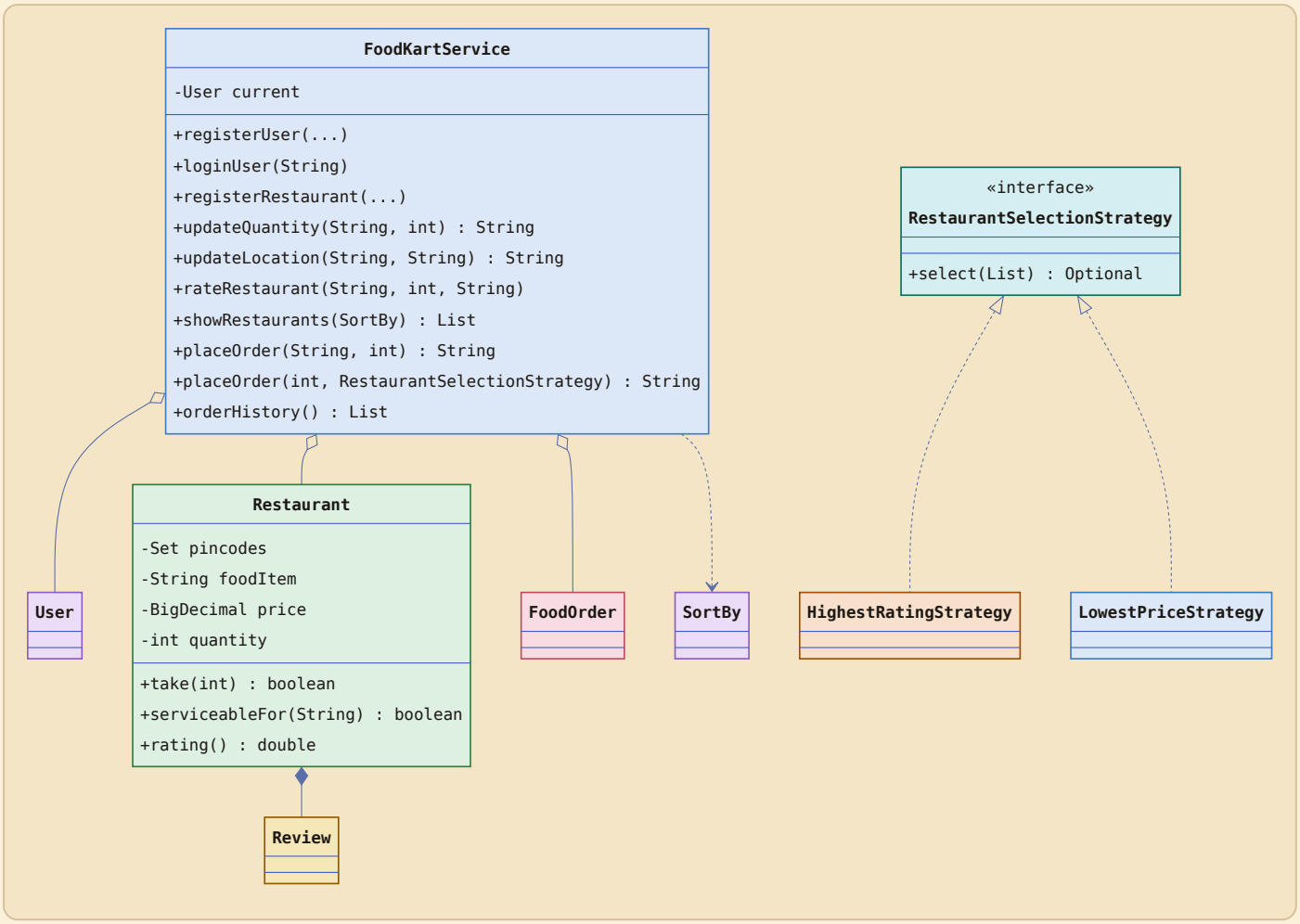
Core entities

Class	Responsibility
`Location`	A point; `distanceTo`.
`Customer`	id, name, phone, address.
`MenuItem`	Name, price, available flag (both changeable by the restaurant).
`Restaurant`	Menu, open flag, **kitchen capacity** (max items cooking at once), `tryReserveKitchen` / `releaseKitchen`.
`Cart`	One restaurant + item quantities; refuses items from a second restaurant.
`OrderItem`	Snapshot of item name, **unit price at order time**, quantity.
`OrderStatus`	State machine: PENDING → CONFIRMED → PREPARING → READY_FOR_PICKUP → OUT_FOR_DELIVERY → DELIVERED, or CANCELLED.
`Order`	Customer, restaurant, items, delivery fee, payment, status history, assigned agent.
`PaymentMethod` → `UpiPayment`, `CardPayment`, `CashOnDelivery`	Sealed interface; COD is not paid upfront.
`DeliveryAgent`	Location, `AtomicBoolean available` (a true/false flag that can be checked-and-changed atomically), deliveries today; `tryClaim()` is a CAS.
`DeliveryAssignmentStrategy` → `NearestAgentStrategy`, `FairShareStrategy`	Ranks free riders for an order.
`OrderListener` / `Notifier`	Observer for status changes.
`FoodDeliveryService`	Facade: browse, place, accept, prepare, ready, deliver, cancel; the waiting-for-rider queue.
(FoodKart) `User`, `Restaurant`, `Review`, `FoodOrder`, `SortBy`, `RestaurantSelectionStrategy`, `FoodKartService`	The machine-coding model; `Restaurant.take(q)` is the atomic capacity check.

Class diagram

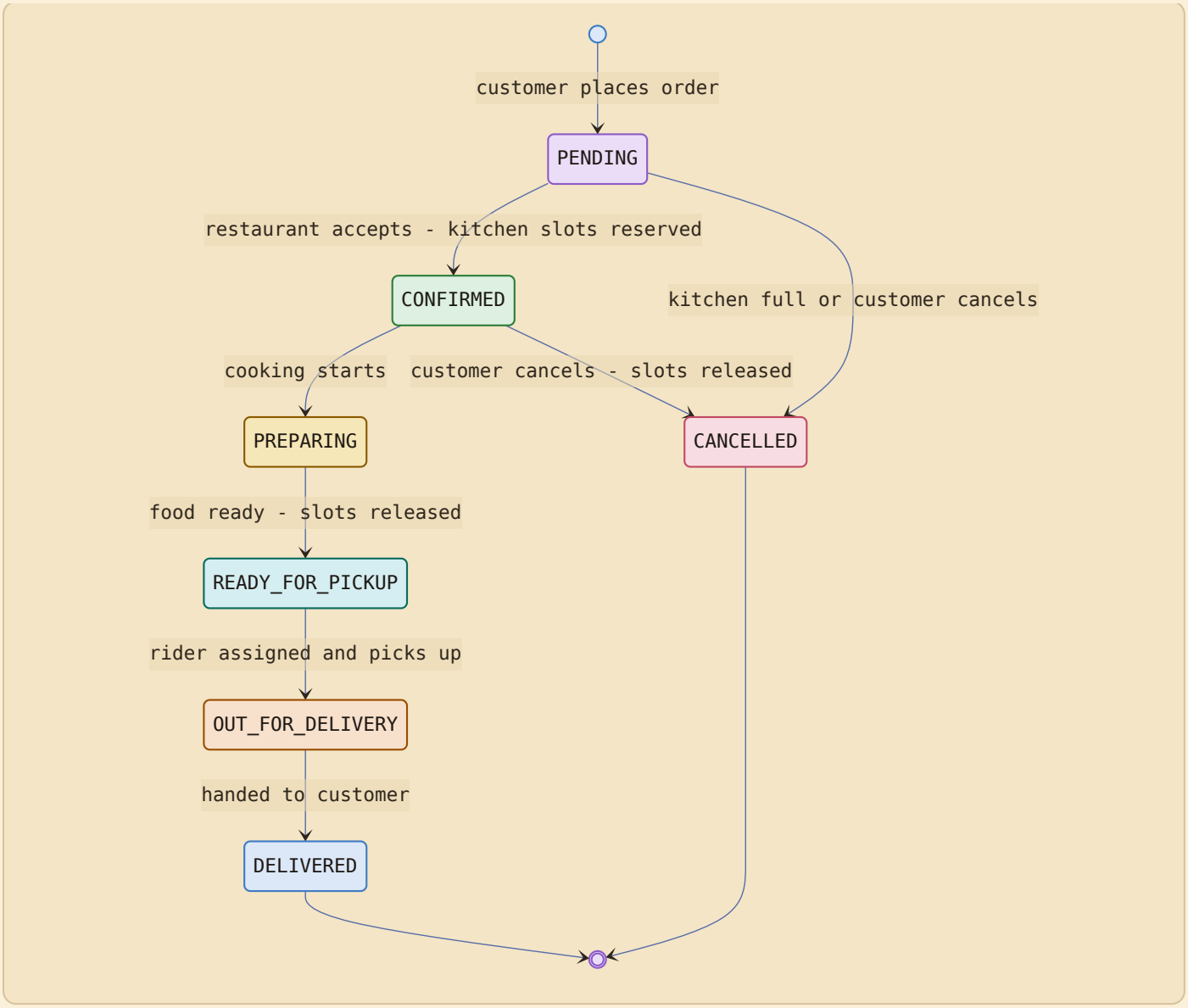


FoodKart (Part B):



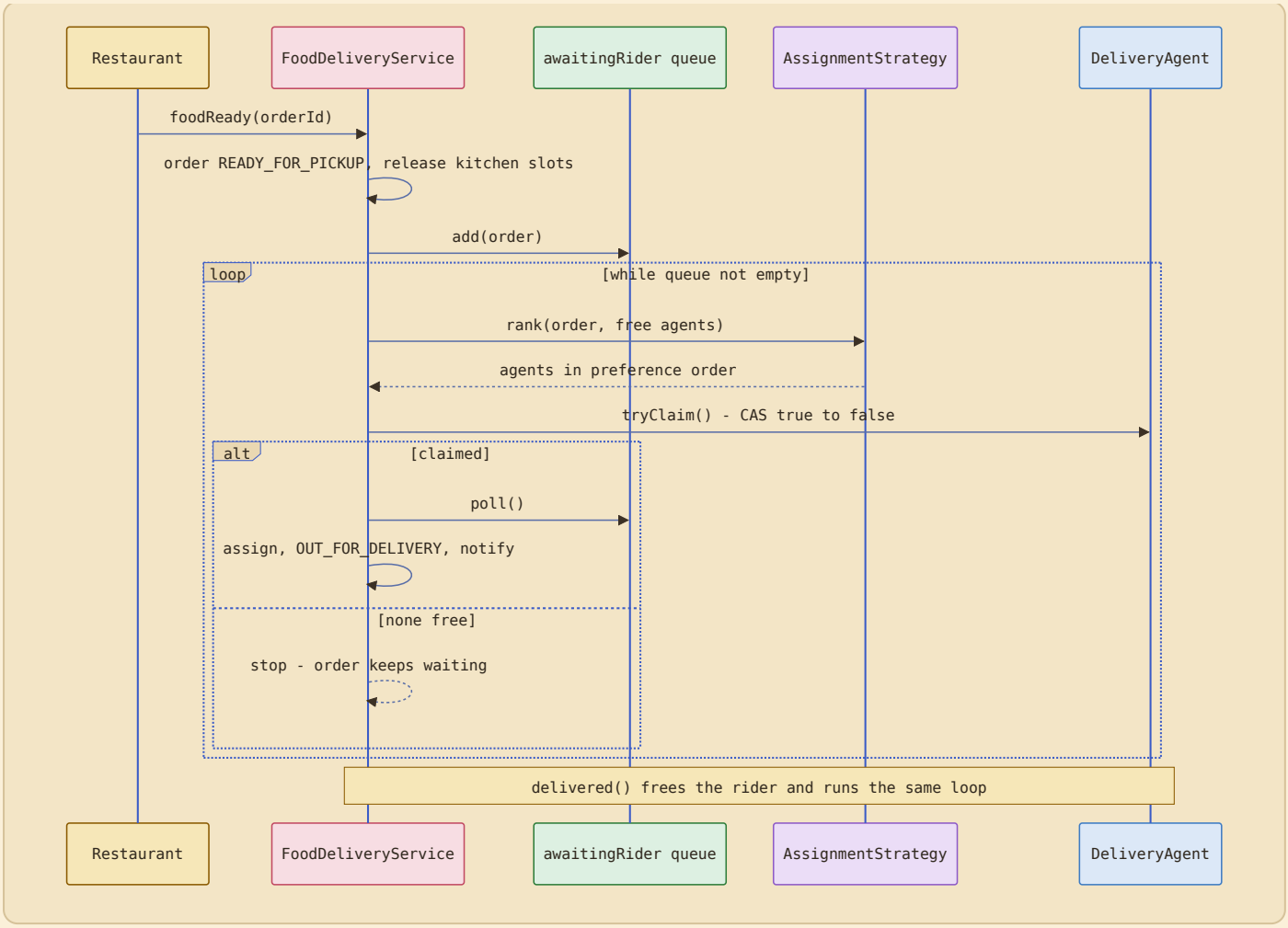
Key flows

Order tracking state machine



(Real apps often split `READY_FOR_PICKUP → OUT_FOR_DELIVERY` into "rider assigned", "arrived at restaurant", "picked up". Adding them is a local enum change.)

Food ready → rider assignment



FoodKart `place_order(name, qty)`: user logged in? → restaurant exists? → delivers to the user's pincode? → `take(qty)` atomically (quantity ≥ qty, then subtract) → record in history → "Order Placed Successfully." Otherwise "Cannot place order".

FoodKart selection variant: filter restaurants that deliver to the user's pincode and have `quantity ≥ qty` → ask the strategy to pick one → `take(qty)`; if another user took the capacity in between, drop that restaurant and ask again (bounded retries).

Design decisions & patterns used

Decision	Why	Alternative rejected
Strategy for rider assignment (`rank` returns a preference list)	Nearest vs fair-share vs rating-based are swappable at run time (`setStrategy`, step 7). Returning a *ranked list* lets the service try the next rider if the first was just taken (the free list is read before claiming, so it can be stale).	Returning one rider (fails if that rider was claimed a moment earlier).
CAS on the rider (`AtomicBoolean.compareAndSet(true,false)`)	The rider guards itself: whoever calls `tryClaim()` — this dispatcher, a second dispatcher instance, or a future "rider accepts" button — exactly one `compareAndSet` succeeds. In this code `dispatchWaiting` is also `synchronized` (to drain the queue in FIFO order), so the CAS is the safety net that keeps working even when claims come from outside that lock.	Check `isAvailable()` then set (a race as soon as two callers exist).
Kitchen capacity reserved on accept, released on ready	Models the rule "a restaurant cannot accept more items while its kitchen is full" from the capacity-based food-ordering variant.	Unlimited orders per restaurant.
Waiting queue for riders	No free rider is normal at peak; FIFO keeps it fair; freeing a rider re-runs dispatch.	Failing the order.
State machine in `OrderStatus`	Tracking states and legal moves in one place; cancel after cooking starts is impossible.	Scattered `if`s.
`OrderItem` copies the price	A later menu price change (step 3) cannot alter a placed order.	Referencing `MenuItem` directly.
Sealed `PaymentMethod` + records	A closed, known set; COD overrides `paidUpfront()` so refunds skip it.	`String paymentType`.

Decision	Why	Alternative rejected
Observer (`OrderListener`)	Customer app, restaurant tablet and rider app all react to the same events.	The service printing/sending messages itself.
One restaurant per cart	Matches real apps; a single kitchen, single pickup, single fee.	Multi-restaurant carts (needs split orders).
FoodKart: `SortBy` enum holds its comparator (an object that decides which of two items comes first)	`show_restaurant(rating	price)`becomes`sorted(sortBy.comparator)`; both descending as the statement says; name as tie-breaker for stable output.
FoodKart: owner recorded at registration	The statement says restaurants are registered "in the context of the logged-in user"; only that owner may update quantity or location.	Anyone updating any restaurant.
FoodKart: `Restaurant.take(q)` synchronized	"Check quantity then subtract" is atomic — two users cannot both take the last plates.	Separate `getQuantity()` + `setQuantity()`.

Java implementation

Part A — Food delivery platform (`lld.ch19.fooddelivery`)

Files: `Model.java` (location, customer, menu item, restaurant, cart, order item, status machine, payments, order, delivery agent), `FoodDeliveryService.java` (strategies, listener, notifier, service), `Main.java`.

```java
// File: lld/ch19/fooddelivery/FoodDeliveryService.java
package lld.ch19.fooddelivery;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.util.ArrayList;
import java.util.Comparator;
import java.util.List;
import java.util.Map;
import java.util.Optional;
import java.util.Queue;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.ConcurrentLinkedQueue;
import java.util.concurrent.CopyOnWriteArrayList;
import java.util.concurrent.atomic.AtomicInteger;

/** Strategy: which free rider gets this order? Returns riders in preference order. */
interface DeliveryAssignmentStrategy {
    List<DeliveryAgent> rank(Order order, List<DeliveryAgent> free);
}

/** Closest rider to the restaurant - fastest pickup. */
final class NearestAgentStrategy implements DeliveryAssignmentStrategy {
    @Override public List<DeliveryAgent> rank(Order o, List<DeliveryAgent> free) {
        Location r = o.restaurant().location();
        return free.stream().sorted(Comparator.comparingDouble(a ->
a.location().distanceTo(r))).toList();
    }
}

/** Fewest deliveries today first (spreads earnings), nearest as tie-breaker. */
final class FairShareStrategy implements DeliveryAssignmentStrategy {
    @Override public List<DeliveryAgent> rank(Order o, List<DeliveryAgent> free) {
        Location r = o.restaurant().location();
        return free.stream().sorted(Comparator.comparingInt(DeliveryAgent::deliveriesToday)
                .thenComparingDouble(a -> a.location().distanceTo(r))).toList();
    }
}

/** Observer: customer app, restaurant tablet and rider app all track the order. */
interface OrderListener { void onStatus(Order order, OrderStatus status); }

final class Notifier implements OrderListener {
    @Override public void onStatus(Order o, OrderStatus s) {
        String who = switch (s) {
            case PENDING -> "restaurant " + o.restaurant().name() + ": new order";
            case OUT_FOR_DELIVERY -> "customer " + o.customer().name() + ": " +
o.agent().map(DeliveryAgent::name).orElse("?") + " is on the way";
            default -> "customer " + o.customer().name();
        };
        System.out.println("    [" + o.id() + " " + s + "] -> " + who);
    }
}

/** Facade: browse, order, accept, prepare, dispatch, deliver, cancel. */
final class FoodDeliveryService {
    private static final BigDecimal BASE_FEE = new BigDecimal("20.00");
    private static final BigDecimal PER_KM = new BigDecimal("8.00");

    private final Map<String, Restaurant> restaurants = new ConcurrentHashMap<>();
    private final List<DeliveryAgent> agents = new CopyOnWriteArrayList<>();
    private final Map<String, Order> orders = new ConcurrentHashMap<>();
```

```
    private final Queue<Order> awaitingRider = new ConcurrentLinkedQueue<>();
    private final List<OrderListener> listeners = new CopyOnWriteArrayList<>();
    private final AtomicInteger seq = new AtomicInteger(100);
    private volatile DeliveryAssignmentStrategy strategy;

    FoodDeliveryService(DeliveryAssignmentStrategy strategy) { this.strategy = strategy; }

    void setStrategy(DeliveryAssignmentStrategy s) { strategy = s; }
    void subscribe(OrderListener l) { listeners.add(l); }
    void addRestaurant(Restaurant r) { restaurants.put(r.id(), r); }
    void addAgent(DeliveryAgent a) { agents.add(a); }

    /** Open restaurants within the radius, nearest first. */
    List<Restaurant> browse(Customer c, double radiusKm) {
        return restaurants.values().stream().filter(Restaurant::isOpen)
                .filter(r -> r.location().distanceTo(c.address()) <= radiusKm)
                .sorted(Comparator.comparingDouble(r ->
r.location().distanceTo(c.address()))).toList();
    }

    Order placeOrder(Customer c, Cart cart, PaymentMethod pay) {
        if (cart.isEmpty()) throw new IllegalStateException("empty cart");
        Restaurant r = cart.restaurant();
        if (!r.isOpen()) throw new IllegalStateException(r.name() + " is closed");
        List<OrderItem> items = new ArrayList<>();
        cart.quantities().forEach((itemId, qty) -> {
            MenuItem m = r.item(itemId).orElseThrow();
            if (!m.available()) throw new IllegalStateException(m.name() + " is sold out");
            items.add(new OrderItem(itemId, m.name(), m.price(), qty));
        });
        BigDecimal km = BigDecimal.valueOf(r.location().distanceTo(c.address())).setScale(1,
RoundingMode.HALF_UP);
        BigDecimal fee = BASE_FEE.add(PER_KM.multiply(km)).setScale(2, RoundingMode.HALF_EVEN);
        Order o = new Order("OD" + seq.incrementAndGet(), c, r, items, fee, pay);
        orders.put(o.id(), o);
        cart.clear();
        publish(o, OrderStatus.PENDING);
        return o;
    }

    /** The restaurant accepts only if its kitchen has room; otherwise the order is auto-cancelled. */
    boolean restaurantAccepts(String orderId) {
        Order o = orders.get(orderId);
        synchronized (o) {                              // same lock as cancel(): capacity can never
leak
            if (o.status() != OrderStatus.PENDING) return false;
            if (!o.restaurant().tryReserveKitchen(o.itemCount())) {
                System.out.println("    (" + o.restaurant().name() + " kitchen full: needs " +
o.itemCount()
                        + " item slots, free " + o.restaurant().freeCapacity() + ")");
                transition(o, OrderStatus.CANCELLED);
                return false;
            }
            transition(o, OrderStatus.CONFIRMED);
            return true;
        }
    }

    void startPreparing(String orderId) { transition(orders.get(orderId), OrderStatus.PREPARING); }

    /** Food is ready: kitchen capacity comes back, and we look for a rider. */
    void foodReady(String orderId) {
```

```
        Order o = orders.get(orderId);
        transition(o, OrderStatus.READY_FOR_PICKUP);
        o.restaurant().releaseKitchen(o.itemCount());
        awaitingRider.add(o);
        dispatchWaiting();
    }

    void delivered(String orderId) {
        Order o = orders.get(orderId);
        transition(o, OrderStatus.DELIVERED);
        DeliveryAgent a = o.agent().orElseThrow();
        a.moveTo(o.customer().address());
        a.completeDelivery();
        dispatchWaiting();                              // a freed rider takes the oldest waiting
order
    }

    boolean cancel(String orderId) {
        Order o = orders.get(orderId);
        synchronized (o) {
            if (!o.status().canMoveTo(OrderStatus.CANCELLED)) return false;
            boolean kitchenHeld = o.status() == OrderStatus.CONFIRMED;
            transition(o, OrderStatus.CANCELLED);
            if (kitchenHeld) o.restaurant().releaseKitchen(o.itemCount());
        }
        if (o.payment().paidUpfront()) System.out.println("    refund " + o.total() + " to " +
o.payment().describe());
        return true;
    }

    Order order(String id) { return orders.get(id); }

    /** Hand waiting orders (oldest first) to free riders. CAS on the rider prevents double booking.
*/
    private synchronized void dispatchWaiting() {
        Order o;
        while ((o = awaitingRider.peek()) != null) {
            List<DeliveryAgent> free = agents.stream().filter(DeliveryAgent::isAvailable).toList();
            Optional<DeliveryAgent> got = Optional.empty();
            for (DeliveryAgent a : strategy.rank(o, free)) if (a.tryClaim()) { got = Optional.of(a);
break; }
            if (got.isEmpty()) {
                System.out.println("    (" + o.id() + " waiting for a free rider)");
                return;
            }
            awaitingRider.poll();
            o.assign(got.get());
            transition(o, OrderStatus.OUT_FOR_DELIVERY);
        }
    }

    private void transition(Order o, OrderStatus s) {
        o.moveTo(s);
        listeners.forEach(l -> l.onStatus(o, s));
    }

    private void publish(Order o, OrderStatus s) { listeners.forEach(l -> l.onStatus(o, s)); }
}
```

```java
// File: lld/ch19/fooddelivery/Main.java
package lld.ch19.fooddelivery;

import java.math.BigDecimal;

public class Main {
    public static void main(String[] args) {
        FoodDeliveryService app = new FoodDeliveryService(new NearestAgentStrategy());
        app.subscribe(new Notifier());

        Restaurant dosa = new Restaurant("R1", "Dosa Corner", new Location(1, 1), 4);
        dosa.addItem(new MenuItem("D1", "Masala Dosa", new BigDecimal("120.00")));
        dosa.addItem(new MenuItem("D2", "Filter Coffee", new BigDecimal("40.00")));
        Restaurant pizza = new Restaurant("R2", "Pizza Hub", new Location(5, 5), 10);
        pizza.addItem(new MenuItem("P1", "Margherita", new BigDecimal("299.00")));
        Restaurant far = new Restaurant("R3", "Far Away Grill", new Location(30, 30), 10);
        app.addRestaurant(dosa);
        app.addRestaurant(pizza);
        app.addRestaurant(far);
        DeliveryAgent ravi = new DeliveryAgent("A1", "Ravi", new Location(0, 0));
        DeliveryAgent meena = new DeliveryAgent("A2", "Meena", new Location(6, 6));
        app.addAgent(ravi);
        app.addAgent(meena);

        Customer asha = new Customer("C1", "Asha", "98000", new Location(2, 2));
        Customer vik = new Customer("C2", "Vikram", "98001", new Location(4, 4));

        System.out.println("== 1. Browse within 10 km ==");
        app.browse(asha, 10).forEach(r -> System.out.println("  " + r.name()));

        System.out.println("\n== 2. One restaurant per cart ==");
        Cart cart = new Cart();
        cart.add(dosa, "D1", 2);
        cart.add(dosa, "D2", 1);
        try { cart.add(pizza, "P1", 1); } catch (IllegalStateException e) { System.out.println("  " +
e.getMessage()); }

        System.out.println("\n== 3. Place, accept, prepare, ready, deliver ==");
        Order o1 = app.placeOrder(asha, cart, new UpiPayment("asha@upi"));
        System.out.println("  subtotal " + o1.subtotal() + " + delivery " + o1.deliveryFee() + " = " +
o1.total());
        dosa.item("D1").orElseThrow().setPrice(new BigDecimal("150.00"));    // later price change...
        System.out.println("  price changed after ordering; order still " + o1.total());
        app.restaurantAccepts(o1.id());
        app.startPreparing(o1.id());
        app.foodReady(o1.id());
        System.out.println("  rider: " + o1.agent().orElseThrow() + " (nearest to Dosa Corner)");

        System.out.println("\n== 4. Kitchen capacity: 4 items max at Dosa Corner ==");
        Cart c2 = new Cart();
        c2.add(dosa, "D1", 3);
        Order o2 = app.placeOrder(vik, c2, new CashOnDelivery());
        app.restaurantAccepts(o2.id());                                    // 3 of 4 slots used
        Cart c3 = new Cart();
        c3.add(dosa, "D2", 2);
        Order o3 = app.placeOrder(asha, c3, new CardPayment("4242"));
        app.restaurantAccepts(o3.id());                                    // needs 2, only 1 free

        System.out.println("\n== 5. Only one rider free: second ready order waits ==");
        app.startPreparing(o2.id());
        app.foodReady(o2.id());                                            // Meena takes it
        Cart c4 = new Cart();
```

```
        c4.add(pizza, "P1", 1);
        Order o4 = app.placeOrder(vik, c4, new UpiPayment("vik@upi"));
        app.restaurantAccepts(o4.id());
        app.startPreparing(o4.id());
        app.foodReady(o4.id());                                          // nobody free
        app.delivered(o1.id());                                          // Ravi frees up -> takes
o4
        System.out.println("  o4 rider: " + o4.agent().orElseThrow());

        System.out.println("\n== 6. Cancel rules ==");
        System.out.println("  cancel o2 (out for delivery)? " + app.cancel(o2.id()));
        Cart c5 = new Cart();
        c5.add(pizza, "P1", 2);
        Order o5 = app.placeOrder(asha, c5, new CardPayment("4242"));
        app.restaurantAccepts(o5.id());
        System.out.println("  cancel o5 (confirmed)? " + app.cancel(o5.id()));

        System.out.println("\n== 7. Swap strategy at run time: fair share ==");
        app.delivered(o2.id());
        app.delivered(o4.id());
        System.out.println("  deliveries today: Ravi=" + ravi.deliveriesToday() + ", Meena=" +
meena.deliveriesToday()
                + " (both now at Vikram's address)");
        app.setStrategy(new FairShareStrategy());
        Cart c6 = new Cart();
        c6.add(dosa, "D2", 1);
        Order o6 = app.placeOrder(asha, c6, new UpiPayment("asha@upi"));
        app.restaurantAccepts(o6.id());
        app.startPreparing(o6.id());
        app.foodReady(o6.id());
        System.out.println("  o6 rider: " + o6.agent().orElseThrow() + " (fewer deliveries today)");
        System.out.println("  o1 history: " + o1.history());
    }
}
```

```java
// File: lld/ch19/fooddelivery/Model.java
package lld.ch19.fooddelivery;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.util.ArrayList;
import java.util.Collection;
import java.util.EnumSet;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;
import java.util.Optional;
import java.util.Set;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.atomic.AtomicBoolean;
import java.util.concurrent.atomic.AtomicInteger;

/** A point on a simple km grid (real systems use lat/long and a road-distance service). */
record Location(double x, double y) {
    double distanceTo(Location o) { return Math.hypot(x - o.x, y - o.y); }
}

record Customer(String id, String name, String phone, Location address) { }

final class MenuItem {
    private final String id;
    private final String name;
    private volatile BigDecimal price;
    private volatile boolean available = true;

    MenuItem(String id, String name, BigDecimal price) { this.id = id; this.name = name; this.price =
price; }
    String id() { return id; }
    String name() { return name; }
    BigDecimal price() { return price; }
    boolean available() { return available; }
    void setPrice(BigDecimal p) { price = p; }
    void setAvailable(boolean a) { available = a; }
}

/**
 * A restaurant manages its own menu and has a kitchen capacity: at most N items being
 * prepared at once. Capacity is taken when an order is accepted and returned when the
 * food is ready (dispatched).
 */
final class Restaurant {
    private final String id;
    private final String name;
    private final Location location;
    private final int kitchenCapacity;
    private final Map<String, MenuItem> menu = new ConcurrentHashMap<>();
    private volatile boolean open = true;
    private int itemsInKitchen;                         // guarded by this

    Restaurant(String id, String name, Location location, int kitchenCapacity) {
        this.id = id; this.name = name; this.location = location; this.kitchenCapacity =
kitchenCapacity;
    }

    String id() { return id; }
    String name() { return name; }
    Location location() { return location; }
```

```
    boolean isOpen() { return open; }
    void setOpen(boolean o) { open = o; }
    void addItem(MenuItem m) { menu.put(m.id(), m); }
    Optional<MenuItem> item(String itemId) { return Optional.ofNullable(menu.get(itemId)); }
    Collection<MenuItem> menu() { return menu.values(); }

    synchronized boolean tryReserveKitchen(int items) {
        if (itemsInKitchen + items > kitchenCapacity) return false;
        itemsInKitchen += items;
        return true;
    }

    synchronized void releaseKitchen(int items) { itemsInKitchen -= items; }

    synchronized int freeCapacity() { return kitchenCapacity - itemsInKitchen; }
}

/** A customer's cart. One restaurant per cart, as in Swiggy/Zomato. */
final class Cart {
    private Restaurant restaurant;
    private final Map<String, Integer> quantities = new LinkedHashMap<>();

    synchronized void add(Restaurant r, String itemId, int qty) {
        if (restaurant != null && !restaurant.id().equals(r.id()))
            throw new IllegalStateException("cart has items from " + restaurant.name() + " - clear it
first");
        r.item(itemId).orElseThrow(() -> new IllegalArgumentException("no item " + itemId));
        restaurant = r;
        quantities.merge(itemId, qty, Integer::sum);
    }

    synchronized void clear() { restaurant = null; quantities.clear(); }
    synchronized Restaurant restaurant() { return restaurant; }
    synchronized Map<String, Integer> quantities() { return Map.copyOf(quantities); }
    synchronized boolean isEmpty() { return quantities.isEmpty(); }
}

/** Price is copied at order time: a later menu price change never alters a placed order. */
record OrderItem(String itemId, String name, BigDecimal unitPrice, int quantity) {
    BigDecimal total() { return unitPrice.multiply(BigDecimal.valueOf(quantity)); }
}

/** The order-tracking state machine. */
enum OrderStatus {
    PENDING, CONFIRMED, PREPARING, READY_FOR_PICKUP, OUT_FOR_DELIVERY, DELIVERED, CANCELLED;

    private Set<OrderStatus> next() {
        return switch (this) {
            case PENDING -> EnumSet.of(CONFIRMED, CANCELLED);
            case CONFIRMED -> EnumSet.of(PREPARING, CANCELLED);
            case PREPARING -> EnumSet.of(READY_FOR_PICKUP);
            case READY_FOR_PICKUP -> EnumSet.of(OUT_FOR_DELIVERY);
            case OUT_FOR_DELIVERY -> EnumSet.of(DELIVERED);
            case DELIVERED, CANCELLED -> EnumSet.noneOf(OrderStatus.class);
        };
    }

    boolean canMoveTo(OrderStatus t) { return next().contains(t); }
}

/** Strategy: how the customer pays. */
sealed interface PaymentMethod permits UpiPayment, CardPayment, CashOnDelivery {
```

```
    String describe();
    default boolean paidUpfront() { return true; }
}

record UpiPayment(String vpa) implements PaymentMethod {
    public String describe() { return "UPI " + vpa; }
}

record CardPayment(String last4) implements PaymentMethod {
    public String describe() { return "card ****" + last4; }
}

record CashOnDelivery() implements PaymentMethod {
    public String describe() { return "cash on delivery"; }
    public boolean paidUpfront() { return false; }
}

final class Order {
    private final String id;
    private final Customer customer;
    private final Restaurant restaurant;
    private final List<OrderItem> items;
    private final BigDecimal deliveryFee;
    private final PaymentMethod payment;
    private final List<OrderStatus> history = new ArrayList<>();
    private OrderStatus status = OrderStatus.PENDING;
    private DeliveryAgent agent;

    Order(String id, Customer c, Restaurant r, List<OrderItem> items, BigDecimal fee, PaymentMethod
payment) {
        this.id = id; this.customer = c; this.restaurant = r; this.items = List.copyOf(items);
        this.deliveryFee = fee; this.payment = payment;
        history.add(status);
    }

    String id() { return id; }
    Customer customer() { return customer; }
    Restaurant restaurant() { return restaurant; }
    List<OrderItem> items() { return items; }
    PaymentMethod payment() { return payment; }
    int itemCount() { return items.stream().mapToInt(OrderItem::quantity).sum(); }
    BigDecimal subtotal() { return items.stream().map(OrderItem::total).reduce(BigDecimal.ZERO,
BigDecimal::add); }
    BigDecimal total() { return subtotal().add(deliveryFee).setScale(2, RoundingMode.HALF_EVEN); }
    BigDecimal deliveryFee() { return deliveryFee; }
    synchronized OrderStatus status() { return status; }
    synchronized List<OrderStatus> history() { return List.copyOf(history); }
    synchronized Optional<DeliveryAgent> agent() { return Optional.ofNullable(agent); }
    synchronized void assign(DeliveryAgent a) { agent = a; }

    synchronized void moveTo(OrderStatus t) {
        if (!status.canMoveTo(t)) throw new IllegalStateException(id + ": " + status + " -> " + t + "
not allowed");
        status = t;
        history.add(t);
    }
}

/** A rider. available is an AtomicBoolean so two orders can never claim the same rider. */
final class DeliveryAgent {
    private final String id;
    private final String name;
```

```
    private volatile Location location;
    private final AtomicBoolean available = new AtomicBoolean(true);
    private final AtomicInteger deliveriesToday = new AtomicInteger();

    DeliveryAgent(String id, String name, Location location) { this.id = id; this.name = name;
this.location = location; }
    String id() { return id; }
    String name() { return name; }
    Location location() { return location; }
    void moveTo(Location l) { location = l; }
    boolean isAvailable() { return available.get(); }
    int deliveriesToday() { return deliveriesToday.get(); }

    /** Compare-and-set: exactly one caller wins. */
    boolean tryClaim() { return available.compareAndSet(true, false); }

    void completeDelivery() { deliveriesToday.incrementAndGet(); available.set(true); }

    @Override public String toString() { return name; }
}
```

Output:

```
== 1. Browse within 10 km ==
  Dosa Corner
  Pizza Hub

== 2. One restaurant per cart ==
  cart has items from Dosa Corner - clear it first

== 3. Place, accept, prepare, ready, deliver ==
    [OD101 PENDING] -> restaurant Dosa Corner: new order
  subtotal 280.00 + delivery 31.20 = 311.20
  price changed after ordering; order still 311.20
    [OD101 CONFIRMED] -> customer Asha
    [OD101 PREPARING] -> customer Asha
    [OD101 READY_FOR_PICKUP] -> customer Asha
    [OD101 OUT_FOR_DELIVERY] -> customer Asha: Ravi is on the way
  rider: Ravi (nearest to Dosa Corner)

== 4. Kitchen capacity: 4 items max at Dosa Corner ==
    [OD102 PENDING] -> restaurant Dosa Corner: new order
    [OD102 CONFIRMED] -> customer Vikram
    [OD103 PENDING] -> restaurant Dosa Corner: new order
    (Dosa Corner kitchen full: needs 2 item slots, free 1)
    [OD103 CANCELLED] -> customer Asha

== 5. Only one rider free: second ready order waits ==
    [OD102 PREPARING] -> customer Vikram
    [OD102 READY_FOR_PICKUP] -> customer Vikram
    [OD102 OUT_FOR_DELIVERY] -> customer Vikram: Meena is on the way
    [OD104 PENDING] -> restaurant Pizza Hub: new order
    [OD104 CONFIRMED] -> customer Vikram
    [OD104 PREPARING] -> customer Vikram
    [OD104 READY_FOR_PICKUP] -> customer Vikram
    (OD104 waiting for a free rider)
    [OD101 DELIVERED] -> customer Asha
    [OD104 OUT_FOR_DELIVERY] -> customer Vikram: Ravi is on the way
  o4 rider: Ravi

== 6. Cancel rules ==
  cancel o2 (out for delivery)? false
    [OD105 PENDING] -> restaurant Pizza Hub: new order
    [OD105 CONFIRMED] -> customer Asha
    [OD105 CANCELLED] -> customer Asha
    refund 651.60 to card ****4242
  cancel o5 (confirmed)? true

== 7. Swap strategy at run time: fair share ==
    [OD102 DELIVERED] -> customer Vikram
    [OD104 DELIVERED] -> customer Vikram
  deliveries today: Ravi=2, Meena=1 (both now at Vikram's address)
    [OD106 PENDING] -> restaurant Dosa Corner: new order
    [OD106 CONFIRMED] -> customer Asha
    [OD106 PREPARING] -> customer Asha
    [OD106 READY_FOR_PICKUP] -> customer Asha
    [OD106 OUT_FOR_DELIVERY] -> customer Asha: Meena is on the way
  o6 rider: Meena (fewer deliveries today)
  o1 history: [PENDING, CONFIRMED, PREPARING, READY_FOR_PICKUP, OUT_FOR_DELIVERY, DELIVERED]
```

Reading the output. Step 1 hides the far-away grill (outside 10 km). Step 2 refuses a second restaurant in one cart. Step 3: the delivery fee is 20 + 8 × 1.4 km = 31.20; changing the dosa price afterwards does not change the order. Step 4: Dosa Corner holds 4 item slots; an order for 3 fits,

the next order for 2 is auto-cancelled. Step 5: Meena takes OD102; OD104 waits because Ravi is still out; as soon as OD101 is delivered, Ravi is claimed for OD104. Step 6: an order out for delivery cannot be cancelled; a confirmed card order can, and is refunded. Step 7: with both riders at the same spot, the fair-share strategy picks Meena (1 delivery) over Ravi (2).

Part B — FoodKart (`lld.ch19.foodkart`)

Files: `FoodKart.java` (gender, user, review, order, restaurant, sort, selection strategies, service), `Main.java` (driver replaying the problem's sample test case, then extras).

```java
// File: lld/ch19/foodkart/FoodKart.java
package lld.ch19.foodkart;

import java.math.BigDecimal;
import java.util.Comparator;
import java.util.LinkedHashSet;
import java.util.List;
import java.util.Map;
import java.util.Optional;
import java.util.Set;
import java.util.TreeSet;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.CopyOnWriteArrayList;
import java.util.concurrent.atomic.AtomicInteger;

enum Gender { M, F, OTHER }

record User(String name, Gender gender, String phone, String pincode) { }

record Review(String userPhone, int rating, String comment) {
    Review {
        if (rating < 1 || rating > 5) throw new IllegalArgumentException("rating must be 1..5");
    }
}

record FoodOrder(int id, String userPhone, String restaurant, String item, int quantity, BigDecimal
total) { }

/** A restaurant serves ONE dish to MANY pincodes. Quantity is its remaining capacity. */
final class Restaurant {
    private final String name;
    private final String ownerPhone;
    private final String foodItem;
    private final BigDecimal price;
    private volatile Set<String> pincodes;
    private int quantity;                                          // guarded by this
    private final List<Review> reviews = new CopyOnWriteArrayList<>();

    Restaurant(String name, String ownerPhone, Set<String> pincodes, String foodItem, BigDecimal
price, int quantity) {
        this.name = name; this.ownerPhone = ownerPhone; this.pincodes = Set.copyOf(pincodes);
        this.foodItem = foodItem; this.price = price; this.quantity = quantity;
    }

    String name() { return name; }
    String ownerPhone() { return ownerPhone; }
    String foodItem() { return foodItem; }
    BigDecimal price() { return price; }
    Set<String> pincodes() { return pincodes; }
    void setPincodes(Set<String> p) { pincodes = Set.copyOf(p); }
    synchronized int quantity() { return quantity; }
    synchronized void addQuantity(int q) { quantity += q; }

    /** Check-and-take in one step, so two users cannot both get the last plates. */
    synchronized boolean take(int q) {
        if (q <= 0 || q > quantity) return false;
        quantity -= q;
        return true;
    }

    /** Serviceable = delivers to this pincode AND has non-zero quantity (from the problem statement).
*/
```

```
    boolean serviceableFor(String pincode) { return pincodes.contains(pincode) && quantity() > 0; }

    void addReview(Review r) { reviews.add(r); }

    double rating() { return reviews.stream().mapToInt(Review::rating).average().orElse(0.0); }

    String describe() { return name + ", " + String.join("/", new TreeSet<>(pincodes)) + ", " +
foodItem + " - " + quantity(); }
}

/** show_restaurant(rating|price): both sorted in DESCENDING order, as the problem states. */
enum SortBy {
    RATING(Comparator.comparingDouble(Restaurant::rating).reversed()),
    PRICE(Comparator.comparing(Restaurant::price).reversed());

    final Comparator<Restaurant> comparator;
    SortBy(Comparator<Restaurant> c) { comparator = c.thenComparing(Restaurant::name); }
}

/** Strategy: when the user does not name a restaurant, which one should we pick? */
interface RestaurantSelectionStrategy {
    Optional<Restaurant> select(List<Restaurant> canFulfil);
}

final class HighestRatingStrategy implements RestaurantSelectionStrategy {
    @Override public Optional<Restaurant> select(List<Restaurant> c) {
        return c.stream().max(Comparator.comparingDouble(Restaurant::rating)
                .thenComparing(Restaurant::quantity));              // tie: the one with most capacity
left
    }
}

final class LowestPriceStrategy implements RestaurantSelectionStrategy {
    @Override public Optional<Restaurant> select(List<Restaurant> c) {
        return c.stream().min(Comparator.comparing(Restaurant::price)
                .thenComparing(Comparator.comparingDouble(Restaurant::rating).reversed()));
    }
}

/** The service behind every command in the machine-coding round. In-memory only. */
final class FoodKartService {
    private final Map<String, User> users = new ConcurrentHashMap<>();
    private final Map<String, Restaurant> restaurants = new ConcurrentHashMap<>();
    private final Map<String, List<FoodOrder>> history = new ConcurrentHashMap<>();
    private final AtomicInteger orderSeq = new AtomicInteger();
    private volatile User current;                                  // one logged-in user at a time

    void registerUser(String name, Gender g, String phone, String pincode) {
        if (users.putIfAbsent(phone, new User(name, g, phone, pincode)) != null)
            throw new IllegalArgumentException("phone already registered: " + phone);
    }

    void loginUser(String phone) {
        User u = users.get(phone);
        if (u == null) throw new IllegalArgumentException("unknown user " + phone);
        current = u;                                                // previous user is logged out
    }

    void registerRestaurant(String name, String pincodesSlashSeparated, String item, BigDecimal price,
int qty) {
        User owner = requireLogin();
        Restaurant r = new Restaurant(name, owner.phone(), parsePincodes(pincodesSlashSeparated),
```

```
item, price, qty);
        if (restaurants.putIfAbsent(name, r) != null) throw new IllegalArgumentException("restaurant
exists: " + name);
    }

    String updateQuantity(String name, int toAdd) {
        Restaurant r = ownedByCurrentUser(name);
        if (toAdd <= 0) throw new IllegalArgumentException("quantity to add must be positive");
        r.addQuantity(toAdd);
        return r.describe();
    }

    String updateLocation(String name, String pincodesSlashSeparated) {
        Restaurant r = ownedByCurrentUser(name);
        r.setPincodes(parsePincodes(pincodesSlashSeparated));
        return r.describe();
    }

    void rateRestaurant(String name, int rating, String comment) {
        restaurant(name).addReview(new Review(requireLogin().phone(), rating, comment == null ? "" :
comment));
    }

    List<String> showRestaurants(SortBy sortBy) {
        String pin = requireLogin().pincode();
        return restaurants.values().stream().filter(r -> r.serviceableFor(pin))
                .sorted(sortBy.comparator).map(r -> r.name() + ", " + r.foodItem()).toList();
    }

    String placeOrder(String name, int qty) {
        User u = requireLogin();
        Restaurant r = restaurants.get(name);
        if (r == null || !r.pincodes().contains(u.pincode()) || !r.take(qty)) return "Cannot place
order";
        record(u, r, qty);
        return "Order Placed Successfully.";
    }

    /** Variant: the system picks the restaurant. Only those that deliver here AND have capacity for
qty qualify. */
    String placeOrder(int qty, RestaurantSelectionStrategy strategy) {
        User u = requireLogin();
        List<Restaurant> canFulfil = restaurants.values().stream()
                .filter(r -> r.pincodes().contains(u.pincode()) && r.quantity() >= qty).toList();
        for (int attempt = 0; attempt < 3; attempt++) {           // retry if a racer took the
capacity
            Optional<Restaurant> pick = strategy.select(canFulfil);
            if (pick.isEmpty()) return "Cannot place order";
            if (pick.get().take(qty)) {
                record(u, pick.get(), qty);
                return "Order Placed Successfully from " + pick.get().name() + ".";
            }
            canFulfil = canFulfil.stream().filter(r -> r != pick.get()).toList();
        }
        return "Cannot place order";
    }

    List<FoodOrder> orderHistory() { return List.copyOf(history.getOrDefault(requireLogin().phone(),
List.of())); }

    private void record(User u, Restaurant r, int qty) {
        FoodOrder o = new FoodOrder(orderSeq.incrementAndGet(), u.phone(), r.name(), r.foodItem(),
```

```
qty,
                r.price().multiply(BigDecimal.valueOf(qty)));
        history.computeIfAbsent(u.phone(), k -> new CopyOnWriteArrayList<>()).add(o);
    }

    private User requireLogin() {
        User u = current;
        if (u == null) throw new IllegalStateException("no user logged in");
        return u;
    }

    private Restaurant restaurant(String name) {
        Restaurant r = restaurants.get(name);
        if (r == null) throw new IllegalArgumentException("no restaurant " + name);
        return r;
    }

    private Restaurant ownedByCurrentUser(String name) {
        Restaurant r = restaurant(name);
        if (!r.ownerPhone().equals(requireLogin().phone())) throw new IllegalStateException("only the
owner can update " + name);
        return r;
    }

    private static Set<String> parsePincodes(String s) {
        Set<String> out = new LinkedHashSet<>();
        for (String p : s.split("/")) if (!p.isBlank()) out.add(p.trim());
        if (out.isEmpty()) throw new IllegalArgumentException("at least one pincode needed");
        return out;
    }

}
```

```java
// File: lld/ch19/foodkart/Main.java
package lld.ch19.foodkart;

import java.math.BigDecimal;

/** Driver: replays the sample test case from the problem statement, then the extras. */
public class Main {
    public static void main(String[] args) {
        FoodKartService fk = new FoodKartService();

        System.out.println("== Sample test case ==");
        fk.registerUser("Pralove", Gender.M, "phoneNumber-1", "HSR");
        fk.registerUser("Nitesh", Gender.M, "phoneNumber-2", "BTM");
        fk.registerUser("Vatsal", Gender.M, "phoneNumber-3", "BTM");

        fk.loginUser("phoneNumber-1");
        fk.registerRestaurant("Food Court-1", "BTM/HSR", "NI Thali", new BigDecimal("100"), 5);
        fk.registerRestaurant("Food Court-2", "BTM", "Burger", new BigDecimal("120"), 3);
        fk.loginUser("phoneNumber-2");
        fk.registerRestaurant("Food Court-3", "HSR", "SI Thali", new BigDecimal("150"), 1);
        fk.loginUser("phoneNumber-3");

        System.out.println("show_restaurant(price):");
        fk.showRestaurants(SortBy.PRICE).forEach(s -> System.out.println("  " + s));
        System.out.println("place_order(Food Court-1, 2): " + fk.placeOrder("Food Court-1", 2));
        System.out.println("place_order(Food Court-2, 7): " + fk.placeOrder("Food Court-2", 7));
        fk.rateRestaurant("Food Court-2", 3, "Good Food");
        fk.rateRestaurant("Food Court-1", 5, "Nice Food");
        System.out.println("show_restaurant(rating):");
        fk.showRestaurants(SortBy.RATING).forEach(s -> System.out.println("  " + s));

        fk.loginUser("phoneNumber-1");
        System.out.println("update_quantity(Food Court-2, 5): " + fk.updateQuantity("Food Court-2",
5));
        System.out.println("update_location(Food Court-2, BTM/HSR): " + fk.updateLocation("Food Court-
2", "BTM/HSR"));

        System.out.println("\n== Extras ==");
        fk.loginUser("phoneNumber-2");
        try { fk.updateQuantity("Food Court-1", 10); } catch (IllegalStateException e) {
System.out.println("non-owner update: " + e.getMessage()); }
        try { fk.rateRestaurant("Food Court-1", 6, null); } catch (IllegalArgumentException e) {
System.out.println("rating 6: " + e.getMessage()); }
        fk.rateRestaurant("Food Court-1", 4, null);             // rating without a comment
        System.out.println("order from a restaurant that does not deliver to BTM: " +
fk.placeOrder("Food Court-3", 1));
        fk.registerRestaurant("Food Court-4", "BTM", "Dosa", new BigDecimal("80"), 4);

        System.out.println("\n== Selection strategy variant (user in BTM, wants 4 plates) ==");
        fk.loginUser("phoneNumber-3");
        System.out.println("highest rating: " + fk.placeOrder(4, new HighestRatingStrategy()));
        System.out.println("lowest price  : " + fk.placeOrder(4, new LowestPriceStrategy()));
        System.out.println("lowest price  : " + fk.placeOrder(4, new LowestPriceStrategy()));
        System.out.println("lowest price  : " + fk.placeOrder(4, new LowestPriceStrategy()) + "  (no
one has 4 left)");

        System.out.println("\n== Order history of phoneNumber-3 ==");
        fk.orderHistory().forEach(o -> System.out.println("  " + o));
    }
}
```

Output:

```
== Sample test case ==
show_restaurant(price):
  Food Court-2, Burger
  Food Court-1, NI Thali
place_order(Food Court-1, 2): Order Placed Successfully.
place_order(Food Court-2, 7): Cannot place order
show_restaurant(rating):
  Food Court-1, NI Thali
  Food Court-2, Burger
update_quantity(Food Court-2, 5): Food Court-2, BTM, Burger - 8
update_location(Food Court-2, BTM/HSR): Food Court-2, BTM/HSR, Burger - 8

== Extras ==
non-owner update: only the owner can update Food Court-1
rating 6: rating must be 1..5
order from a restaurant that does not deliver to BTM: Cannot place order

== Selection strategy variant (user in BTM, wants 4 plates) ==
highest rating: Order Placed Successfully from Food Court-2.
lowest price  : Order Placed Successfully from Food Court-4.
lowest price  : Order Placed Successfully from Food Court-2.
lowest price  : Cannot place order  (no one has 4 left)

== Order history of phoneNumber-3 ==
  FoodOrder[id=1, userPhone=phoneNumber-3, restaurant=Food Court-1, item=NI Thali, quantity=2,
total=200]
  FoodOrder[id=2, userPhone=phoneNumber-3, restaurant=Food Court-2, item=Burger, quantity=4,
total=480]
  FoodOrder[id=3, userPhone=phoneNumber-3, restaurant=Food Court-4, item=Dosa, quantity=4, total=320]
  FoodOrder[id=4, userPhone=phoneNumber-3, restaurant=Food Court-2, item=Burger, quantity=4,
total=480]
```

Reading the output. The "Sample test case" block reproduces every expected line from the problem statement: price-sorted list (Burger 120 before Thali 100), the successful and failed orders, the rating-sorted list, and the owner's quantity and location updates (`Burger - 8`). The extras show owner-only updates, rating validation, a rating without comment, and a restaurant that does not deliver to the user's pincode. The variant shows the two strategies choosing **different** restaurants (highest rating → Food Court-2; lowest price → Food Court-4), capacity filtering (Food Court-1 has only 3 plates, so it is never picked for 4), and a clean refusal once no one has 4 plates left.

Concurrency & edge cases

Case	Handling
Two claims for the same rider	`tryClaim()` CAS: only one succeeds; the loser moves to the next rider in its ranked list. (Inside one service the synchronized dispatcher already serialises claims; the CAS also covers claims made outside it.)
Two dispatch loops run at once	`dispatchWaiting` is `synchronized`, so the waiting queue is drained by one thread at a time (FIFO fairness preserved).
Kitchen over-booked	`tryReserveKitchen` is `synchronized` (check + add in one step).
Cancel races with accept	Both lock the `Order`; accept re-checks `PENDING`, cancel releases slots only if they were reserved (`CONFIRMED`). No leak, no double release.
Item sold out after it was added to the cart	`placeOrder` re-checks `available` and rejects.
Restaurant closes	`browse` hides it; `placeOrder` refuses.
COD cancelled	No refund (nothing was paid).
FoodKart: no user logged in	`requireLogin()` throws a clear `IllegalStateException`.
FoodKart: duplicate phone / restaurant name	Rejected with `putIfAbsent`.
FoodKart: rating outside 1–5	The `Review` record's compact constructor (a record's short validation block that runs on creation) rejects it.
FoodKart: strategy picks a restaurant whose capacity was just taken	`take()` fails; that restaurant is dropped and the strategy asked again (bounded retries).

Real-world example

In Swiggy and Zomato the order screen walks through the same states — order placed, restaurant confirmed, food being prepared, rider assigned / picked up, delivered — and if you add a dish from a second restaurant, the app asks whether to **reset your cart**, which is our one-restaurant `Cart` rule. The rider side is a dispatch problem: these platforms assign riders using distance, rider load and fairness signals, and may batch two orders on one trip — all variations of `DeliveryAssignmentStrategy`. FoodKart itself is a Flipkart machine-coding question: a 90-minute, in-memory, driver-class exercise that rewards clean separation (service, model, strategy) over clever algorithms.

Extensions & follow-up questions

- **"Batch two orders for one rider."** A strategy that accepts a rider already carrying one order if the second pickup is on the way; the agent then has a list of orders and a capacity (e.g. 2).
- **"ETA."** An `EtaEstimator` strategy: prep time (per restaurant load) + travel time (distance / speed, or a maps API — an online service other programs call).
- **"Surge / distance-based fees."** Extract `DeliveryFeePolicy` (Strategy): base + per-km today, add surge when waiting orders exceed free riders.
- **"Rider rejects an assignment."** Put the rider back to available, add the order back to the head of the queue, and exclude that rider for this order.
- **"Multiple dishes per restaurant in FoodKart."** Replace `foodItem/price/quantity` with a `Map<String, MenuItem>`; `take` becomes per item and all-or-nothing for multi-item orders.
- **"Order split across restaurants (Flipkart variant)."** The selection strategy returns a *set* of restaurants that together cover all items; reserve capacity in each, roll back all on any

failure.

- **"Live tracking."** The rider app pushes location updates; an Observer forwards them to the customer's app (WebSocket).

In the interview

Design round (45 min): clarify (5) → entities + one-restaurant cart (8) → **order state machine** (7) → **assignment strategy + CAS claim + waiting queue** (12) → payments/notifications (5) → concurrency and extensions (8).

Machine-coding round (FoodKart, 90 min): model classes first (User, Restaurant, Review, Order) → service with one method per command → the driver replaying the sample → sort enum → owner checks and validation → selection strategy → order history. Keep it runnable at every step; print exactly the expected lines.

> **30-second answer:** "Restaurants own their menus and a kitchen capacity reserved on accept and released when food is ready. Orders copy item prices and move through a state machine from PENDING to DELIVERED. When food is ready the order joins a FIFO queue; a pluggable assignment strategy ranks free riders and we claim one with a compare-and-set, so a rider can never get two orders. Every status change is published to observers for the customer, restaurant and rider apps."

Follow-ups with answers

- *Why release kitchen capacity on "ready" and not on "delivered"?* The kitchen is free once the food leaves it; delivery time is the rider's constraint, not the kitchen's.
- *Why does the strategy return a list?* Because between ranking and claiming, the top rider may be claimed by someone else (another dispatcher or the rider's own app); the list lets us fall through without re-ranking.
- *How would you scale assignment across cities?* Partition by city/zone; each zone has its own dispatcher (single writer), so there is no cross-zone contention (threads or servers fighting over the same lock or data).

Self-check

1. **Q:** Which states allow a customer cancel, and why not later? **A:** PENDING and CONFIRMED; once cooking starts the food cost is already spent.
2. **Q:** What stops two orders getting the same rider? **A:** `AtomicBoolean.compareAndSet(true, false)` in `tryClaim()`.
3. **Q:** In FoodKart, what makes a restaurant "serviceable"? **A:** It delivers to the user's pincode and has quantity > 0.
4. **Q:** Why does `show_restaurant(price)` list Burger (120) before NI Thali (100)? **A:** The statement asks for descending order.
5. **Q:** How does the selection variant avoid choosing a restaurant that cannot fulfil the order? **A:** It filters by `quantity ≥ qty` before the strategy chooses, and `take()` re-checks atomically.
6. **Q:** Which patterns appear? **A:** Strategy (assignment, selection, payment), Observer (notifications), state machine (order status), Facade (service).

TL;DR

- One-restaurant cart; order copies prices; **state machine** PENDING → … → DELIVERED.
- **Kitchen capacity** reserved on accept, released on ready; **riders** claimed with **CAS**, chosen by a swappable **strategy**, with a FIFO queue when none are free.

- FoodKart: pincode-based serviceability, one dish per restaurant, average rating, `SortBy` enum, atomic `take`, and a **selection strategy** filtered by capacity.

Design a Grocery Store (POS & Checkout)

A **point-of-sale (POS) system** is the software behind a supermarket till: the cashier (or a self-checkout kiosk) scans items, the screen shows a running total, offers are applied, the customer pays in one or more ways, a receipt is printed, and stock goes down. Amazon (§19.1) was about an online cart that becomes an order; a grocery till is different in three ways that make it a good interview question:

- **Items are weighed as well as counted.** Loose produce is priced per kilogram; the scale sends a weight and the price is computed at the till.
- **Offers interact.** "Buy 2 get 1 free", "3 for ₹80", "10% off produce" and a basket coupon can all apply to one basket — and the rules for combining them decide what the customer pays, what tax is due, and how much a return refunds.
- **The till must never block a physical sale.** If the system says milk stock is 0 but a customer is holding a carton, the *count* is wrong, not the customer.

Money in this section is Indian rupees (₹; written `Rs` in the program output) and uses `BigDecimal`, following the chapter's four rules.

Problem & clarifying questions

Question	Assumed answer
Who uses it?	Cashiers at several registers (lanes) in one store; a supervisor for voids and refunds. Self-checkout uses the same register API.
How are items identified?	Packaged goods by **barcode** (EAN-13, the standard 13-digit product barcode); loose produce by a 4-digit **PLU** (price look-up) code plus the scale's weight.
Are shelf prices tax-inclusive?	Yes (as in India, the UK and the EU): the receipt shows how much tax the price *contains*. A US-style store adds sales tax at the till — the design must support both.
Which offers?	Buy X get Y free, multi-buy ("3 for ₹80"), percent off a department, and a basket coupon ("₹50 off when you spend ₹500").
Can offers stack?	Each product gets at most **one** item offer — the best for the customer. A basket coupon applies on top, to what is still owed.
How do people pay?	Cash (with change), card, UPI — and **split tender** (part UPI, part cash).
Can the customer change their mind?	Void a line while scanning; go back to scanning if nothing is paid yet; cancel the whole sale (card/UPI payments reversed).
Returns?	Yes, with the receipt: refund what the customer **actually paid** for that item, never more units than bought.
Stock?	Decrease on sale, increase on return; alert when low; never refuse a sale because of the stock count.

Requirements

Functional

1. Catalog lookup by barcode or PLU; weighed items priced by weight.
2. A **sale** collects line items; lines can be voided (kept on the receipt, marked VOID).

3. Item offers and basket coupons, with clear stacking rules.
4. Totals with tax per tax category (inclusive or exclusive policy).
5. Split payment: cash (with change), card, UPI; a declined card changes nothing.
6. Immutable **receipt** with lines, discounts, tax breakdown, tenders and change.
7. Returns against a receipt, refunding the net price paid, at most the quantity bought.
8. Stock updates and **low-stock / negative-stock alerts**.

Non-functional

- **Exact money**: `BigDecimal`, 2 decimal places, `HALF_UP` rounding (a 5 or more rounds up), and parts (per-product discounts, refunds) that always add up to the total.
- **Several registers at once** share the catalog, the stock and the receipts safely.
- **Deterministic pricing**: the same basket always prices the same way (a **pure function**: its result depends only on its inputs, and it changes nothing outside itself) — needed for audits and for returns.
- **Auditable**: voids stay on the receipt; prices are copied into lines at scan time.

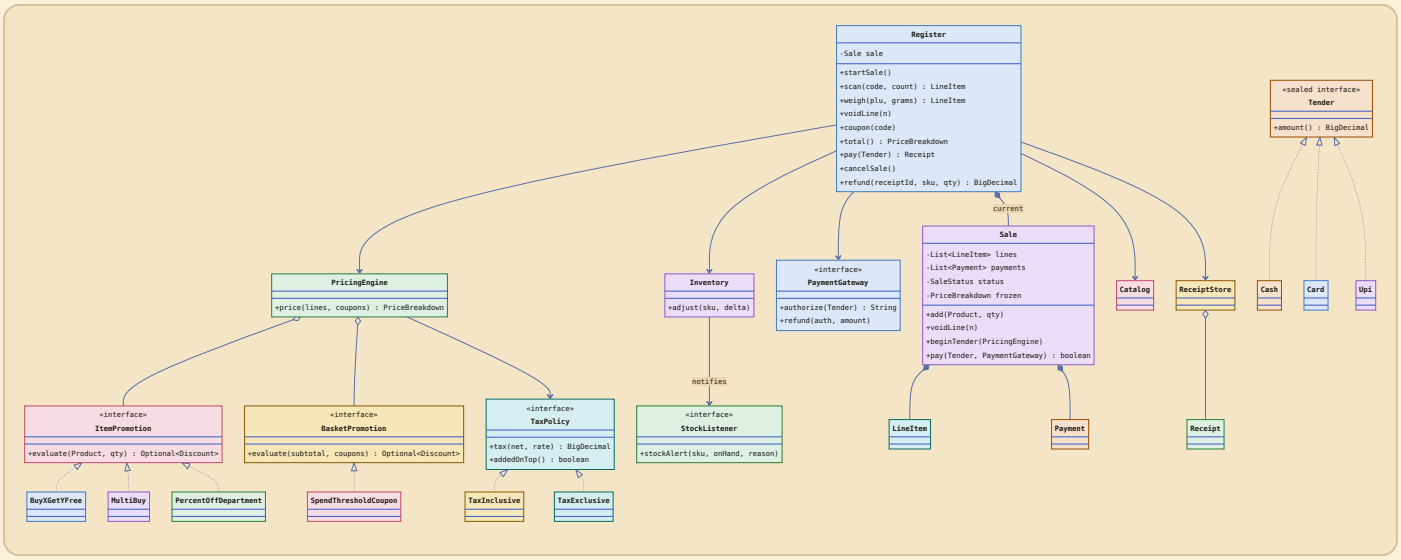
Use cases

Actor	Use cases
Cashier / self-checkout	start sale, scan, weigh, void line, add coupon, total, take payment, cancel sale
Customer	pay by cash / card / UPI, return items with the receipt
Supervisor	approve voids and returns (not modelled), reprice products
Payment network (system)	authorise and reverse card / UPI payments
Purchasing / store manager (subscriber)	receive low-stock and negative-stock alerts

Core entities

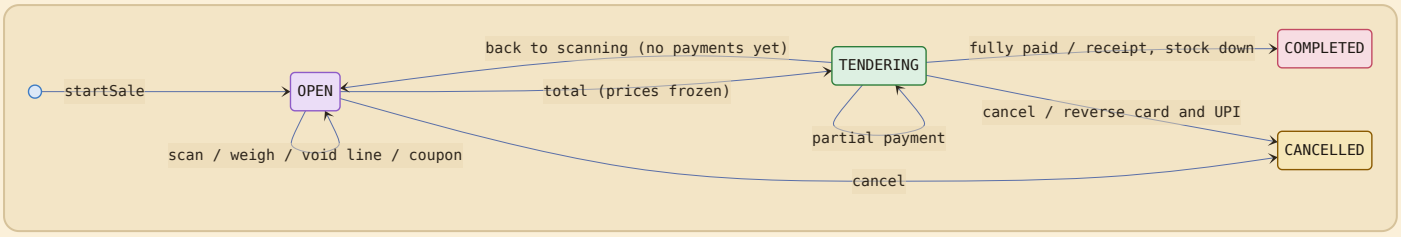
Class	Kind	Responsibility
`Product`	record	SKU (stock-keeping unit — the store's id for one distinct product), barcode/PLU, name, department, `PricingUnit` (EACH or PER_KG), unit price, `TaxCategory`.
`Catalog`	class	Lookup by code; `reprice` replaces the record (old sales keep old prices).
`LineItem`	record	One scan: product copy, quantity (count or kg), gross amount, voided flag.
`Sale` + `SaleStatus`	class + enum	OPEN → TENDERING → COMPLETED, or → CANCELLED; lines, coupons, payments, frozen price breakdown.
`ItemPromotion` + `BuyXGetYFree`, `MultiBuy`, `PercentOffDepartment`	interface + records	**Strategy** for item offers.
`BasketPromotion` + `SpendThresholdCoupon`	interface + record	**Strategy** for basket offers.
`TaxPolicy` + `TaxInclusive`, `TaxExclusive`	interface + classes	**Strategy** for how tax relates to the shelf price.
`PricingEngine` → `PriceBreakdown`	class → record	Pure function: lines + coupons → discounts, net per SKU, tax per category, total.
`Tender` (sealed) `Cash`, `Card`, `Upi`; `Payment`	records	How the customer pays; the part applied to the bill.
`PaymentGateway`	interface	Port (an interface through which the core code talks to an outside system) to the card / UPI networks (authorise, refund).
`Receipt`, `ReceiptStore`	record, class	Immutable proof of purchase; returns ledger per receipt and SKU.
`Inventory`, `StockListener`	class, interface	Shared stock; **Observer** for edge-triggered alerts (fired once when a line is crossed, not on every sale below it).
`Register`	class	**Facade** for one lane: the only API the till UI calls.

Class diagram

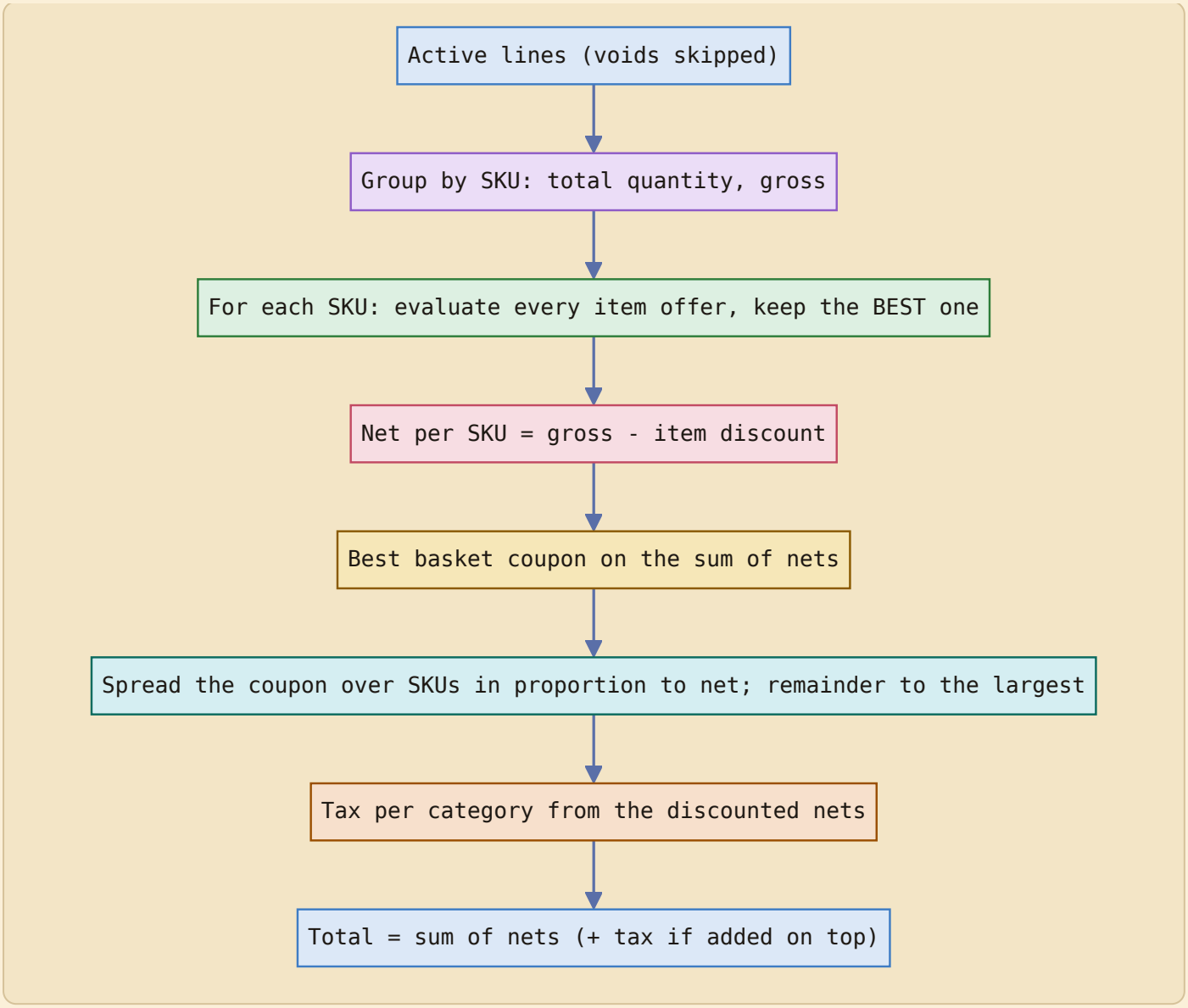


Key flows

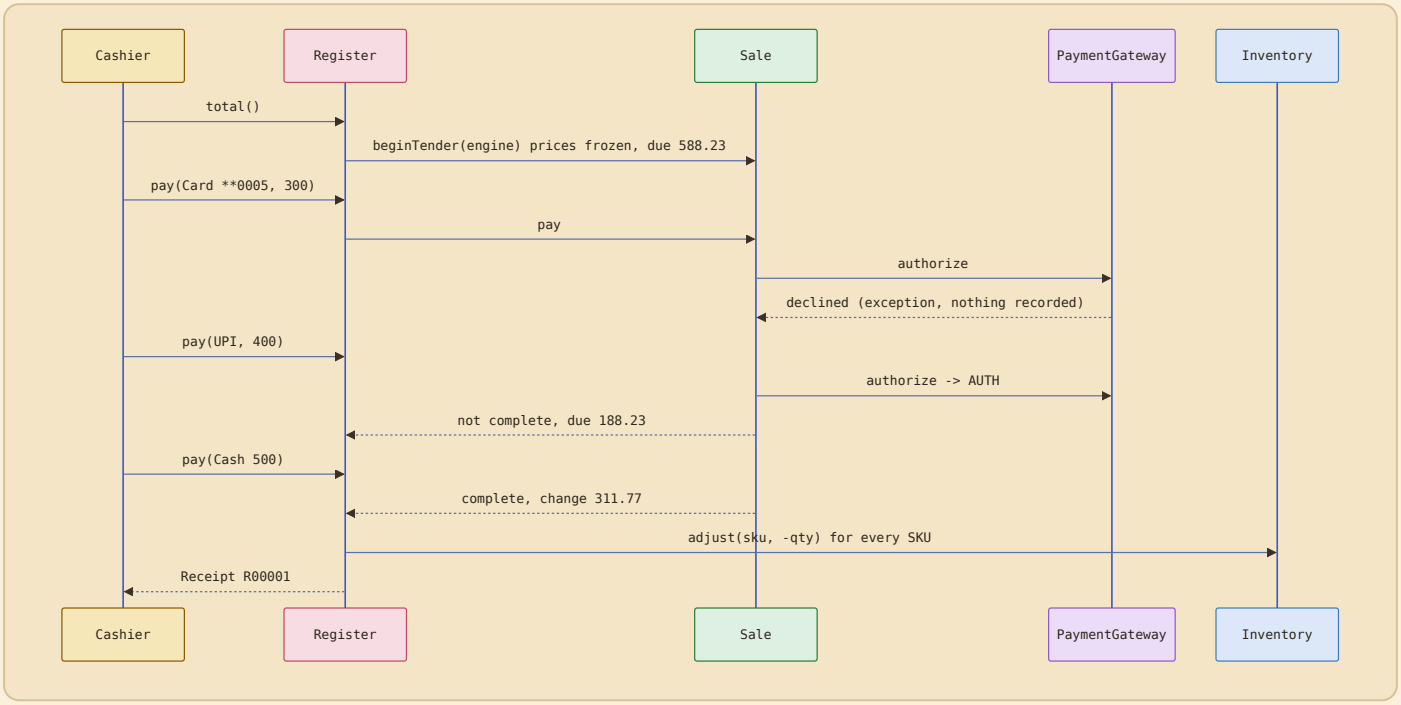
Sale lifecycle.



How one basket is priced (the hard method, `PricingEngine.price`):



Split payment.



Design decisions & patterns used

Decision	Pattern	Why	Alternative rejected
Each offer is a small class behind `ItemPromotion` / `BasketPromotion`	**Strategy**	Marketing adds offers weekly; a new offer is one new class, the engine does not change	A `switch` over offer types inside the pricing code
Best offer per SKU wins; basket coupon after item offers	Explicit stacking policy	Stacking rules are business decisions that must be written down once; "best for the customer" is simple to explain at the till	Applying every offer in list order (results depend on the order; can go below zero)
`PricingEngine` is a **pure function** of lines and coupons	Stateless service	Same basket → same price; one engine shared by all registers (stateless: it keeps no data between calls); easy to test	Mutating discounts into the line items as they are scanned
Prices **frozen** when payment starts	Snapshot	A price change or an offer expiring mid-payment must not change the amount due	Re-pricing on every payment
Coupon **spread over SKUs**, remainder to the largest	Allocation with exact rounding	Tax is per category and returns refund per item, so every rupee of discount must belong to some item — and the parts must add up exactly	Keeping the coupon as one unallocated number
Tax as a `TaxPolicy`	**Strategy**	Inclusive (India/EU) vs exclusive (US) is a store-level choice	Two versions of the engine
Tenders as a **sealed** interface	Sealed types	The receipt printer's `switch` is checked by the compiler	`String paymentType`
`Register` as the only API the UI calls	**Facade**	The UI does not need to know about engines, stores and gateways	UI code calling each part directly
Sale status in an enum with `canMoveTo`	Enum state machine	"Scan after payment started" and "pay a cancelled sale" fail loudly	Boolean flags (`isPaying`, `isDone`)
Stock may go **negative**, alerts are **edge-triggered**	**Observer**	The shelf is the truth; negative stock is a signal for a recount. Each crossing alerts once	Refusing the sale; alerting on every sale below the line
Voided lines are kept and printed	Audit trail (a permanent record of every action)	Voids are a common way to hide theft; supervisors review them	Deleting the line

Java implementation

Package `lld.ch19.grocery`.

```java
// File: lld/ch19/grocery/Catalog.java
package lld.ch19.grocery;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.util.Map;
import java.util.Optional;
import java.util.concurrent.ConcurrentHashMap;

/** Money helpers: BigDecimal, 2 decimal places, HALF_UP (the usual retail rounding). Never double. */
final class Money {
    private Money() { }
    static final BigDecimal ZERO = of("0");
    static BigDecimal of(String s) { return new BigDecimal(s).setScale(2, RoundingMode.HALF_UP); }
    static BigDecimal round(BigDecimal b) { return b.setScale(2, RoundingMode.HALF_UP); }
    static String fmt(BigDecimal b) { return "Rs " + round(b).toPlainString(); }
}

enum Department { PRODUCE, DAIRY, SNACKS, HOUSEHOLD }

/** How the shelf price is quoted: per piece, or per kilogram (weighed at the till). */
enum PricingUnit { EACH, PER_KG }

/** Tax slab. Rates are illustrative (loosely modelled on India's GST slabs), not tax advice. */
enum TaxCategory {
    EXEMPT("0"), REDUCED("5"), STANDARD("18");
    final BigDecimal ratePercent;
    TaxCategory(String r) { ratePercent = new BigDecimal(r); }
}

/** code = barcode (EAN-13) for packaged goods, or a 4-digit PLU for loose produce. */
record Product(String sku, String code, String name, Department dept, PricingUnit unit,
               BigDecimal unitPrice, TaxCategory tax) {
    Product {
        if (unitPrice.signum() <= 0) throw new PosException("Price must be positive: " + name);
        unitPrice = Money.round(unitPrice);
    }
}

/** Look-up by scanned code. Read far more than written, so a ConcurrentHashMap shared by all
registers. */
final class Catalog {
    private final Map<String, Product> byCode = new ConcurrentHashMap<>();

    void add(Product p) {
        if (byCode.putIfAbsent(p.code(), p) != null) throw new PosException("Duplicate code " +
p.code());
    }

    /** Price changes replace the record; open sales keep the price they already scanned. */
    void reprice(String code, BigDecimal newPrice) {
        byCode.computeIfPresent(code, (c, p) -> new Product(p.sku(), c, p.name(), p.dept(), p.unit(),
newPrice, p.tax()));
    }

    Optional<Product> find(String code) { return Optional.ofNullable(byCode.get(code)); }
}

final class PosException extends RuntimeException {
    PosException(String message) { super(message); }
}
```

```java
// File: lld/ch19/grocery/Promotions.java
package lld.ch19.grocery;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.util.Optional;
import java.util.Set;

/** One discount line on the receipt. sku == null means a basket-level discount. */
record Discount(String description, String sku, BigDecimal amount) { }

/** Strategy for an item-level offer. Called once per SKU with the total quantity of that SKU in the
basket. */
interface ItemPromotion {
    Optional<Discount> evaluate(Product p, BigDecimal quantity);
}

/** "Buy 2, get 1 free": in every group of (buy + free) units, the free units cost nothing. */
record BuyXGetYFree(String sku, int buy, int free) implements ItemPromotion {
    @Override public Optional<Discount> evaluate(Product p, BigDecimal qty) {
        if (!p.sku().equals(sku)) return Optional.empty();
        int groups = qty.intValue() / (buy + free);
        if (groups == 0) return Optional.empty();
        BigDecimal off = p.unitPrice().multiply(BigDecimal.valueOf((long) groups * free));
        return Optional.of(new Discount("Buy " + buy + " get " + free + " free: " + p.name(), sku,
Money.round(off)));
    }
}

/** "3 for Rs 100": each full group of n units costs a fixed price. */
record MultiBuy(String sku, int n, BigDecimal groupPrice) implements ItemPromotion {
    @Override public Optional<Discount> evaluate(Product p, BigDecimal qty) {
        if (!p.sku().equals(sku)) return Optional.empty();
        int groups = qty.intValue() / n;
        BigDecimal saving = p.unitPrice().multiply(BigDecimal.valueOf(n)).subtract(groupPrice);
        if (groups == 0 || saving.signum() <= 0) return Optional.empty();
        return Optional.of(new Discount(n + " for " + Money.fmt(groupPrice) + ": " + p.name(), sku,
                Money.round(saving.multiply(BigDecimal.valueOf(groups)))));
    }
}

/** "10% off all produce" — applies to weighed items too. */
record PercentOffDepartment(Department dept, int percent) implements ItemPromotion {
    @Override public Optional<Discount> evaluate(Product p, BigDecimal qty) {
        if (p.dept() != dept) return Optional.empty();
        BigDecimal gross = Money.round(p.unitPrice().multiply(qty));
        BigDecimal off = gross.multiply(BigDecimal.valueOf(percent)).divide(BigDecimal.valueOf(100),
2, RoundingMode.HALF_UP);
        return Optional.of(new Discount(percent + "% off " + dept + ": " + p.name(), p.sku(), off));
    }
}

/** Strategy for a basket-level offer, evaluated after item offers, on what the customer still owes.
*/
interface BasketPromotion {
    Optional<Discount> evaluate(BigDecimal subtotalAfterItemOffers, Set<String> couponsPresented);
}

/** "Rs 50 off when you spend Rs 500 or more" — needs the coupon code to be scanned. */
record SpendThresholdCoupon(String code, BigDecimal minSpend, BigDecimal amountOff) implements
BasketPromotion {
    @Override public Optional<Discount> evaluate(BigDecimal subtotal, Set<String> coupons) {
```

```
        if (!coupons.contains(code) || subtotal.compareTo(minSpend) < 0) return Optional.empty();
        return Optional.of(new Discount("Coupon " + code, null, amountOff.min(subtotal)));
    }
}
```

```java
// File: lld/ch19/grocery/Pricing.java
package lld.ch19.grocery;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.util.ArrayList;
import java.util.Comparator;
import java.util.EnumMap;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;
import java.util.Optional;
import java.util.Set;

/** Strategy: are shelf prices tax-inclusive (India, EU, UK) or is tax added at the till (US sales tax)? */
interface TaxPolicy {
    BigDecimal tax(BigDecimal net, BigDecimal ratePercent);
    boolean addedOnTop();
}

final class TaxInclusive implements TaxPolicy {
    /** The price already contains the tax: tax = price x r / (100 + r). */
    @Override public BigDecimal tax(BigDecimal net, BigDecimal r) {
        return net.multiply(r).divide(BigDecimal.valueOf(100).add(r), 2, RoundingMode.HALF_UP);
    }
    @Override public boolean addedOnTop() { return false; }
}

final class TaxExclusive implements TaxPolicy {
    @Override public BigDecimal tax(BigDecimal net, BigDecimal r) {
        return net.multiply(r).divide(BigDecimal.valueOf(100), 2, RoundingMode.HALF_UP);
    }
    @Override public boolean addedOnTop() { return true; }
}

/** All scanned lines of one SKU added together. */
record SkuTotal(Product product, BigDecimal quantity, BigDecimal gross) { }

/** Everything the receipt needs. netBySku = what the customer really paid per SKU, after every discount. */
record PriceBreakdown(List<SkuTotal> items, List<Discount> discounts, Map<String, BigDecimal> netBySku,
                      BigDecimal gross, BigDecimal discountTotal, Map<TaxCategory, BigDecimal> taxByCategory,
                      BigDecimal total) { }

/**
 * Pure function of (lines, coupons): no state, so every register can share one engine and
 * the same basket always prices the same way. Rules:
 *  1. each SKU gets at most ONE item offer — the best one for the customer;
 *  2. then the best basket coupon, on what is still owed;
 *  3. basket discounts are spread over SKUs in proportion to their value (needed for tax and refunds);
 *  4. tax is computed per tax category on the discounted amounts.
 */
final class PricingEngine {
    private final List<ItemPromotion> itemOffers;
    private final List<BasketPromotion> basketOffers;
    private final TaxPolicy taxPolicy;

    PricingEngine(List<ItemPromotion> itemOffers, List<BasketPromotion> basketOffers, TaxPolicy
```

```
taxPolicy) {
        this.itemOffers = List.copyOf(itemOffers);
        this.basketOffers = List.copyOf(basketOffers);
        this.taxPolicy = taxPolicy;
    }

    PriceBreakdown price(List<LineItem> lines, Set<String> coupons) {
        Map<String, SkuTotal> bySku = new LinkedHashMap<>();
        for (LineItem l : lines) {
            if (l.voided()) continue;
            bySku.merge(l.product().sku(), new SkuTotal(l.product(), l.quantity(), l.gross()),
                    (a, b) -> new SkuTotal(a.product(), a.quantity().add(b.quantity()),
a.gross().add(b.gross())));
        }
        List<Discount> discounts = new ArrayList<>();
        Map<String, BigDecimal> net = new LinkedHashMap<>();
        for (SkuTotal t : bySku.values()) {
            Optional<Discount> best = itemOffers.stream()
                    .map(p -> p.evaluate(t.product(), t.quantity()))
                    .flatMap(Optional::stream)
                    .max(Comparator.comparing(Discount::amount));
            BigDecimal off = best.map(d -> d.amount().min(t.gross())).orElse(Money.ZERO);
            best.ifPresent(d -> discounts.add(new Discount(d.description(), d.sku(), off)));
            net.put(t.product().sku(), t.gross().subtract(off));
        }
        BigDecimal subtotal = sum(net.values());
        Optional<Discount> basket = basketOffers.stream()
                .map(b -> b.evaluate(subtotal, coupons))
                .flatMap(Optional::stream)
                .max(Comparator.comparing(Discount::amount));
        basket.ifPresent(d -> { discounts.add(d); spread(d.amount(), net); });

        Map<TaxCategory, BigDecimal> netByTax = new EnumMap<>(TaxCategory.class);
        bySku.values().forEach(t -> netByTax.merge(t.product().tax(), net.get(t.product().sku()),
BigDecimal::add));
        Map<TaxCategory, BigDecimal> taxByCategory = new EnumMap<>(TaxCategory.class);
        netByTax.forEach((cat, amt) -> taxByCategory.put(cat, taxPolicy.tax(amt, cat.ratePercent)));

        BigDecimal gross = sum(bySku.values().stream().map(SkuTotal::gross).toList());
        BigDecimal netTotal = sum(net.values());
        BigDecimal total = taxPolicy.addedOnTop() ? netTotal.add(sum(taxByCategory.values())) :
netTotal;
        return new PriceBreakdown(List.copyOf(bySku.values()), List.copyOf(discounts),
Map.copyOf(net),
                gross, gross.subtract(netTotal), taxByCategory, total);
    }

    /** Proportional allocation; the rounding remainder goes to the largest SKU so the parts add up
exactly. */
    private static void spread(BigDecimal amount, Map<String, BigDecimal> net) {
        BigDecimal base = sum(net.values());
        if (base.signum() == 0) return;
        BigDecimal given = Money.ZERO;
        String largest = null;
        for (var e : net.entrySet()) {
            BigDecimal share = amount.multiply(e.getValue()).divide(base, 2, RoundingMode.HALF_UP);
            e.setValue(e.getValue().subtract(share));
            given = given.add(share);
            if (largest == null || e.getValue().compareTo(net.get(largest)) > 0) largest = e.getKey();
        }
        net.merge(largest, given.subtract(amount), BigDecimal::add);
    }
```

```
    private static BigDecimal sum(Iterable<BigDecimal> xs) {
        BigDecimal s = Money.ZERO;
        for (BigDecimal x : xs) s = s.add(x);
        return s;
    }
}
```

```java
// File: lld/ch19/grocery/Sale.java
package lld.ch19.grocery;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.LinkedHashSet;
import java.util.List;
import java.util.Set;

/** One scan. The product (and so its price) is copied in at scan time; later reprices do not change
it. */
record LineItem(int lineNo, Product product, BigDecimal quantity, BigDecimal gross, boolean voided) {
    LineItem asVoided() { return new LineItem(lineNo, product, quantity, gross, true); }
}

enum SaleStatus {
    OPEN, TENDERING, COMPLETED, CANCELLED;

    boolean canMoveTo(SaleStatus next) {
        return switch (this) {
            case OPEN -> next == TENDERING || next == CANCELLED;
            case TENDERING -> next == COMPLETED || next == OPEN || next == CANCELLED;
            case COMPLETED, CANCELLED -> false;
        };
    }
}

/** How the customer pays. Sealed: the compiler knows every kind of tender. */
sealed interface Tender permits Cash, Card, Upi { BigDecimal amount(); }
record Cash(BigDecimal amount) implements Tender { }
record Card(String last4, BigDecimal amount) implements Tender { }
record Upi(String vpa, BigDecimal amount) implements Tender { }

/** applied = the part of the tender that went towards the bill (cash can be more; the rest is
change). */
record Payment(Tender tender, BigDecimal applied, String authCode) { }

/** Strategy / port to the card and UPI networks. Cash needs no gateway. */
interface PaymentGateway {
    String authorize(Tender t);          // returns an auth code or throws PosException("declined
...")
    void refund(String authCode, BigDecimal amount);
}

/** One customer's basket at one register, from first scan to receipt. */
final class Sale {
    final String id;
    private final List<LineItem> lines = new ArrayList<>();
    private final Set<String> coupons = new LinkedHashSet<>();
    private final List<Payment> payments = new ArrayList<>();
    private SaleStatus status = SaleStatus.OPEN;
    private PriceBreakdown frozen;       // prices are frozen when payment starts
    private BigDecimal change = Money.ZERO;

    Sale(String id) { this.id = id; }

    SaleStatus status() { return status; }
    List<LineItem> lines() { return List.copyOf(lines); }
    List<Payment> payments() { return List.copyOf(payments); }
    Set<String> coupons() { return Set.copyOf(coupons); }
    BigDecimal change() { return change; }
    PriceBreakdown breakdown() { return frozen; }
```

```
    LineItem add(Product p, BigDecimal qty) {
        require(SaleStatus.OPEN, "scan");
        if (qty.signum() <= 0) throw new PosException("Quantity must be positive");
        LineItem l = new LineItem(lines.size() + 1, p, qty, Money.round(p.unitPrice().multiply(qty)),
false);
        lines.add(l);
        return l;
    }

    void voidLine(int lineNo) {
        require(SaleStatus.OPEN, "void a line");
        if (lineNo < 1 || lineNo > lines.size()) throw new PosException("No line " + lineNo);
        LineItem l = lines.get(lineNo - 1);
        if (l.voided()) throw new PosException("Line " + lineNo + " is already void");
        lines.set(lineNo - 1, l.asVoided());          // keep the line: the audit trail shows the void
    }

    void addCoupon(String code) { require(SaleStatus.OPEN, "add a coupon"); coupons.add(code); }

    PriceBreakdown beginTender(PricingEngine engine) {
        require(SaleStatus.OPEN, "start payment");
        if (lines.stream().allMatch(LineItem::voided)) throw new PosException("Nothing to pay for");
        frozen = engine.price(lines, coupons);
        move(SaleStatus.TENDERING);
        return frozen;
    }

    /** Back to scanning ("I forgot the milk") — only while nothing has been paid. */
    void reopen() {
        if (!payments.isEmpty()) throw new PosException("Payments already taken; cancel them first");
        frozen = null;
        move(SaleStatus.OPEN);
    }

    BigDecimal due() {
        BigDecimal paid = payments.stream().map(Payment::applied).reduce(Money.ZERO, BigDecimal::add);
        return frozen.total().subtract(paid);
    }

    /** Returns true when the bill is fully paid and the sale is complete. */
    boolean pay(Tender t, PaymentGateway gateway) {
        require(SaleStatus.TENDERING, "pay");
        BigDecimal due = due();
        if (t.amount().signum() <= 0) throw new PosException("Tender must be positive");
        String auth = "CASH";
        BigDecimal applied = t.amount();
        if (t instanceof Cash) {
            if (t.amount().compareTo(due) > 0) { change = t.amount().subtract(due); applied = due; }
        } else {
            if (t.amount().compareTo(due) > 0) throw new PosException("Card/UPI cannot be more than
the amount due " + Money.fmt(due));
            auth = gateway.authorize(t);                // throws if declined: nothing recorded
        }
        payments.add(new Payment(t, applied, auth));
        if (due().signum() == 0) { move(SaleStatus.COMPLETED); return true; }
        return false;
    }

    /** Cancel the whole sale; card/UPI payments already taken are reversed. */
    void cancel(PaymentGateway gateway) {
        if (!status.canMoveTo(SaleStatus.CANCELLED)) throw new PosException("Cannot cancel a " +
```

```
status + " sale");
        for (Payment p : payments) if (!(p.tender() instanceof Cash)) gateway.refund(p.authCode(),
p.applied());
        move(SaleStatus.CANCELLED);
    }

    private void require(SaleStatus s, String action) {
        if (status != s) throw new PosException("Cannot " + action + " when the sale is " + status);
    }

    private void move(SaleStatus next) {
        if (!status.canMoveTo(next)) throw new PosException(status + " -> " + next + " not allowed");
        status = next;
    }
}
```

```java
// File: lld/ch19/grocery/Register.java
package lld.ch19.grocery;

import java.math.BigDecimal;
import java.math.RoundingMode;
import java.util.List;
import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;
import java.util.concurrent.CopyOnWriteArrayList;
import java.util.concurrent.atomic.AtomicLong;

/** Observer: purchasing, the store manager's dashboard, shrinkage reports. */
interface StockListener {
    void stockAlert(String sku, BigDecimal onHand, String reason);
}

/**
 * Store stock shared by every register. A till never refuses a sale because the SYSTEM says zero:
 * the item is physically in the customer's hand, so the count is what is wrong. Stock may go
 * negative, and that is reported as a discrepancy for a recount.
 */
final class Inventory {
    private final Map<String, BigDecimal> onHand = new ConcurrentHashMap<>();
    private final Map<String, BigDecimal> reorderPoint = new ConcurrentHashMap<>();
    private final List<StockListener> listeners = new CopyOnWriteArrayList<>();

    void set(String sku, String qty, String reorderAt) {
        onHand.put(sku, new BigDecimal(qty));
        reorderPoint.put(sku, new BigDecimal(reorderAt));
    }
    void addListener(StockListener l) { listeners.add(l); }
    BigDecimal onHand(String sku) { return onHand.getOrDefault(sku, BigDecimal.ZERO); }

    /** delta < 0 for a sale, > 0 for a return. merge() is atomic per key, so registers never lose
updates. */
    void adjust(String sku, BigDecimal delta) {
        BigDecimal after = onHand.merge(sku, delta, BigDecimal::add);
        BigDecimal before = after.subtract(delta);
        BigDecimal point = reorderPoint.getOrDefault(sku, BigDecimal.ZERO);
        // edge-triggered: alert only when THIS change crosses the line, so each crossing alerts
exactly once
        if (before.compareTo(point) > 0 && after.compareTo(point) <= 0)
            listeners.forEach(l -> l.stockAlert(sku, after, "LOW - reorder"));
        if (before.signum() >= 0 && after.signum() < 0)
            listeners.forEach(l -> l.stockAlert(sku, after, "NEGATIVE - recount the shelf"));
    }
}

/** Immutable proof of purchase. */
record Receipt(String id, String registerId, List<LineItem> lines, PriceBreakdown breakdown,
               List<Payment> payments, BigDecimal change) {

    String render() {
        StringBuilder sb = new StringBuilder();
        sb.append("  ---------------- RECEIPT ").append(id).append(" (").append(registerId).append(")
----------------\n");
        for (LineItem l : lines) {
            String qty = l.product().unit() == PricingUnit.PER_KG
                    ? l.quantity().setScale(3, RoundingMode.UNNECESSARY) + " kg @ " +
Money.fmt(l.product().unitPrice()) + "/kg"
                    : l.quantity().toPlainString() + " x " + Money.fmt(l.product().unitPrice());
            sb.append("  %2d %-18s %-24s %12s%s%n".formatted(l.lineNo(), l.product().name(), qty,
```

```
Money.fmt(l.gross()),
                    l.voided() ? "  VOID" : ""));
        }
        sb.append("  %-46s %12s%n".formatted("Gross", Money.fmt(breakdown.gross())));
        for (Discount d : breakdown.discounts())
            sb.append("  %-46s %12s%n".formatted("  " + d.description(), "-" +
Money.fmt(d.amount())));
        sb.append("  %-46s %12s%n".formatted("TOTAL", Money.fmt(breakdown.total())));
        breakdown.taxByCategory().forEach((cat, tax) ->
                sb.append("  %-46s %12s%n".formatted("  incl. tax " + cat + " @" + cat.ratePercent +
"%", Money.fmt(tax))));
        for (Payment p : payments) sb.append("  %-46s %12s%n".formatted("Paid " +
describe(p.tender()), Money.fmt(p.tender().amount())));
        if (change.signum() > 0) sb.append("  %-46s %12s%n".formatted("Change", Money.fmt(change)));
        return sb.toString();
    }

    private static String describe(Tender t) {
        return switch (t) {
            case Cash c -> "cash";
            case Card c -> "card **" + c.last4();
            case Upi u -> "UPI " + u.vpa();
        };
    }
}

/** Receipts from every register, plus what has already been returned against each (shared, thread-
safe). */
final class ReceiptStore {
    private final Map<String, Receipt> receipts = new ConcurrentHashMap<>();
    private final Map<String, BigDecimal> returned = new ConcurrentHashMap<>();   // key receiptId/sku
    private final AtomicLong nextId = new AtomicLong(1);

    String newId() { return "R%05d".formatted(nextId.getAndIncrement()); }
    void save(Receipt r) { receipts.put(r.id(), r); }
    Receipt get(String id) {
        Receipt r = receipts.get(id);
        if (r == null) throw new PosException("Unknown receipt " + id);
        return r;
    }

    /** Atomically claims qty units for return (two counters cannot refund the same unit twice);
returns the total claimed so far. */
    BigDecimal claimReturn(Receipt r, String sku, BigDecimal qty, BigDecimal bought) {
        return returned.compute(r.id() + "/" + sku, (k, already) -> {
            BigDecimal next = (already == null ? BigDecimal.ZERO : already).add(qty);
            if (next.compareTo(bought) > 0)
                throw new PosException("Only " +
bought.subtract(next.subtract(qty)).stripTrailingZeros().toPlainString()
                        + " of " + sku + " left to return on " + r.id());
            return next;
        });
    }
}

/**
 * Facade for one checkout lane. A register is driven by one cashier (one thread); the things it
 * shares with other registers — catalog, inventory, receipts — are the thread-safe ones.
 */
final class Register {
    private final String id;
    private final Catalog catalog;
```

```
    private final PricingEngine engine;
    private final Inventory inventory;
    private final ReceiptStore receipts;
    private final PaymentGateway gateway;
    private Sale sale;
    private int saleNo = 1;

    Register(String id, Catalog catalog, PricingEngine engine, Inventory inventory, ReceiptStore
receipts, PaymentGateway gateway) {
        this.id = id; this.catalog = catalog; this.engine = engine;
        this.inventory = inventory; this.receipts = receipts; this.gateway = gateway;
    }

    void startSale() {
        if (sale != null && (sale.status() == SaleStatus.OPEN || sale.status() ==
SaleStatus.TENDERING))
            throw new PosException("Finish or cancel sale " + sale.id + " first");
        sale = new Sale(id + "-S" + saleNo++);
    }

    LineItem scan(String code) { return scan(code, 1); }

    LineItem scan(String code, int count) {
        Product p = lookup(code);
        if (p.unit() != PricingUnit.EACH) throw new PosException(p.name() + " is sold by weight: put
it on the scale");
        return current().add(p, BigDecimal.valueOf(count));
    }

    /** Loose produce: the cashier keys the PLU and the scale sends the weight in grams. */
    LineItem weigh(String plu, int grams) {
        Product p = lookup(plu);
        if (p.unit() != PricingUnit.PER_KG) throw new PosException(p.name() + " is not sold by
weight");
        if (grams <= 0) throw new PosException("Scale reading must be positive");
        return current().add(p, BigDecimal.valueOf(grams).movePointLeft(3));
    }

    void voidLine(int lineNo) { current().voidLine(lineNo); }
    void coupon(String code) { current().addCoupon(code); }
    PriceBreakdown total() { return current().beginTender(engine); }
    void backToScanning() { current().reopen(); }
    void cancelSale() { current().cancel(gateway); }

    /** Returns the receipt when the bill is fully paid, else null (more tenders needed). */
    Receipt pay(Tender t) {
        Sale s = current();
        if (!s.pay(t, gateway)) return null;
        for (SkuTotal item : s.breakdown().items()) inventory.adjust(item.product().sku(),
item.quantity().negate());
        Receipt r = new Receipt(receipts.newId(), id, s.lines(), s.breakdown(), s.payments(),
s.change());
        receipts.save(r);
        return r;
    }

    /** Refund what the customer actually paid per unit (after offers and coupons), not the shelf
price. */
    BigDecimal refund(String receiptId, String sku, int qty) {
        Receipt r = receipts.get(receiptId);
        SkuTotal bought = r.breakdown().items().stream().filter(t ->
t.product().sku().equals(sku)).findFirst()
```

```
                .orElseThrow(() -> new PosException(sku + " is not on receipt " + receiptId));
        BigDecimal q = BigDecimal.valueOf(qty);
        BigDecimal after = receipts.claimReturn(r, sku, q, bought.quantity());
        // refund = share(after) - share(before): the parts always add up to exactly what was paid
        BigDecimal net = r.breakdown().netBySku().get(sku);
        BigDecimal refund = share(net, after, bought.quantity()).subtract(share(net,
after.subtract(q), bought.quantity()));
        inventory.adjust(sku, q);
        return refund;
    }

    private static BigDecimal share(BigDecimal net, BigDecimal units, BigDecimal of) {
        return net.multiply(units).divide(of, 2, RoundingMode.HALF_UP);
    }

    private Product lookup(String code) {
        if (code.length() == 13 && !Ean13.valid(code)) throw new PosException("Bad barcode " + code +
": check digit wrong, rescan");
        return catalog.find(code).orElseThrow(() -> new PosException("Unknown code " + code + ": call
a supervisor"));
    }

    private Sale current() {
        if (sale == null) throw new PosException("No sale in progress");
        return sale;
    }
}

/** EAN-13 check digit: weights 1,3,1,3... on the first 12 digits; the 13th makes the sum a multiple
of 10. */
final class Ean13 {
    private Ean13() { }
    static String withCheckDigit(String first12) {
        int sum = 0;
        for (int i = 0; i < 12; i++) sum += (first12.charAt(i) - '0') * (i % 2 == 0 ? 1 : 3);
        return first12 + (10 - sum % 10) % 10;
    }
    static boolean valid(String code) {
        return code.chars().allMatch(Character::isDigit) && withCheckDigit(code.substring(0,
12)).equals(code);
    }
}
```

```java
// File: lld/ch19/grocery/Main.java
package lld.ch19.grocery;

import java.math.BigDecimal;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.concurrent.CountDownLatch;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.atomic.AtomicInteger;

public class Main {
    /** Fake card/UPI network: declines one blocked card, approves everything else. */
    static final class FakeGateway implements PaymentGateway {
        private final AtomicInteger seq = new AtomicInteger(1000);
        @Override public String authorize(Tender t) {
            if (t instanceof Card c && c.last4().equals("0005")) throw new PosException("Card **0005
declined by issuer");
            return "AUTH" + seq.incrementAndGet();
        }
        @Override public void refund(String auth, BigDecimal amount) {
            System.out.println("   gateway: reversed " + Money.fmt(amount) + " on " + auth);
        }
    }

    static final String MILK = Ean13.withCheckDigit("890100000001"), PANEER =
Ean13.withCheckDigit("890100000002"),
            CHIPS = Ean13.withCheckDigit("890100000003"), BISCUITS =
Ean13.withCheckDigit("890100000004"),
            DETERGENT = Ean13.withCheckDigit("890100000005"), BANANAS = "4011", TOMATOES = "4664";

    public static void main(String[] args) throws Exception {
        Catalog catalog = new Catalog();
        catalog.add(new Product("BAN", BANANAS, "Bananas", Department.PRODUCE, PricingUnit.PER_KG,
Money.of("60"), TaxCategory.EXEMPT));
        catalog.add(new Product("TOM", TOMATOES, "Tomatoes", Department.PRODUCE, PricingUnit.PER_KG,
Money.of("40"), TaxCategory.EXEMPT));
        catalog.add(new Product("MLK", MILK, "Milk 1L", Department.DAIRY, PricingUnit.EACH,
Money.of("68"), TaxCategory.EXEMPT));
        catalog.add(new Product("PNR", PANEER, "Paneer 200g", Department.DAIRY, PricingUnit.EACH,
Money.of("90"), TaxCategory.REDUCED));
        catalog.add(new Product("CHP", CHIPS, "Chips 50g", Department.SNACKS, PricingUnit.EACH,
Money.of("20"), TaxCategory.STANDARD));
        catalog.add(new Product("BIS", BISCUITS, "Biscuits", Department.SNACKS, PricingUnit.EACH,
Money.of("30"), TaxCategory.REDUCED));
        catalog.add(new Product("DET", DETERGENT, "Detergent 1kg", Department.HOUSEHOLD,
PricingUnit.EACH, Money.of("245"), TaxCategory.STANDARD));

        List<ItemPromotion> offers = List.of(
                new BuyXGetYFree("CHP", 2, 1),                        // saves 20 on 3 packs
                new MultiBuy("CHP", 3, Money.of("50")),               // saves 10 on 3 packs: loses to
the one above
                new MultiBuy("BIS", 3, Money.of("80")),
                new PercentOffDepartment(Department.PRODUCE, 10));
        List<BasketPromotion> coupons = List.of(new SpendThresholdCoupon("SAVE50", Money.of("500"),
Money.of("50")));
        PricingEngine engine = new PricingEngine(offers, coupons, new TaxInclusive());

        Inventory stock = new Inventory();
        for (String sku : List.of("BAN", "TOM", "MLK", "CHP", "BIS", "DET")) stock.set(sku, "200",
"20");
```

```
        stock.set("PNR", "100", "20");
        List<String> alerts = Collections.synchronizedList(new ArrayList<>());
        stock.addListener((sku, left, why) -> alerts.add(sku + " " + why + " (on hand " +
left.stripTrailingZeros().toPlainString() + ")"));
        ReceiptStore receipts = new ReceiptStore();
        FakeGateway gateway = new FakeGateway();
        Register lane1 = new Register("LANE-1", catalog, engine, stock, receipts, gateway);

        System.out.println("== 1. Scan a basket ==");
        lane1.startSale();
        lane1.weigh(BANANAS, 1245);
        lane1.weigh(TOMATOES, 500);
        lane1.scan(MILK);
        lane1.scan(MILK);
        lane1.scan(PANEER);
        lane1.scan(CHIPS, 3);
        lane1.scan(BISCUITS, 4);
        lane1.scan(DETERGENT);
        lane1.voidLine(4);                                          // customer puts one milk back
        for (String bad : List.of(Ean13.withCheckDigit("890100000009"), "8901000000018", BANANAS)) {
            try { lane1.scan(bad); } catch (PosException e) { System.out.println("   error: " +
e.getMessage()); }
        }
        lane1.coupon("SAVE50");

        System.out.println("\n== 2. Total, then split payment: declined card, UPI, cash with change
==");
        PriceBreakdown bill = lane1.total();
        System.out.println("   amount due " + Money.fmt(bill.total()));
        try { lane1.scan(MILK); } catch (PosException e) { System.out.println("   error: " +
e.getMessage()); }
        try { lane1.pay(new Card("0005", Money.of("300"))); } catch (PosException e) {
System.out.println("   error: " + e.getMessage()); }
        try { lane1.pay(new Upi("asha@okbank", Money.of("5000"))); } catch (PosException e) {
System.out.println("   error: " + e.getMessage()); }
        System.out.println("   UPI 400 -> receipt? " + lane1.pay(new Upi("asha@okbank",
Money.of("400"))));
        Receipt r1 = lane1.pay(new Cash(Money.of("500")));
        System.out.print(r1.render());

        System.out.println("\n== 3. Returns refund what was really paid ==");
        System.out.println("   return 1 of 3 chips -> refund " + Money.fmt(lane1.refund(r1.id(),
"CHP", 1)));
        System.out.println("   return 2 more chips -> refund " + Money.fmt(lane1.refund(r1.id(),
"CHP", 2))
                + "  (all three add up to the " + Money.fmt(r1.breakdown().netBySku().get("CHP")) + "
paid)");
        try { lane1.refund(r1.id(), "CHP", 1); } catch (PosException e) { System.out.println("
error: " + e.getMessage()); }
        try { lane1.refund(r1.id(), "PNR", 2); } catch (PosException e) { System.out.println("
error: " + e.getMessage()); }

        System.out.println("\n== 4. Price change during a sale; going back to scanning; cancelling
==");
        lane1.startSale();
        lane1.scan(MILK);
        catalog.reprice(MILK, Money.of("72"));
        lane1.scan(MILK);
        SkuTotal milk = lane1.total().items().get(0);
        System.out.println("   2 milk scanned around a reprice: gross " + Money.fmt(milk.gross()) + "
= Rs 68 (old price kept) + Rs 72 (new)");
        lane1.backToScanning();
```

```
        lane1.scan(PANEER);
        System.out.println("   after adding paneer, due " + Money.fmt(lane1.total().total()));
        lane1.pay(new Card("4242", Money.of("100")));
        try { lane1.backToScanning(); } catch (PosException e) { System.out.println("   error: " +
e.getMessage()); }
        lane1.cancelSale();
        try { lane1.pay(new Cash(Money.of("10"))); } catch (PosException e) { System.out.println("
error: " + e.getMessage()); }

        System.out.println("\n== 5. Same basket, US-style tax added at the till ==");
        Sale same = new Sale("X");
        r1.lines().forEach(l -> { if (!l.voided()) same.add(l.product(), l.quantity()); });
        same.addCoupon("SAVE50");
        PriceBreakdown us = same.beginTender(new PricingEngine(offers, coupons, new TaxExclusive()));
        System.out.println("   inclusive total " + Money.fmt(r1.breakdown().total()) + " vs exclusive
total " + Money.fmt(us.total())
                + " (tax " + us.taxByCategory() + ")");

        System.out.println("\n== 6. Four registers sell 30 paneer each at once; stock was 100 ==");
        CountDownLatch go = new CountDownLatch(1);
        try (ExecutorService pool = Executors.newFixedThreadPool(4)) {
            for (int i = 1; i <= 4; i++) {
                Register lane = new Register("LANE-" + (i + 1), catalog, engine, stock, receipts,
gateway);
                pool.submit(() -> {
                    go.await();
                    for (int k = 0; k < 30; k++) {
                        lane.startSale();
                        lane.scan(PANEER);
                        lane.total();
                        lane.pay(new Cash(Money.of("100")));
                    }
                    return null;
                });
            }
            go.countDown();
        }
        System.out.println("   paneer on hand: " + stock.onHand("PNR") + "  (sold 120 + 1 earlier; the
till never refuses a real item)");
        alerts.stream().sorted().forEach(a -> System.out.println("   alert: " + a));
    }
}
```

How the code works, part by part

- **Scanning.** `Register.scan` checks the EAN-13 check digit (a last digit computed from the others, so a misread code fails the sum) first (a misread barcode is caught before lookup), then refuses weighed products; `weigh` converts grams to kilograms with `movePointLeft(3)` so 1245 g becomes exactly `1.245`. `Sale.add` copies the product — and therefore its price — into the new `LineItem`.
- **Pricing.** `PricingEngine.price` skips voided lines, groups by SKU, asks every item offer for a discount and keeps the largest (capped at the gross), applies the best basket coupon to what is left, spreads that coupon across SKUs, and computes tax per category on the discounted amounts. With `TaxInclusive`, tax = amount × r ÷ (100 + r); with `TaxExclusive`, tax = amount × r ÷ 100 and is added to the total.
- **Paying.** `beginTender` freezes the breakdown. `pay` treats cash specially: if it is more than the amount due, the extra becomes change. Card and UPI may not exceed the amount due and must be authorised; a declined authorisation throws before anything is recorded. When the amount due reaches zero, the sale completes and the register writes the receipt and reduces stock.

- **Returns.** `ReceiptStore.claimReturn` uses `ConcurrentHashMap.compute` so "check how many were already returned, then add" is atomic. The refund for a return is `share(after) − share(before)`, where `share(k)` is the net paid for the first *k* units; that way three separate one-item returns add up to exactly what was paid.
- **Stock.** `Inventory.adjust` uses `merge`, which is atomic per key; it compares the value before and after *this* change, so each crossing of the reorder point — the stock level at which more should be ordered — (or of zero) alerts exactly once, however many registers sell at the same time.

Output:

```
== 1. Scan a basket ==
   error: Unknown code 8901000000095: call a supervisor
   error: Bad barcode 8901000000018: check digit wrong, rescan
   error: Bananas is sold by weight: put it on the scale

== 2. Total, then split payment: declined card, UPI, cash with change ==
   amount due Rs 588.23
   error: Cannot scan when the sale is TENDERING
   error: Card **0005 declined by issuer
   error: Card/UPI cannot be more than the amount due Rs 588.23
   UPI 400 -> receipt? null
  ---------------- RECEIPT R00001 (LANE-1) ----------------
   1 Bananas            1.245 kg @ Rs 60.00/kg       Rs 74.70
   2 Tomatoes           0.500 kg @ Rs 40.00/kg       Rs 20.00
   3 Milk 1L            1 x Rs 68.00                 Rs 68.00
   4 Milk 1L            1 x Rs 68.00                 Rs 68.00  VOID
   5 Paneer 200g        1 x Rs 90.00                 Rs 90.00
   6 Chips 50g          3 x Rs 20.00                 Rs 60.00
   7 Biscuits           4 x Rs 30.00                Rs 120.00
   8 Detergent 1kg      1 x Rs 245.00               Rs 245.00
  Gross                                            Rs 677.70
    10% off PRODUCE: Bananas                        -Rs 7.47
    10% off PRODUCE: Tomatoes                       -Rs 2.00
    Buy 2 get 1 free: Chips 50g                    -Rs 20.00
    3 for Rs 80.00: Biscuits                       -Rs 10.00
    Coupon SAVE50                                  -Rs 50.00
  TOTAL                                            Rs 588.23
    incl. tax EXEMPT @0%                            Rs 0.00
    incl. tax REDUCED @5%                           Rs 8.78
    incl. tax STANDARD @18%                        Rs 40.07
  Paid UPI asha@okbank                             Rs 400.00
  Paid cash                                        Rs 500.00
  Change                                           Rs 311.77

== 3. Returns refund what was really paid ==
   return 1 of 3 chips -> refund Rs 12.29
   return 2 more chips -> refund Rs 24.58  (all three add up to the Rs 36.87 paid)
   error: Only 0 of CHP left to return on R00001
   error: Only 1 of PNR left to return on R00001

== 4. Price change during a sale; going back to scanning; cancelling ==
   2 milk scanned around a reprice: gross Rs 140.00 = Rs 68 (old price kept) + Rs 72 (new)
   after adding paneer, due Rs 230.00
   error: Payments already taken; cancel them first
   gateway: reversed Rs 100.00 on AUTH1002
   error: Cannot pay when the sale is CANCELLED

== 5. Same basket, US-style tax added at the till ==
   inclusive total Rs 588.23 vs exclusive total Rs 644.73 (tax {EXEMPT=0.00, REDUCED=9.22,
STANDARD=47.28})

== 6. Four registers sell 30 paneer each at once; stock was 100 ==
   paneer on hand: -21  (sold 120 + 1 earlier; the till never refuses a real item)
   alert: PNR LOW - reorder (on hand 20)
   alert: PNR NEGATIVE - recount the shelf (on hand -1)
```

Reading the output: the misread barcode (wrong check digit), an unknown code and scanning bananas without the scale are all refused. On the receipt, chips qualify for both "buy 2 get 1 free" (saves ₹20) and "3 for ₹50" (saves ₹10); only the better one is applied. The ₹50 coupon applied because the basket was still over ₹500 after item offers. The total ₹588.23 contains ₹8.78 + ₹40.07 of tax; the

same basket under a US-style policy costs ₹644.73. The declined card and the over-large UPI amount changed nothing; UPI ₹400 plus cash ₹500 gave ₹311.77 change. Returning the chips one, then two, refunded ₹12.29 + ₹24.58 = ₹36.87 — exactly the net paid after the offer and the coupon share, not the shelf price of ₹60. The re-priced milk shows that a line keeps the price it was scanned at. Four registers selling 30 paneer each drove stock from 99 to −21 without refusing anyone, and the low and negative alerts each fired once.

Concurrency & edge cases

Case	Handling
Four registers sell the same product at once	`Inventory.adjust` uses `ConcurrentHashMap.merge` (atomic per key); no lost updates
Stock count says zero but the item is in hand	Sale goes through; stock goes negative; one "recount" alert
Alert spam when stock is low	Edge-triggered: alert only on the change that crosses the line
Two service desks refund the same receipt line at once	`ReceiptStore.claimReturn` uses `compute`: check-and-add is atomic; the second over-claim is refused
Price changes during a sale	Lines keep the price copied at scan; the frozen breakdown fixes the amount due
Offer changes during payment	Breakdown frozen at `total()`; going back to scanning re-prices
Card declined	Exception from the gateway before anything is recorded
Cancel after a card payment	`cancel` reverses every non-cash payment through the gateway
Rounding pennies in coupon allocation and refunds	Remainder to the largest SKU; refunds as differences of cumulative shares
One register used by one cashier	`Register` and `Sale` are single-threaded by design; shared objects (`Catalog`, `Inventory`, `ReceiptStore`) are the thread-safe ones

Real-world example

Supermarket tills identify loose produce by **PLU codes** — 4011 is the standard code for yellow bananas, and tomatoes have their own codes in the 4000 range — and read the weight from an integrated scale, exactly like `weigh(plu, grams)`. Packaged goods carry EAN-13 barcodes whose last digit is a check digit, which is why a smudged barcode is rejected rather than misread as another product. Receipts in tax-inclusive countries (for example GST — Goods and Services Tax — invoices in India or VAT — value-added tax — receipts in the UK) print how much tax the total *contains*, while US receipts add sales tax below the subtotal — our two `TaxPolicy` strategies. Retailers also run periodic "cycle counts" to fix the kind of negative stock our `Inventory` reports.

Extensions & follow-up questions

- **"Loyalty points and member prices."** A `CustomerContext` passed to the engine; member-only offers check it. Points are earned on the *net* paid (after discounts), from the receipt.
- **"Weighed items with a pre-packed label."** In-store printed barcodes that encode the price or weight in the digits (a variable-measure barcode); the scanner decodes them into a `LineItem` directly.
- **"Offers that span products (buy pasta + sauce, save ₹20)."** A `BundlePromotion` that looks at several SKUs; the "one offer per unit" rule then needs a small optimiser (try the combinations, keep the best) instead of best-per-SKU.
- **"Refund to the original payment method."** Store tender splits on the receipt; refund card/UPI parts through the gateway first, then cash.

- **"The network is down."** Offline mode: queue card authorisations (with a floor limit — a maximum amount accepted without online approval), keep selling, reconcile (match up the records) later; the catalog and offers are cached locally in each register.
- **"End-of-day."** A `Shift` per cashier: opening float (the cash put in the drawer at the start), cash counted at close, expected cash = float + cash sales − change − cash refunds.
- **"Self-checkout."** The same `Register` facade with a weight-check step at the bagging area (weight mismatch → assistance needed).

In the interview

Plan for 45 minutes: clarify offers, tax, tenders, returns (5) → entities and the Register facade (8) → sale state machine (5) → **pricing engine** with the stacking rules (12) → split payment and receipt (7) → returns, stock and concurrency (8).

> **30-second answer:** "A Register facade drives one Sale through OPEN, TENDERING, COMPLETED or CANCELLED. Lines copy the price at scan time; produce is priced by weight from the scale. A pure pricing engine groups lines by SKU, gives each SKU its single best offer, applies a basket coupon, spreads it over SKUs so tax and refunds are exact, and computes tax through a pluggable inclusive or exclusive policy. The price is frozen at total; tenders are a sealed type, cash can give change, card and UPI go through a gateway. On completion an immutable receipt is stored and shared stock is reduced atomically, with edge-triggered alerts — and the till never refuses a real item."

Follow-ups with answers

- *Why not apply all offers that match?* Stacking can give away more than intended or even a negative price; the business has to choose a rule, and "best single offer per product, then one basket coupon" is simple and explainable.
- *Why spread the coupon over items?* Tax differs per category and returns must refund the price actually paid for that item.
- *Why is the pricing engine stateless?* So every register can share it and the same basket always gives the same answer — also what makes returns reproducible.

Self-check

1. **Q:** Three chips qualify for "buy 2 get 1 free" and "3 for ₹50". What is applied? **A:** Only the larger discount (₹20), because each SKU gets its single best item offer.
2. **Q:** Why does a return of one chip refund ₹12.29, not ₹20? **A:** The refund is the net price paid — after the offer and the item's share of the coupon.
3. **Q:** What stops two service desks refunding the same unit? **A:** `ConcurrentHashMap.compute` makes "check already returned, add" one atomic step.
4. **Q:** Why is stock allowed to go negative? **A:** The physical item is in the customer's hand; the count is wrong, so we sell and raise a recount alert.
5. **Q:** When are prices frozen, and why? **A:** At `total()`, so price or offer changes during payment cannot change the amount due.
6. **Q:** Which patterns appear? **A:** Strategy (offers, tax, payment gateway), Observer (stock alerts), Facade (Register), enum state machine (sale), sealed types (tenders).

TL;DR

- **Register facade** → one **Sale** state machine; lines copy prices; produce priced by **weight** (PLU + scale).
- **Pure pricing engine**: best item offer per SKU → basket coupon → spread over SKUs → tax per category via a **TaxPolicy** strategy.

- **Freeze** prices at total; **split tender** with a sealed `Tender`; declined cards change nothing; cancel reverses non-cash payments.
- Returns refund the **net paid**, claimed atomically; stock updates are atomic, may go negative, alerts are **edge-triggered**.

Chapter summary

Problem	Hard part	The move that solves it
Amazon	Guests vs members; last-item races; cancel rules	Ordering only on `Member` (LSP); atomic all-or-nothing stock reservation; order state machine
Brokerage	Four order types; tax lots; over-spending	Two abstract questions on `Order`; `StockLot` + FIFO `LotSale`; reserved funds
Stock exchange	Fair matching	Per-symbol `TreeMap<price, FIFO queue>`; price then time; one lock per symbol
Auction	Concurrent bids	All bid rules in one per-listing critical section; notify after unlocking
Music streaming	Shuffle/repeat without losing order	Play order as a strategy-produced permutation of indexes; plan rules in the enum
Food delivery / FoodKart	Capacity and riders under concurrency	Reserve kitchen slots atomically; CAS-claim riders; strategies for assignment and selection
Grocery POS	Weighed items, stacking offers, split tender, returns	Pure pricing engine (best offer per SKU, coupon spread, tax per category); frozen price; atomic return claims; edge-triggered stock alerts

Patterns that kept coming back: Strategy (payments, play order, rider assignment, restaurant selection, trade-price policy, grocery offers and tax), Observer (notifications, fills, plays), state machines in enums (orders, auctions, player), Template Method (statements), Facade (every `*Service`), and one concurrency rule — **make check-then-act atomic** (lock, or compare-and-set).

Sources combined in this chapter: the DesignGurus *Grokking the Object-Oriented Design Interview* lessons on Amazon and the stock brokerage (requirements, actors, staged class design and diagrams); the awesome-low-level-design problem statements for the auction system, music streaming service and food delivery service; and the kumaransg/LLD machine-coding statements for the Navi stock exchange and Flipkart's FoodKart. All Java code is original. The grocery point-of-sale case is designed from first principles, using public retail conventions (GS1 EAN-13 check digits, IFPS PLU codes for produce, tax-inclusive vs sales-tax receipts).